MATH2601 – Higher Linear Algebra

Chapter 1. Groups and Fields

1.1 Groups

Definition 1.1 A group G is a non-empty set with a binary operation $*$ defined on it, such that

(1) for all $a, b \in G$, the composition $a * b$ is uniquely defined and $a * b \in G$ (closure);

(2) $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$ (associativity);

(3) there is an element $e \in G$ such that $a * e = a = e * a$ for all $a \in G$ (e is called the identity of the group);

(4) for each $a \in G$ there is an element $a' \in G$ such that $a * a' = e = a' * a$ (we call a' an inverse of a).

If G is a finite set then the order of G is $|G|$, the number of elements in G.

Technically, the group is the pair $(G, *)$. We sometimes say that G is a group under the operation $*$.

Definition 1.2 A group G is abelian if the operation $*$ is commutative: $a * b = b * a$ for all $a, b \in G$.

Notes:

A binary operation is just a function $* : G \times G \to G$, but we write $a * b$ instead of $*(a, b)$. Property (1) in Definition 1.1 just says that $*$ is a well-defined binary operation on G.

Fact: the identity e is unique and the inverse of each $a \in G$ is unique. (We state this properly in a lemma soon.)

We often write $a * a * \cdots * a = a^n$ if there are n copies of a on the left hand side.

If the operation $*$ is some kind of multiplication $\times$ then we often write ab instead of $a \times b$, write the identity as 1 and write a^{-1} for the inverse of a.

If the operation $*$ is some kind of addition $+$, we usually write 0 for the identity and $-a$ for the inverse of a (and call it the negative of a.

Then we can write $a + a + \cdots + a = na$ if there are n copies of a on the left hand side. Here $n \in \mathbb{N}$ and $a \in G$. (The notation na is just shorthand for repeated addition, not multiplication!)

The group $\{e\}$ with one element is called the trivial group. This is the smallest possible group, since G must be nonempty.

Example 1.1

(a) $(\mathbb{Z}, +)$ is an abelian group under integer addition. The identity is 0 and a has inverse $-a$ for all $a \in \mathbb{Z}$.

(b) $\mathbb{Z}$ is not a group under multiplication - why not?

(c) Are there any subsets of $\mathbb{R}$ that form a non-trivial group under the usual multplication in $\mathbb{R}$?

(d) For any $m \in \mathbb{N}$, the set $\mathbb{Z}_m = \{0, 1, 2, \ldots, m-1\}$ of remainders modulo m is a group under addition modulo m.

(e) If p is prime then $\mathbb{Z}_p^* = \mathbb{Z}_p - \{0\}$ is a group under multiplication modulo p.
(This does not work for non-primes: why not?)

All these examples are abelian. How can we find some non-abelian groups?

Example 1.2
For any set S, the set F of bijective functions $f : S \to S$ is a group under composition.

Proof:

$\square$

Exercise: Find a set S such that $(F, \circ)$ is not abelian.

Groups arise from the study of symmetries.

Example 1.3

Let T be an equilateral triangle in $\mathbb{R}^2$. A symmetry of T is a map from $\mathbb{R}^2$ to $\mathbb{R}^2$ which preserves distances and maps T to itself. Let G be the set of symmetries of T.

- The composition of two elements of G is in G (closure).
- The composition is associative (see Example 1.2).
- There is an identity element (the identity map).
- Every element of G has an inverse given by reversing the mapping.

Hence G forms a group under composition. It is the dihedral group of order 6, usually written as D_6.

You can check that $|G| = 6$ and every element of G is the identity map, a rotation or a reflection in $\mathbb{R}^2$.

Lemma 1.1 Let $(G, *)$ be a group.
(a) There is only one identity element in G.
(b) Each element of G has only one inverse.
(c) $(a^{-1})^{-1} = a$ for all $a \in G$.
(d) $(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$.
(e) Let $a, b, c, \in G$. If $a * b = a * c$ then $b = c$.
Similarly, if $b * a = c * a$ then $b = c$. (Cancellation law.)

Proof.

$\square$

Example 1.4 Let m be a postive integer and consider the set

$$C_m = \{e, a, a^2, a^3, \ldots, a^{m-1}\},$$

where a is some (undefined) symbol. Define a binary operation $*$ on C_m by

$$a^k * a^\ell = \begin{cases} a^{k+\ell} & \text{if } k + \ell < m, \\ a^{k+\ell-m} & \text{otherwise,} \end{cases}$$

with the convention that $a^0 = e$.
This is the usual multiplication of powers with the additional assumption that $a^m = e$.

Exercise: Show that $(C_m, *)$ is an abelian group. It is the cyclic group of order m.

We often write C_m as $\langle a : a^m = e\rangle$ and say it is generated by a.

Permutation Groups

Let $[n] = \{1, 2, \ldots, n\}$. There are $n!$ ways to rearrange, or permute, the elements of $[n]$.

For example, there are 6 permutations of $[3]$:

$$1\,2\,3; \qquad 1\,3\,2; \qquad 2\,1\,3; \qquad 2\,3\,1; \qquad 3\,1\,2; \qquad 3\,2\,1.$$

Each permutation of $[n]$ is a bijection $\sigma : [n] \to [n]$. For example, the last permutation in the above list is the function

$$\sigma : 1 \mapsto 3, \quad 2 \mapsto 2, \quad 3 \mapsto 1$$

which we can also write as

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

Proposition 1.2 The set $\mathcal{S}_n$ of all permutations of n objects forms a group under composition. This group has order $n!$.

Proof. This follows from Example 1.2 and by counting the number of permutations of $[n]$. $\square$

We call $\mathcal{S}_n$ the symmetric group on n objects.

Later in the course we use permutations to define the determinant of a matrix.

Small groups can be pictured using a multiplication table, where the row element is multiplied on the left of the column element.

$$\begin{array}{c|cc} + & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array} \qquad\qquad \begin{array}{c|ccc} * & e & a & b \\ \hline e & e & a & b \\ a & a & b & e \\ b & b & e & a \end{array} \tag{1}$$

The multiplication table on the left shows $\mathbb{Z}_2$ under addition modulo 2.

The multiplication table on the right is for a group of order 3 with identity element e. Since $b = a^2$ and $ab = a^3 = e$, this group is C_3, the cyclic group of order 3.

In a multiplication table of a finite group G, each row must be a permutation of the elements of G.

- If we had a repetition in a row, for example $xa = xb$, then the cancellation rule will give $a = b$. The same holds for columns.

So each element occurs at most once in each row, and hence it must occur exactly once, since G is finite. The same is true for columns.

- Furthermore: if $ab = b$ then multiplying by b^{-1} gives $a = e$.

Hence the row corresponding to the identity is the only one that contains any fixed elements (and similarly for columns).

With these rules, the tables of (1) are the only possibilities for a group of order 2 or order 3.

Exercise. Let $G = \{I, A, B, C\}$ where

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \ A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \ B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \ C = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Show that G is an abelian group under matrix multiplication and write out its multiplication table.

Solution: Write out the multiplication table first:

	I	A	B	C
I				
A				
B				
C				

If the multiplication table is symmetric (as a matrix) then the group is abelian.

1.2 Fields

A field is a set of numbers (like $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$) where we can add, subtract, multiply and divide in a nice way.

Definition 1.3 A field $(\mathbb{F}, +, \times)$ is a set $\mathbb{F}$ together with two binary operations, addition $(+)$ and multiplication $(\times)$ defined on $\mathbb{F}$, such that

(a) $(\mathbb{F}, +)$ is an abelian group;

(b) $\mathbb{F}^* = \mathbb{F} - \{0\}$ is an abelian group under multiplication (where 0 is the additive identity);

(c) The distributive laws hold: for all $a, b, c \in \mathbb{F}$,

$a \times (b + c) = a \times b + a \times c$ and $(a + b) \times c = a \times c + b \times c$.

We usually just refer to "the field $\mathbb{F}$" if the operations are the obvious ones.

Notes:

We write 0 for the additive identity in $\mathbb{F}$ and 1 for the multiplicative identity in $\mathbb{F}$.

The smallest possible field has two elements: it is $\mathbb{F} = \{0, 1\}$ where $1 + 1 = 0$.

The standard examples of fields are the sets $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ with the familar operations.

For convenience, we write $a - b$ instead of $a + (-b)$, and write a/c instead of ac^{-1} when $c \neq 0$.

Lemma 1.1 implies that $-(-a) = a$. The cancellation laws imply that if $ab = ac$ and $a \neq 0$ then $b = c$.

Example 1.6 Let p be any prime.
From the results of Example 1.1, to prove that the set $\mathbb{Z}_p$ is a field we only need to check the distributive laws.

But proving that $a(b+c) = ab + ac$ in $\mathbb{Z}_p$ is the same as proving that

$$a(b+c) - ab - ac \equiv 0 \pmod{p}$$

which follows trivially from the distributive laws in $\mathbb{Z}$.
The other cancellation law follows similarly. $\diamond$

Fact: If $\mathbb{F}$ is a finite field (that is, a field with a finite number of elements) then $|\mathbb{F}| = p^k$ for some prime p and positive integer k. The prime p is called characteristic of the field.

This is the Galois field of size p^k, written as $\mathrm{GF}(p^k)$.
Note: If $\mathrm{GF}(p^k) = \mathbb{Z}_{p^k}$ then $k = 1$.

Lemma 1.3 Let $\mathbb{F}$ be a field and let $a, b, c \in \mathbb{F}$. Then

(a) $a0 = 0$;

(b) $a(-b) = -(ab)$;

(c) $a(b - c) = ab - ac$;

(d) if $ab = 0$ then either $a = 0$ or $b = 0$.

Proof: (of d) – the others are left as an exercise for you.

$\square$

1.3 Subgroups and subfields

When you define a new kind of mathematical object, like a group, you can also define sub-objects and (later) maps between objects. We start with a subgroup.

Definition 1.3

Let $(G, *)$ be a group and let H be a non-empty subset of G. If H is a group under the restriction of $*$ to H then we say that H is a subgroup of G.

We write this as $H \leq G$ and say that H inherits the group structure from G.

Exercise.

Prove that any subgroup of an abelian group is abelian.

Lemma 1.4 (The Subgroup Lemma)
Let $(G, *)$ be a group and let H be a non-empty subset of G.

Then H is a subgroup of G if and only if

(a) $a * b \in H$ for all $a, b \in H$ (closed under $*$);
(b) $a^{-1} \in H$ for all $a \in H$ (closed under inverses).

Proof:

$\square$

Example 1.7

(a) Every non-trivial group G has at least two subgroups: $\{e\}$ and G.

(b) For any $m \in \mathbb{N}$ let $m\mathbb{Z}$ be the set of all multiples of m. Then $(m\mathbb{Z}, +)$ is a subgroup of $(\mathbb{Z}, +)$.

Conversely, every subgroup of $(\mathbb{Z}, +)$ is $m\mathbb{Z}$ for some $m \in \mathbb{N}$.

(c) For $m \in \mathbb{Z}^+$ consider $C_m = \langle a : a^m = e \rangle$, the cyclic group of order m.

Take any integer k with $1 < k < m$ and let $H_k = \langle a^k \rangle$ be the set of all powers of a^k. Then H_k is a subgroup of C_m.

Clearly the order of H_k is at most m.

In fact, the order of H_k is strictly less than m if and only if k and m have a common divisor $d > 1$. Exercise: try with C_6.

Example 1.8 Let n be any positive integer.
The set of invertible $n \times n$ matrices over field $\mathbb{F}$ is a group under matrix multiplication, called the general linear group and denoted $\mathrm{GL}(n, \mathbb{F})$.

It is a special case of Example 1.2 with $S = \mathbb{F}^n$, and is non-abelian if $n > 1$.

The groups $\mathrm{GL}(n, \mathbb{R})$ and $\mathrm{GL}(n, \mathbb{C})$ are especially important in this course. They have many important subgroups, such as

- the special linear groups $\mathrm{SL}(n, \mathbb{R})$ and $\mathrm{SL}(n, \mathbb{C})$ of matrices with determinant 1.
- the group $\mathrm{O}(n)$ of orthogonal matrices, $\mathrm{O}(n) \leq GL(n, \mathbb{R})$.
- the group $\mathrm{SO}(n) = \mathrm{O}(n) \cap \mathrm{SL}(n, \mathbb{R})$ of special orthogonal matrices.

Definition 1.4 If $(\mathbb{F}, +, \times)$ is a field and $\mathbb{E} \subseteq \mathbb{F}$ is also a field under the same operations restricted to $\mathbb{E}$, then we say that $(\mathbb{E}, +, \times)$ is a subfield of $(\mathbb{F}, +, \times)$. We write this as $\mathbb{E} \leq \mathbb{F}$.

Lemma 1.5 (The Subfield Lemma) Let $\mathbb{E} \neq \{0\}$ be a non-empty subset of field $\mathbb{F}$.

Then $\mathbb{E}$ is a subfield of $\mathbb{F}$ if and only if for all $a, b \in \mathbb{E}$:

$$a + b \in \mathbb{E}, \quad -b \in \mathbb{E}, \quad a \times b \in \mathbb{E}, \quad b^{-1} \in \mathbb{E} \text{ if } \ b \neq 0\,.$$

Proof. The distributive laws are inherited from $\mathbb{F}$ to $\mathbb{E}$.

The rest of the proof follows from applying the Subgroup Lemma (Lemma 1.4) to both $(\mathbb{E}, +)$ and $(\mathbb{E}^*, \times)$. $\square$

Example 1.9 Let α be any irrational real or complex number. We define $\mathbb{Q}(\alpha)$ to be the smallest field which contains both $\mathbb{Q}$ and α. These fields are very important in number theory.

Fact: $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$.

It is not too hard to prove that $\{a + b\sqrt{2}\ :\ a, b \in \mathbb{Q}\}$ is a subfield of $\mathbb{R}$. (Closure under division takes the most thought.)

The hard part is to prove that $\{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is the smallest field which contains both $\mathbb{Q}$ and $\sqrt{2}$.

1.4 Morphisms

Now we want to define nice maps between our objects.

Definition 1.5 Let $(G, *)$ and $(H, \circ)$ be two groups.
A (group) homomorphism from G to H is a map $\phi : G \to H$ that respects the two group operations; that is,

$$\phi(a * b) = \phi(a) \circ \phi(b) \quad \text{for all} \quad a, b \in G\,.$$

A bijective homomorphism $\phi : G \to H$ is called an isomorphism. If there is an isomorphism between G and H then we say that G and H are isomorphic.

Question: How would you define a field homomorphism?

If two groups are isomorphic, then they are really the same group, at least as far as group theory is concerned.

Fact: isomorphism is an equivalence relation on groups. (Exercise for those of you who know what an equivalence relation is: prove the above fact.)

Example 1.10 Let $m \geq 2$ be any integer. Recall the set $m\mathbb{Z}$ from Example 1.7. Define $\phi : (\mathbb{Z}, +) \to (m\mathbb{Z}, +)$ by $\phi(a) = ma$.

Show that ϕ is a group isomorphism.
Proof.

$\square$

Example 1.11
For any $m \in \mathbb{Z}^+$ let Z_m be the set of all mth roots of unity. Then Z_m under multiplication is a subgroup of $\mathbb{C}^* = \mathbb{C} - \{0\}$.

Let $z_m = e^{2\pi i/m}$. From first year, we know that

$$Z_m = \{1, z_m, z_m^2, \ldots, z_m^{m-1}\}.$$

Now consider the bijection $\phi : C_m \to Z_m$ given by $\phi(a^k) = z_m^k$. We see that ϕ is a homomorphism: if $p + q < m$ then

The calculations for the $p + q > m$ are left as an exercise.

Hence C_m and Z_m are isomorphic.

Lemma 1.6 Let $(G, *)$ and $(H, \circ)$ be two groups and let $\phi : G \to H$ be a homomorphism between them. Then

(a) ϕ maps the identity of G to the identity of H.

(b) $\phi(a^{-1}) = (\phi(a))^{-1}$ for all $a \in G$.

(c) if ϕ is an isomorphism from G to H then ϕ^{-1} is an isomorphism from H to G.

Proof: (of (a))

Exercise: prove (b) and (c). $\square$

Definition 1.6 Let $\phi : G \to H$ be a group homomorphism and let e' be the identity of H.

The kernel of ϕ is the set

$$\ker(\phi) = \{g \in G \, : \, \phi(g) = e'\}.$$

The image of ϕ is the set

$$\mathrm{im}(\phi) = \{h \in H \, : \, h = \phi(g) \text{ for some } g \in G\}.$$

Equivalently, we can define the image as follows:

$$\mathrm{im}(\phi) = \{\phi(g) \, : \, g \in G\}.$$

Lemma 1.7 Let $\phi : G \to H$ be a group homomorphism. Then $\ker(\phi)$ is a subgroup of G and $\operatorname{im}(\phi)$ is a subgroup of H.
Proof: (for kernel)

Exercise: Complete the proof: that is, prove that $\operatorname{im}(\phi) \leq H$.

$\square$

Lemma 1.8
A homomorphism ϕ is one-to-one if and only if $\ker(\phi) = \{e\}$, where e is the identity of G.

If ϕ is one-to-one then $\mathrm{im}(\phi)$ is isomorphic to G.

Proof. $\square$

A common use of group homomorphisms is to look for a homomorphism $\phi : G \to \mathrm{GL}(n, \mathbb{F})$ for some $n \in \mathbb{Z}^+$ and some field $\mathbb{F}$.

The group $\mathrm{im}(\phi)$ is called a (linear) representation of G on $\mathbb{F}^n$.

If ϕ is one-to-one then the representation is said to be faithful. Example 1.11 gave a faithful representations of C_m on $\mathbb{C}$.

Example 1.12 Let $C_4 = \{e, a, a^2, a^3\}$, so $a^4 = e$.

Let $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and define $\phi : C_4 \to \mathrm{GL}(2, \mathbb{R})$ by $\phi(a^k) = J^k$.

Exercise: Prove that ϕ is a faithful representation of C_4 on $\mathbb{R}^2$.

People are also interested in one-to-one homomorphisms $\phi : G \to \mathcal{S}_n$ for some $n \in \mathbb{Z}^+$, when G is a finite group. This gives a permutation representation of G as a subgroup of $\mathcal{S}_n$.

British mathematician Arthur Cayley (1821–1895) proved that every finite group G has a permutation representation as a subgroup of $\mathcal{S}_n$ for some $n \leq |G|$.

So the symmetric groups $\mathcal{S}_n$ and their subgroups capture all the other finite groups!

Key points for Chapter 1

- Group, abelian group, p. 2, p. 3
- Cyclic group, p. 9
- Symmetric group, p. 11
- Field, p. 15
- Subgroups, p. 19
- Subgroup Lemma, p. 20
- Matrix groups, p. 22
- Homomorphism, isomorphism p. 25
- Kernel and image, p. 29
- Linear representation, p. 32

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Chapter 2. Vector Spaces

The concept of a vector space is a natural and important generalisation of $\mathbb{R}^n$.

Whenever you can add objects and multiply objects by scalars then it is natural to ask "is the set a vector space?"

For example, we can think of a field $\mathbb{F}$ as a vector space over one of its subfields.

Fact: $\mathbb{C}$ is a vector space over $\mathbb{R}$.

Fact: The set

$$\mathbb{Q}(i) = \{\alpha + \beta i \,:\, \alpha, \beta \in \mathbb{Q}\}$$

is a subfield of $\mathbb{C}$ which can be thought of as a vector space over $\mathbb{Q}$.

2.1 Vector spaces

Definition 2.1 Let $\mathbb{F}$ be a field. A vector space over the field $\mathbb{F}$ consists of an abelian group $(V, +)$, together with a function from $\mathbb{F} \times V$ to V, $(\alpha, \boldsymbol{v}) \mapsto \alpha\boldsymbol{v}$, called scalar multiplication, such that

(a) $\alpha(\beta\boldsymbol{v})=(\alpha\beta)\boldsymbol{v}$ for all $\alpha, \beta \in \mathbb{F}$ and all $\boldsymbol{v} \in V$.

(b) $1\boldsymbol{v} = \boldsymbol{v}$ for all $\boldsymbol{v} \in V$.

(c) $\alpha(\mathbf{u} + \mathbf{v})=\alpha\mathbf{u} + \alpha\mathbf{v}$ for all $\alpha \in \mathbb{F}$ and all $\boldsymbol{u}, \boldsymbol{v} \in V$.

(d) $(\alpha + \beta)\mathbf{u}= \alpha\mathbf{u} + \beta\mathbf{u}$ for all $\alpha, \beta \in \mathbb{F}$ and all $\boldsymbol{u} \in V$.

Notes.

- There are ten axioms here: 5 from the abelian group, closure of scalar multiplication and four axioms (a)–(d).
- We call the addition in V vector addition if we need to distinguish it from the addition in $\mathbb{F}$.

- Since V is a group, V cannot be empty.
- I will use **boldface** letters for vectors, e.g. $\boldsymbol{u}$, $\boldsymbol{v}$ (or underlining when writing by hand). This helps us to distinguish elements of V from elements of $\mathbb{F}$, which we call scalars.

The identity element of $(V, +)$ is written as $\mathbf{0}$ to distinguish it from the zero scalar, $0 \in \mathbb{F}$.

- The two distributive rules (c) and (d) are, respectively distributivity of scalar multiplication over vector addition and distributivity of field addition over scalar multiplication.
- All the results we had for abelian groups in Chapter 1 (e.g. uniqueness of zero vector and negatives, cancellation law) apply to vector addition.

Now we explore how vector addition and scalar multiplication combine.

Lemma 2.1 Let V be a vector space over field $\mathbb{F}$.
For all $\boldsymbol{v}, \boldsymbol{w} \in V$ and $\lambda \in \mathbb{F}$:

(a) $0\boldsymbol{v}=\mathbf{0}$ and $\lambda\mathbf{0} = \mathbf{0}$.

(b) $(-1)\boldsymbol{v}=-\boldsymbol{v}$ for all $\boldsymbol{v} \in V$.

(c) if $\lambda\boldsymbol{v} = \mathbf{0}$ then $\lambda = 0$ or $\boldsymbol{v} = \mathbf{0}$.

(d) if $\lambda\boldsymbol{v} = \lambda\boldsymbol{w}$ and $\lambda \neq 0$ then $\boldsymbol{v} = \boldsymbol{w}$.

Proof. (For (c))

Exercise: Prove the other statements.

$\square$

Standard examples of vector spaces

Example: The set $\mathbb{F}^n$ consists of all *n*-tuples of elements of $\mathbb{F}$:

$$\mathbb{F}^n = \left\{ \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} : \alpha_i \in \mathbb{F} \right\}.$$

If $\boldsymbol{x} = (\alpha_i)_{1 \le i \le n}$ and $\boldsymbol{y} = (\beta_i)_{1 \le i \le n}$ are elements of $\mathbb{F}^n$, then vector addition on $\mathbb{F}^n$ is defined as $\boldsymbol{x} + \boldsymbol{y} = (\alpha_i + \beta_i)_{1 \le i \le n}$.

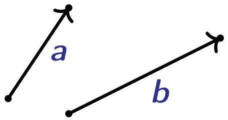
Scalar multiplication on $\mathbb{F}^n$ is $\lambda \mathbf{x} = (\lambda \alpha_i)_{1 \le i \le n}$.

With these operations, $\mathbb{F}^n$ is a vector space over $\mathbb{F}$.

You should already be friends with $\mathbb{R}^n$ from first year.

Example: Geometric vectors

Geometric vectors are ordered pairs of points in $\mathbb{R}^n$, joined by labelled arrows.



We add these objects by placing them head to tail.
Multiplying by a scalar λ is just stretching the vector's length by a factor of λ, while preserving the direction if $\lambda \geq 0$, or reversing it if $\lambda < 0$.

The set of all geometric vectors in $\mathbb{R}^n$ does not form a vector space. (Why not?)

However, if you define two geometric vectors to be equivalent if one is a translation of the other, then the set of equivalence classes of geometric vectors is a vector space.

Example: Matrices
For any $p, q \in \mathbb{Z}^+$, let $M_{p,q}(\mathbb{F})$ be the set of $p \times q$ matrices with elements from $\mathbb{F}$.

Then $M_{p,q}(\mathbb{F})$ is a vector space over $\mathbb{F}$ with the usual addition of matrices as the vector addition, and scalar multiplication performed by multiplying each element of the matrix by the scalar.

Example: Polynomials
The set of all polynomials with coefficients in $\mathbb{F}$, denoted $\mathcal{P}(\mathbb{F})$, is a vector space over $\mathbb{F}$ with operations

$$(f+g)(x) = f(x) + g(x) \quad \text{for all } x \in \mathbb{F},$$
$$(\lambda f)(x) = \lambda f(x) \quad \text{for all } \lambda, x \in \mathbb{F}.$$

Similarly, the set $\mathcal{P}_n(F)$ of polynomials in $\mathcal{P}(\mathbb{F})$ with degree at most n is a vector space over $\mathbb{F}$.

Example: Function Spaces

Let X be a nonempty set and let $\mathbb{F}$ be a field.Then define

$$\mathcal{F}\left[X\right] = \left\{f : X \to \mathbb{F}\right\}.$$

The set $\mathcal{F}\left[X\right]$ is a vector space over $\mathbb{F}$:

- Let the zero in $\mathcal{F}[X]$ be the zero function which maps $x \mapsto 0$ for all $x \in X$.
- Given $f, g \in \mathcal{F}[X]$, define $f + g$ by
$$(f+g)(x) = f(x) + g(x) \text{ for all } x \in X.$$
- Given $f \in \mathcal{F}[X]$ and $\lambda \in \mathbb{F}$, define λf by $(\lambda f)(x) = \lambda\left(f(x)\right)$ for all $x \in X$.

Exercise: How do you define

- $-f$ for $f \in \mathcal{F}[X]$?
- $\lambda(f+g)$ for $f, g \in \mathcal{F}[X]$, $\lambda \in \mathbb{F}$?

Example: A non-standard vector space

Let $V = \mathbb{R}^+ = (0, \infty)$ be the set of positive real numbers.

Define addition $\oplus$ and scalar multiplication $\otimes$ on V by

$$\boldsymbol{v} \oplus \boldsymbol{w} = \boldsymbol{v}\boldsymbol{w}, \qquad \alpha \otimes \boldsymbol{v} = \boldsymbol{v}^{\alpha}.$$

With these operations, V is a vector space over $\mathbb{R}$ whose vector addition is multiplication and whose scalar multiplication is exponentiation. (!!!)

Exercise (if you are feeling brave):
Prove that this is a vector space.

We will revisit this example in the tutorials.

2.2 Subspaces

Definition 2.2. Let V be a vector space over $\mathbb{F}$ and let $U \subseteq V$. Then U is a subspace of V, written $U \leq V$, if U is a vector space over $\mathbb{F}$ with the same addition and scalar multiplication as in V.

Every vector space has $\{\mathbf{0}\}$ (the trivial subspace) and itself as subspaces.

Example: $\mathcal{P}_n(\mathbb{R})$ is a subspace of $\mathcal{P}(\mathbb{R})$, which is a subspace of $\mathcal{F}[\mathbb{R}] = \{f : \mathbb{R} \to \mathbb{R}\}$.

You may recall from first year that to prove that U is a subspace, it enough to check that U is nonempty and closed under the two operations.

We can combine the two closure requirements into one . . .

Lemma 2.2 (Subspace Test Lemma)
Let V be a vector space over the field $\mathbb{F}$ and suppose that U is a nonempty subset of V.
Then U is a subspace of V if and only if $\alpha\mathbf{u} + \mathbf{v} \in U$ for all $\boldsymbol{u}, \boldsymbol{v} \in U$ and $\alpha \in \mathbb{F}$.

Proof:

$\square$

To show that U is nonempty, we often prove that the zero vector $\mathbf{0}$ in V also belongs to U.

Question: where would the proof fail if U was empty?

Example 2.1 Define subsets U_1, U_2 by

$$U_1 = \{\boldsymbol{x} \in \mathbb{R}^3 \,:\, x_1 + x_2 = 1\},$$
$$U_2 = \{\boldsymbol{x} \in \mathbb{R}^3 \,:\, x_1 x_2 = 0\}.$$

Then U_1 is not a subspace because ______________________.

And U_2 is not a subspace because:

$\diamond$

Example 2.2 Let $U = \{\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 \, : \, x_1 + x_2 = 0\}$.

Prove that U is a subspace.

Proof.

$\square$

Example 2.3 Describe all possible subspaces of $\mathbb{R}^3$.

$\square$

Example 2.4 Other examples of important subspaces:

(a) A square matrix A is symmetric if $A^T = A$. The set of $p \times p$ symmetric matrices is a subspace of $M_{p,p}(\mathbb{F})$.

(b) Let X be any set and $Y \subseteq X$. The set

$$\{f \in \mathcal{F}[X] : f(y) = 0 \text{ for all } y \in Y\}$$

is a subspace of $\mathcal{F}[X]$.

(c) For any interval $I \subseteq \mathbb{R}$, the set $C(I)$ of continuous functions on I is a subspace of $\mathcal{F}(I)$.

Similarly, if I is open then

- the set of differentiable functions on I,
- the set of continuously differentiable functions on I,
- the set of twice differentiable functions on I, (etc)

are all subspaces of $C(I)$. Each one is a subspace of the previous one.

Exercise: Prove these statements!

Example 2.5

Sometimes a subset of a vector space V can be a vector space but not a subspace of V!

Consider $\mathbb{C}^2$ as a vector space over $\mathbb{C}$ and let

$$U = \{\boldsymbol{v} \in \mathbb{C}^2 \,:\, v_1 \in \mathbb{R}\}.$$

Then U is not a subspace of $\mathbb{C}^2$ over $\mathbb{C}$. (Why?)

Next, let W be the set $\mathbb{C}^2$ as a vector space over $\mathbb{R}$.
Then U is a subspace of W.

2.3 Linear combinations, spans, independence

Definition 2.3

Let V be a vector space over $\mathbb{F}$. A (finite) linear combination of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n \in V$ is any vector which can be expressed as

$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n \ ,$$

where the α_k are scalars.

If S is a subset of V, then the span of S is

$$\mathrm{span}(S) = \{\text{all finite linear combinations of vectors in } S\} \ .$$

If $\mathrm{span}(S) = V$ then we say that S spans V, or that S is a spanning set for V.

Note that a spanning set can be finite or infinite, but we only take finite linear combinations from it.

We see that $S \subseteq \mathrm{span}(S)$ and $\mathrm{span}(S)$ is a subset of V.

Lemma 2.3 If S is a nonempty subset of a vector space V then $\mathrm{span}(S)$ is a subspace of V.

Proof:

$\square$

Example 2.6

(a) $\mathcal{P}_n(\mathbb{R})=$ $\text{span}(\{1, t, t^2, \ldots, t^n\})$.

(b) The span of two nonzero vectors in $\mathbb{R}^3$ is a plane through the origin, unless the vectors are scalar multiples of each other, in which case it is a line through the origin.

Definition 2.4 Let V be a vector space over a field $\mathbb{F}$.
A subset S of V is linearly independent if for all vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n \in S$ and $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{F}$ with $n \geq 1$,

if $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \cdots + \alpha_n \mathbf{v}_n = \mathbf{0}$ then $\alpha_1 = \cdots = \alpha_n = 0$.

A set which is not linearly independent is said to be linearly dependent.

Notes

(a) S is linearly independent if and only if there is a unique way to write $\mathbf{0}$ as a linear combination of elements of S.

(b) We only consider finite linear combinations.

Lemma 2.4

If $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ is a linearly dependent set of nonzero vectors in V then there is an i with $2 \leq i \leq n$ such that

$$\boldsymbol{v}_i = \sum_{j=1}^{i-1} \beta_j \mathbf{v}_j$$

for some $\beta_1, \ldots, \beta_{i-1} \in \mathbb{F}$.

Proof:

$\square$

Example 2.7 It is easy to see that the set $\{t^2, t^2+t, t^2+t+1\}$ is linearly independent in $\mathcal{P}_2(\mathbb{R})$. (Why?)

For a slightly harder example we prove that $\{p_1,\ p_2,\ p_3\}$ is linearly independent in $\mathcal{P}_2(\mathbb{R})$, where

$$p_1(t) = 1+2t+3t^2\,, \quad p_2(t) = 4+5t+6t^2\,, \quad p_3(t) = -1+2t^2\,.$$

Proof.

Therefore the set $\{p_1, p_2, p_3\}$ is linearly independent. $\square$

Theorem 2.5 In any vector space,

(a) Any subset of a linearly independent set is linearly independent.

(b)(i) If $\boldsymbol{v} \in \text{span}(S)$ and $\boldsymbol{v} \notin S$ then $S \cup \{\boldsymbol{v}\}$ is linearly dependent.

(ii) If S is linearly independent and $S \cup \{\boldsymbol{v}\}$ is linearly dependent then $\mathbf{v} \in \text{span}(S)$.

(c)(i) If $S_1 \subseteq S_2$ then $\text{span}(S_1) \subseteq \text{span}(S_2)$.

(ii) If $S_1 \subseteq \text{span}(S_2)$ then $\text{span}(S_1) \subseteq \text{span}(S_2)$.

(d) $\text{span}(S \cup \{\boldsymbol{v}\}) = \text{span}(S)$ if and only if $\boldsymbol{v} \in \text{span}(S)$.

(e) If S is linearly dependent then there is a vector $\boldsymbol{v} \in S$ such that $\text{span}(S - \{\boldsymbol{v}\}) = \text{span}(S)$.

(f) In $\mathbb{F}^p$, if $P \in \text{GL}(p, \mathbb{F})$ is an invertible matrix and $\{\boldsymbol{v}_i\}$ is linearly independent, then the set $\{P\mathbf{v}_i\}$ is also linearly independent.

Exercise: prove these statements.

2.4 Bases

Definition 2.5 Let $S \subseteq V$. The set S is a basis for V over $\mathbb{F}$ if and only if $V = \text{span}(S)$ and S is a linearly independent set.

A basis is a minimal set of vectors S from which we can form all vectors in V as linear combinations of elements of S.

The following examples are intended to give you some intuitive feel for how you can find a basis of a given vector space.

Exercise: prove in each example that the given set is a basis.

Examples of Bases

Example: $\mathbb{F}^n$ over $\mathbb{F}$

The standard basis of $\mathbb{F}^n$ as a vector space over $\mathbb{F}$ is $\mathcal{B} = \{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\}$ where

$$\boldsymbol{e}_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \begin{array}{l} \\ \\ \\ \leftarrow i\text{th place.} \\ \\ \\ \\ \end{array}$$

We may sometimes use $\boldsymbol{i}$, $\boldsymbol{j}$, $\boldsymbol{k}$ to denote the standard basis of $\mathbb{R}^3$.

Example: Matrix spaces
Define the matrices

$$E_{ij} = (e_{h\ell}) = \begin{cases} 1 & \text{if } h = i \text{ and } \ell = j, \\ 0 & \text{otherwise.} \end{cases}$$

The set

$$\mathcal{B} = \{E_{ij} \,:\, 1 \leq i \leq p,\, 1 \leq j \leq q\}$$

is the standard basis of $M_{p,q}(\mathbb{F})$ as a vector space over $\mathbb{F}$.

Example: Polynomial spaces
The standard basis of $\mathcal{P}_n(\mathbb{F})$ as a vector space over $\mathbb{F}$ is

$$\mathcal{B} = \{1, t, \ldots, t^n\}.$$

The vector space $\mathcal{P}(\mathbb{F})$ does not have a finite basis. The standard basis is $\mathcal{B} = \{1, t, t^2, \ldots\}$.

Example: Function Spaces

The space $\mathcal{F}(X)$ has no obvious basis unless X is finite.

Let $X = \{a_1, \dots, a_n\}$. For $i = 1, \dots n$ define $f_i : X \longrightarrow \mathbb{F}$ by

$$f_i(a_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

The set $\mathcal{B} = \{f_1, f_2, \dots, f_n\}$ is a basis for $\mathcal{F}(X)$.

We call δ_{ij} the Kronecker delta symbol. It is very useful!

Example: Fields

The set $\{1, i\}$ is a basis for $\mathbb{C}$ as a vector space over $\mathbb{R}$.

Similarly, $\mathbb{Q}(\sqrt{2})$ has basis $\{1, \sqrt{2}\}$ over $\mathbb{Q}$.

Do all vector spaces have bases?

Q: How would you find a basis of $\mathbb{R}$ as a vector space over $\mathbb{Q}$?

Q: How would you look for a basis for $C[0, 1]$ over $\mathbb{R}$?

Fact: The statement "every vector space has a basis" is equivalent to the Axiom of Choice. (Difficult proof!)

We will mostly study vector spaces which have a finite basis in this course, so we don't have to worry about this.

Example 2.8

Consider the set of all 2×2 real symmetric matrices.

Any such matrix can be written as

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus any symmetric matrix is a linear combination of

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

Since it is clear this set is linearly independent, it is a basis for the space of symmetric matrices.

2.5 Dimension

Theorem 2.6
Let V be a vector space which has a finite spanning set.
Then V has a finite basis and all bases of V contain the same number of elements.

Proving this result will take a while.
First we prove that we can always find a basis inside a finite spanning set.

Theorem 2.7 Let V be a vector space over $\mathbb{F}$ and let S be a finite spanning set for V. Then S contains a finite basis for V.

Proof.

$\square$

Next we state an important lemma.

Lemma 2.8 (The Exchange Lemma)
Suppose that S is a finite spanning set for V and that T is a (finite) linearly independent subset of V with $|T| \leq |S|$.

Then there is a spanning set S' of V such that

$$T \subseteq S' \quad \text{and} \quad |S'| = |S| \ .$$

Idea of the proof: we inductively swap out (exchange) one element from the finite spanning set S in favour of a suitable element from a smaller linearly independent T.

The proof is important, but a little fiddly. We postpone it until the end of this section.

Corollary 2.9 If S is a finite spanning set for a vector space V and T is a linearly independent subset of V, then T is finite and $|T| \leq |S|$.

Proof.

$\square$

Next we show that if V has a finite spanning set then any linearly independent set can be extended to a basis of V.

Theorem 2.10 Let V be a vector space over $\mathbb{F}$ with a finite spanning set and T a linearly independent subset of V. Then there is a basis B of V which contains T.

Proof:

$\square$

Now we can prove Theorem 2.6:

Theorem 2.6
Let V be a vector space which has a finite spanning set.
Then V has a finite basis and all bases of V contain the same number of elements.

Proof.

$\square$

Because of the previous theorem, the following definition makes sense:

Definition 2.6 The dimension of a vector space V is the size of a basis, if V has a finite basis, or infinity otherwise.

We write $\dim(V) = n$ or $\dim(V) = \infty$.

If we need to emphasise the field, we use the notation $\dim_{\mathbb{F}}(V)$.

Example 2.9

(a) $\dim(\mathbb{F}^n) = n$ and $\dim(M_{m,n}(\mathbb{F})) = mn$.

(b) $\dim(\mathcal{P}_n(\mathbb{F})) = n + 1$

(c) $\dim_{\mathbb{C}} \mathbb{C} = 1$ but $\dim_{\mathbb{R}}(\mathbb{C}) = 2$.

(d) $\dim(\mathcal{P}(\mathbb{F})) = \dim(\mathcal{F}(\mathbb{R})) = \infty$.

(e) Also, $\dim\{\mathbf{0}\} = 0$ in any vector space.

Lemma 2.11 Let V be a finite-dimensional vector space and suppose $\dim(V) = n$.

(a) The number of elements in any spanning set is at least n.

(b) The number of elements in any linearly independent set is at most n.

(c) If $\operatorname{span}(S) = V$ and $|S| = n$ then S is a basis.

(d) If S is linearly independent and $|S| = n$ then S is a basis.

Proof:

$\square$

Exercise: Suppose $\dim(V) = n$ and $U \leq V$. Show that $\dim(U) \leq n$, and that $\dim(U) = n$ if and only if $U = V$.

Now we tie together the ideas of linear combinations, spanning sets and linear independence:

Theorem 2.12. Let V be a finite dimensional vector space over $\mathbb{F}$.

Then $\mathcal{B}=\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ is a basis for V if and only if every $\boldsymbol{x} \in V$ can be written uniquely as $\boldsymbol{x}=\sum_{i=1}^{n} \alpha_i \mathbf{v}_i$ with $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$.

Proof:

$\square$

We now go back to the proof of the Exchange Lemma:

Lemma 2.8 (The Exchange Lemma)
Suppose that S is a finite spanning set for V and that T is a (finite) linearly independent subset of V with $|T| \leq |S|$.

Then there is a spanning set S' of V such that

$$T \subseteq S' \quad \text{and} \quad |S'| = |S| \; .$$

Proof.

2.6 Coordinates

Suppose V is a vector space of dimension n over $\mathbb{F}$ and let $\mathcal{B} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ be an ordered basis for V over $\mathbb{F}$.

If $\boldsymbol{v} \in V$ then $\boldsymbol{v} = \sum_{i=1}^{n} \alpha_i \boldsymbol{v}_i$ with the α_i unique, by Thm 2.12.

We call $\boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$ the coordinate vector of $\boldsymbol{v}$ with respect to $\mathcal{B}$, and refer to the α_i as the coordinates of $\mathbf{v}$.

A useful notation is

$$\boldsymbol{\alpha} = [\mathbf{v}]_{\mathcal{B}} \quad \text{if} \quad \boldsymbol{v} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i.$$

Notes.

(a) $\boldsymbol{u} = \boldsymbol{v}$ if and only if $[\boldsymbol{u}]_{\mathcal{B}} = [\boldsymbol{v}]_{\mathcal{B}}$ for all bases $\mathcal{B}$.

(b) $[\boldsymbol{u} + \boldsymbol{v}]_{\mathcal{B}}=[\boldsymbol{u}]_{\mathcal{B}} + [\boldsymbol{v}]_{\mathcal{B}}$ for any basis $\mathcal{B}$.

(c) $[\lambda \boldsymbol{u}]_{\mathcal{B}} = \lambda\, [\boldsymbol{u}]_{\mathcal{B}}$ for any basis $\mathcal{B}$.

Hence all calculations in any finite-dimensional vector space can be carried out using coordinate vectors.

So, in some sense, every finite-dimensional vector space over $\mathbb{F}$ is just $\mathbb{F}^n$ in disguise for some n.

Example 2.10

Finding coordinates in the standard bases is usually easy.

(a) In $\mathbb{F}^n$ let $\mathcal{S}$ be the standard basis. Then $[\mathbf{v}]_{\mathcal{S}} = \mathbf{v}$.

(But are they conceptually the same thing?).

(b) In $\mathcal{P}_n(\mathbb{F})$ with standard basis $\mathcal{S}$,

$$\text{if} \qquad p(t) = \sum_{i=0}^{n} a_i t^i \quad \text{then} \quad [p]_{\mathcal{S}} = \begin{pmatrix} a_0 \\ \vdots \\ a_n \end{pmatrix}.$$

For example in $\mathcal{P}_2(\mathbb{R})$ we have $[5 - t^2]_{\mathcal{S}} = \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix}$.

The standard way of finding coordinates in other bases is to solve linear equations.

Example 2.11 Find the coordinates of $\begin{pmatrix} 9 \\ 1 \\ 5 \end{pmatrix}$ with respect to the ordered basis $\mathcal{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ of $\mathbb{R}^3$, where

$$\boldsymbol{v}_1 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 2 \\ -5 \\ -3 \end{pmatrix}, \boldsymbol{v}_3 = \begin{pmatrix} 5 \\ -7 \\ -3 \end{pmatrix}.$$

Solution.

We write this in augmented matrix form and row-reduce:

$\diamond$

Note that this calculation confirms that $\mathcal{B}$ is a basis.

You do not always have to solve equations to find coordinates:

Example 2.12 In $\mathcal{P}_2(\mathbb{R})$, consider the ordered basis $\{p_1,\ p_2,\ p_3\}$ where

$$p_1(t) = t(t-1), \quad p_2(t) = t(t+1), \quad p_3(t) = (t+1)(t-1).$$

Find $[5 - t^2]_{\mathcal{B}}$.

2.7 Sums and Direct Sums

Let V be a vector space with subspaces S and T.

Exercise: $S \cap T$ is also a subspace of V. (Why is it never empty)?

In fact, $S \cap T$ is a subspace of both S and T.

However, $S \cup T$ is not in general a subspace of V.

Exercise: Suggest an example which proves this.

How can we combine two subspaces into a larger one?

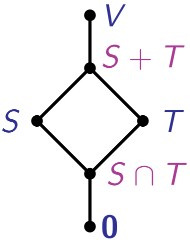
Definition 2.8
The sum $S + T$ of the subspaces S, T is defined as

$$S + T = \{\mathbf{a} + \mathbf{b} : \mathbf{a} \in S, \ \mathbf{b} \in T\}.$$

If $S \cap T = \{\mathbf{0}\}$ then we call the sum a direct sum and denote it as $S \oplus T$.

Exercise: Prove that $S + T$ is a subspace. (Hence $S \oplus T$ is a subspace, when the sum is direct.)

We have a nice Hasse diagram for the subspace relation:



Proposition 2.13. The sum of subspaces S and T is direct if and only if any vector $\boldsymbol{x} \in S + T$ can be written in a unique way as $\boldsymbol{x} = \boldsymbol{a} + \boldsymbol{b}$ with $\boldsymbol{a} \in S$ and $\boldsymbol{b} \in T$.

Exercise: Prove this.

Example 2.13 In $\mathbb{R}^3$, the sum of two different lines through the origin will be a plane through the origin.

Moreover, the lines will intersect only at the origin, so the sum is direct: if $\text{span}\{\boldsymbol{v}\} \neq \text{span}\{\boldsymbol{w}\}$ then

$$\text{span}\{\boldsymbol{v}\} \oplus \text{span}\{\boldsymbol{w}\} = \text{span}\{\boldsymbol{v}, \boldsymbol{w}\} .$$

Two different planes P_1, P_2 through the origin in $\mathbb{R}^3$ must intersect in a line, so the sum $P_1 + P_2$ is not direct.

Here $P_1 + P_2 = \mathbb{R}^3$.

Example 2.14
The vector space $M_{2,2}(\mathbb{R})$ is the direct sum of the subspace S of symmetric matrices and the subspace K of skew-symmetric matrices:

$$S = \left\{ \begin{pmatrix} a & b \\ b & c \end{pmatrix}, a, b, c \in \mathbb{R} \right\}, \qquad K = \left\{ \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix}, b \in \mathbb{R} \right\}.$$

Next, consider the subspaces of upper triangular and lower triangular matrices in $M_{2,2}(\mathbb{R})$:

$$U = \left\{ \begin{pmatrix} a_1 & a_2 \\ 0 & a_3 \end{pmatrix}, a_i \in \mathbb{R} \right\}, \qquad L = \left\{ \begin{pmatrix} b_1 & 0 \\ b_2 & b_3 \end{pmatrix}, b_i \in \mathbb{R} \right\}.$$

Then $M_{22}(\mathbb{R}) = U + L$, but $U \cap L$ is the space of diagonal matrices and so the sum is not direct.

In Example 2.14 we have

$$\dim(S) = \dim(U) = \dim(L) = 3 \text{ and } \dim(K) = 1.$$

Exercise: Write down bases that make this clear.

Since $\dim(M_{2,2}(\mathbb{R})) = 4$, we have

$\dim(S) + \dim(K) = \dim(M_{2,2}(\mathbb{R}))$, but
$\dim(U) + \dim(L) = \dim(M_{2,2}(\mathbb{R})) + 2$.

Q: How can we explain this extra 2?

A: Probably something to do with this: $\dim(U \cap L) = 2$.

Theorem 2.14 Suppose S and T are finite-dimensional subspaces of vector space V. Then

$$\dim(S) + \dim(T) = \dim(S + T) + \dim(S \cap T).$$

We will defer the proof until the end of the section.

The idea of the proof is to start with a basis for $S \cap T$, extend it separately into two new bases, one for S, the other for for T, then show that the union of the bases for S and T gives a basis for $S + T$.

Theorem 2.14 immediately implies the following:

Corollary 2.15 For a direct sum of finite-dimensional subspaces S and T,

$$\dim(S) + \dim(T) = \dim(S \oplus T).$$

Example 2.15 Let $V = \mathbb{R}^{17}$ and let S, T be two subspaces of V with $\dim(S) = 11$ and $\dim(T) = 14$.

What are the possible values for $\dim(S \cap T)$?

Interpret the extreme values.

Solution:

$\diamond$

Example 2.16 Prove that every matrix in $M_{2,2}(\mathbb{R})$ can be written in a unique way as a sum $X + Y$, where X and Y are 2×2 matrices such that

$$X \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{and} \quad Y \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} .$$

Solution: Let a direct sum do the work!

□

Lemma 2.16
Let V be a finite-dimensional vector space and $X \leq V$.
Then there is a subspace Y of V such that $V = X \oplus Y$.

Proof.

$\square$

We call Y a complementary subspace; it is not unique.

Example 2.17 In $\mathbb{R}^2$ let $X = \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} \right\}$.

If Y is any line through the origin distinct from X then
$\mathbb{R}^2 = X \oplus Y$.

The External Direct Sum

The direct sum from Definition 2.8 is sometimes called the internal direct sum, as it applies to subspaces within some larger vector space.

There is also the useful idea of the external direct sum, which is really just a way of turning a Cartesian product of vector spaces into a vector space.

Definition 2.9

Let X and Y be two vector spaces over the same field $\mathbb{F}$.

The Cartesian product $X \times Y$ can be made into a vector space over $\mathbb{F}$ with the operations

$$(\boldsymbol{x}_1, \boldsymbol{y}_1) + (\boldsymbol{x}_2, \boldsymbol{y}_2) = (\boldsymbol{x}_1 + \boldsymbol{x}_2,\ \boldsymbol{y}_1 + \boldsymbol{y}_2), \quad \lambda(\boldsymbol{x}_1, \boldsymbol{y}_1) = (\lambda \boldsymbol{x}_1,\ \lambda \boldsymbol{y}_1).$$

With this structure we call the Cartesian product $X \times Y$ the (external) direct sum of X and Y, and write it as $X \oplus Y$.

Example 2.18 If $X = Y = \mathbb{R}$ then the external direct sum is $X \oplus Y = \mathbb{R}^2$, with the usual operations.

Fact: If X and Y are finite-dimensional then

$$\dim(X \oplus Y) = \dim(X) + \dim(Y).$$

Note: If X and Y are subspaces of some vector space V, we can define their external direct sum, even if $X \cap Y$ is non-trivial.

Fact: The internal and external direct sums of subspaces are "essentially the same" if and only if the subspaces have trivial intersection.

To explain what we mean by "essentially the same" here, we need to turn to the other half of the category of vector spaces: the morphisms.

We finish this section with the deferred proof of Theorem 2.14.

Theorem 2.14 Suppose S and T are finite-dimensional subspaces of vector space V. Then

$$\dim(S) + \dim(T) = \dim(S + T) + \dim(S \cap T).$$

Proof.

□

Key points for Chapter 2

- Vector space, p. 3
- Subspace, p. 11
- Subspace Test Lemma, p. 12
- Linear combination, span, p. 18
- Linear independence, p. 21
- Basis, p. 26
- Dimension, p. 38
- Coordinate vector, coordinates, p. 44
- Sum, direct sum, p. 51
- Complementary subspace, p. 59
- External direct sum, p. 60

MATH2601 – Higher Linear Algebra

Chapter 3. Linear Transformations

3.1 Linear Transformations

Vector spaces have two operations.
We study functions/maps/transformations which respect both operations.

Definition 3.1
Suppose V and W are vector spaces over the field $\mathbb{F}$.
A function $T : V \to W$ is a linear transformation or a linear map (or simply linear) if

- $T(\boldsymbol{u} + \boldsymbol{v}) = T(\boldsymbol{u}) + T(\boldsymbol{v})$ and
- $T(\lambda \boldsymbol{v}) = \lambda T(\boldsymbol{v})$,

for all $\boldsymbol{u}, \boldsymbol{v} \in V$ and all $\lambda \in \mathbb{F}$.

Note: we only define linear maps between vector spaces over the same field.

Lemma 3.1 Let V and W be vector spaces over the field $\mathbb{F}$.

(a) The identity map $\mathrm{id} : V \to V$ defined by $\mathrm{id}(\boldsymbol{v}) = \boldsymbol{v}$ for all $\boldsymbol{v} \in V$ is linear.

(b) If $T : V \to W$ is linear then $T(\mathbf{0}) = \mathbf{0}$ and $T(-\boldsymbol{v}) = -T(\boldsymbol{v})$ for all $\boldsymbol{v} \in V$.

Note: In (b) we could write $T(\mathbf{0}_V) = \mathbf{0}_W$.

Proof.

$\square$

We can test both linearity conditions at once, as we did in the Subspace Test Lemma (Lemma 2.2).

Lemma 3.2 (Linearity Test Lemma)
A function $T : V \to W$ between vector spaces over the same field $\mathbb{F}$ is linear if and only if

$$T(\lambda \boldsymbol{u} + \boldsymbol{v}) = \lambda T(\boldsymbol{u}) + T(\boldsymbol{v})$$

for all $\lambda \in \mathbb{F}$ and $\boldsymbol{u}, \boldsymbol{v} \in V$.

Proof.

$\square$

Example 3.1 Let $V = \mathbb{F}^p$ and $W = \mathbb{F}^q$.
For a fixed $q \times p$ matrix A, define $T : V \to W$ by $T(\boldsymbol{v}) = A\boldsymbol{v}$ for all $\boldsymbol{v} \in V$. Then T is linear since matrix multiplication is linear:

$$\begin{aligned} T(\lambda \boldsymbol{x} + \boldsymbol{y}) &= A(\lambda \boldsymbol{x} + \boldsymbol{y}) \\ &= \lambda A\boldsymbol{x} + A\boldsymbol{y} \\ &= \lambda T(\boldsymbol{x}) + T(\boldsymbol{y}). \end{aligned}$$

Fact: This example captures all linear maps between finite-dimensional vector spaces. (See Theorem 3.14.)

Example 3.2 Many important geometric transformations of $\mathbb{R}^2$ and $\mathbb{R}^3$ (and, if you can imagine it, $\mathbb{R}^n$) are linear: e.g. rotations, reflections and projections.

But translations are not linear. Why?

Theorem 3.3

Let V and W be two vector spaces over a field $\mathbb{F}$.
The set $L(V, W)$ of all linear transformations from V to W is a vector space under the operations defined by

$$(S + T)(\boldsymbol{v}) = S(\boldsymbol{v}) + T(\boldsymbol{v}), \quad (\lambda S)(\boldsymbol{v}) = \lambda S(\boldsymbol{v})$$

for all $\boldsymbol{v} \in V$ and $\lambda \in \mathbb{F}$.

Proof.

$\square$

Lemma 3.4 Let $T : V \to W$ and $S : W \to X$ be linear maps between vector spaces.

Then the composition $S \circ T : V \to X$ is also linear.

Exercise: Prove this lemma.

Lemma 3.5 Let $T : V \to W$ be an invertible linear map between two vector spaces over field $\mathbb{F}$.

Then $T^{-1} : W \to V$ is linear.

Proof.

$\square$

Theorem 3.3 implies that $L(V, V)$, the set of linear maps from V to itself, is a vector space. Also:

Theorem 3.6 The set of invertible linear maps in $L(V, V)$ form a group under composition.

Proof.

$\square$

Example 3.3 On a suitable set of functions, differentiation, definite integration and multiplication by a fixed function are all linear.

Evaluation can also give rise to linear maps:

Example 3.4 Show that $T : \mathcal{P}(\mathbb{R}) \to \mathbb{R}^2$ defined by $T(f) = \begin{pmatrix} f(1) \\ f(2) \end{pmatrix}$ is linear.

Proof.

$\square$

Another important linear transformation is taking coordinates.

Lemma 3.7

Let V be a finite-dimensional vector space over $\mathbb{F}$ with a basis $\mathcal{B} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$.

Then the function $S : V \to \mathbb{F}^p$ defined by $S(\boldsymbol{x}) = [\boldsymbol{x}]_{\mathcal{B}}$ is linear.

Proof:

$\square$

3.2 Kernel and Image

Definition 3.2 Let $T : V \to W$ be a linear transformation.
The kernel (or nullspace) of T is the set

$$\ker T = \{\boldsymbol{v} \in V \ : \ T(\boldsymbol{v}) = \boldsymbol{0}\}.$$

If $U \leq V$ then the image of U is the set

$$T(U) = \{T(\boldsymbol{u}) \ : \ \boldsymbol{u} \in U\}.$$

We also define the image of T (or range of T), denoted $\operatorname{im}(T)$, as the image of all of V; that is, $\operatorname{im}(T) = T(V)$.

Example 3.5

(a) In Example 3.4, recall $T : \mathcal{P}(\mathbb{R}) \to \mathbb{R}^2$ is defined by

$$T(f) = \begin{pmatrix} f(1) \\ f(2) \end{pmatrix}.$$ Find $\operatorname{im}(T)$ and $\ker(T)$.

Solution:

(b) Consider differentiation on quadratic polynomials,

$D : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$. Find $\operatorname{im}(D)$ and $\ker(D)$.

Solution:

Theorem 3.8 Let $T : V \to W$ be a linear transformation between vector spaces over $\mathbb{F}$ and let $U \leq V$. Then

(a) $\ker T$ is a subspace of V.

(b) $T(U)$ is a subspace of W, and so $\operatorname{im}(T) \leq W$.

(c) If U is finite-dimensional then $T(U)$ is finite-dimensional. In particular, if V is finite-dimensional then so is $\operatorname{im}(T)$.

Proof.

Exercise: Prove part (a).

□

Theorem 3.8 means the following now makes sense:

Definition 3.3 If T is a linear transformation then the dimension of the kernel of T is called the nullity of T, and the dimension of the image of T is called the rank of T.

Example 3.6 From Example 3.5 we see that differentiation on $\mathcal{P}_2(\mathbb{R})$ has nullity 1 and rank 2.

Lemma 3.9 A linear map $T : V \to W$ is one-to-one if and only if $\text{nullity}(T) = 0$.

Proof.

$\square$

Theorem 3.10 (Rank–Nullity Theorem) If V is a finite-dimensional vector space over $\mathbb{F}$ and $T : V \to W$ is linear then

$$\mathrm{rank}(T) + \mathrm{nullity}(T) = \dim(V).$$

Proof.

Theorem 3.11 Let V, W be finite-dimensional vector spaces over $\mathbb{F}$ with $\dim(V) = \dim(W)$ and let $T : V \to W$ be linear.

The following are equivalent:

(a) T is invertible (bijective).

(b) T is one-to-one (injective); that is, $\text{nullity}(T) = 0$.

(c) T is onto (surjective); that is, $\text{rank}(T) = \dim(W)$.

Proof.

$\square$

Example 3.7 Define the map $T : \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$ by

$$T(p) = \begin{pmatrix} p(1) \\ p(0) \\ p(-1) \end{pmatrix}.$$ Prove that T is invertible.

Proof.

$\square$

Definition 3.4 An invertible linear map $T : V \to W$ is called an isomorphism of the vector spaces V and W.

If there is an isomorphism between V and W then we say that V and W are isomorphic.

Isomorphism is an equivalence relation on vector spaces.

Reflexive:

Symmetric:

Transitive:

Exercise: Prove this!

Theorem 3.12. Finite-dimensional vector spaces V and W over $\mathbb{F}$ are isomorphic if and only if they have the same dimension.

Proof.

$\square$

In particular, if V is a p-dimensional vector space over $\mathbb{F}$ then V it is isomorphic to $\mathbb{F}^p$.

We proved this by showing that taking coordinates in some chosen basis is an isomorphism.

In fact, all isomorphisms to $\mathbb{F}^p$ can be described this way. Why?

As another special case, note that the vector spaces $\mathbb{F}^p$ and $M_{p,1}(\mathbb{F})$ and $M_{1,p}(\mathbb{F})$ are all isomorphic.

This is why we can treat vectors of p components as if they were $p \times 1$ matrices, and why it doesn't matter (out of context) whether we regard them as row vectors or column vectors.

3.3 Spaces Associated with Matrices

Let A be a $p \times q$ matrix over a field $\mathbb{F}$, and define a map $T : \mathbb{F}^q \to \mathbb{F}^p$ by $T(\boldsymbol{x}) = A\boldsymbol{x}$.

We define the kernel of A to equal the kernel of T, and similarly for the image, nullity and rank of A.

Now suppose A has columns $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_q \in \mathbb{F}^p$. Then

Let the column space of A be the space spanned by the columns of A, denoted by $\mathrm{col}(A)$.

We have just proved that $\mathrm{im}(A) = \mathrm{col}(A)$. Hence $\mathrm{col}(A)$ is a subspace of $\mathbb{F}^p$ and the rank of A equals the dimension of the column space of A.

The Rank-Nullity Theorem for maps immediately gives the Rank-Nullity Theorem for matrices:

$$\operatorname{rank}(A) + \operatorname{nullity}(A) = q \text{ for all } A \in M_{p,q}(\mathbb{F}).$$

Note q is the number of columns of A.

The row space of A is the set of linear combinations of the rows of A, denoted $\operatorname{row}(A)$. Fact: $\operatorname{row}(A)$ is a subspace of $\mathbb{F}^q$.

Note that $\operatorname{row}(A) = \operatorname{col}(A^T) = \operatorname{im}(A^T)$.

The row rank of A is the dimension of the row space of A. How is it related to the usual (column) rank of A?

Theorem 3.13. Let $A \in M_{p,q}(\mathbb{F})$. The spaces $\text{row}(A)$ and $\text{col}(A)$ have the same dimension.

Proof (outline):

$\square$

3.4 The Matrix of a Linear Map

Theorem 3.14 Let V, W be finite-dimensional vector spaces over $\mathbb{F}$. Suppose that $\dim(V) = q$ and V has basis $\mathcal{B}$, and that $\dim(W) = p$ and W has basis $\mathcal{C}$.

If $T : V \to W$ is linear then there is a unique $A \in M_{p,q}(\mathbb{F})$ such that

$$[T(\boldsymbol{v})]_{\mathcal{C}} = A\,[\boldsymbol{v}]_{\mathcal{B}}\,. \tag{1}$$

Conversely, for any $A \in M_{p,q}(\mathbb{F})$, equation (1) defines a unique linear map from V to W.

Note: This theorem says that once we choose bases for the domain V and codomain W of T, we can identify $L(V, W)$ with $M_{p,q}(\mathbb{F})$ using coordinates in those bases.

Proof (of Theorem 3.14).

. . .

Definition 3.5 The matrix A from Theorem 3.14 is called the matrix of T with respect to $\mathcal{B}$ and $\mathcal{C}$.

We denote this matrix by $[T]_{\mathcal{C}}^{\mathcal{B}}$. Then equation (1) can be written as

$$[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C}}^{\mathcal{B}}\,[\mathbf{v}]_{\mathcal{B}}\,.$$

Theorem 3.14 tells us that to find the jth column of $[T]_{\mathcal{C}}^{\mathcal{B}}$, we take the jth element of $\mathcal{B}$ (the basis of the domain V); apply T; and then take the coordinate vector of this image with respect to $\mathcal{C}$ (the basis of the codomain W).

Corollary 3.15 If $\dim(V) = q$ and $\dim(W) = p$ then $\dim(L(V, W)) = pq$.

Example 3.8 Find the matrix of the evaluation map from Example 3.7 (evaluating at 1, 0, -1 in order) with respect to the standard bases in $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^3$.

Solution:

Theorem 3.16 Let $T : V \to W$ and $S : W \to X$ be linear maps between finite-dimensional vector spaces and suppose that V, W and X have bases $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ respectively.

Then the matrix of $S \circ T : V \to X$ is the product of the matrices of T and S, all taken with respect to the appropriate bases:

$$[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = [S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}}.$$

Proof.

□

The matrix of the identity map on a vector space V will be the identity matrix, if we use the same basis in domain and codomain. Combining this with Theorem 3.16 gives:

Corollary 3.17 Let V have basis $\mathcal{B}$ and let W have basis $\mathcal{C}$. If $T : V \to W$ is linear and invertible then the matrix of T^{-1} is the inverse of the matrix of T (with respect to bases $\mathcal{B}$, $\mathcal{C}$). That is,

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = \left([T]_{\mathcal{C}}^{\mathcal{B}}\right)^{-1} .$$

(Note the position of the labels $\mathcal{B}$, $\mathcal{C}$.)

Thus the group of invertible linear maps on an n-dimensional vector space over $\mathbb{F}$ is isomorphic to $\mathrm{GL}(n, \mathbb{F})$.

Exercise: Prove these statements.

Example 3.9 Repeat Example 3.8 but using bases $\mathcal{B} = \{3+2t, 1+2t^2, 1-t-t^2\}$ and $\mathcal{C} = \left\{ \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} \right\}$ in domain and codomain, respectively.

Evaluating T on the elements of $\mathcal{B}$ gives

$$T(3+2t) = \begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, \; T(1+2t^2) = \begin{pmatrix} 3 \\ 1 \\ 3 \end{pmatrix}, \; T(1-t-t^2) = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

Now we have to write these three vectors as linear combinations of the elements of $\mathcal{C}$.

We can do this by solving three sets of three linear equations in three unknowns: simple but tedious. ☹

We can solve the three systems at the same time, since the LHS matrix will be the same each time. Row-reducing to row echelon form gives:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 5 & 3 & -1 \\ 1 & 1 & -1 & 3 & 1 & 1 \\ -1 & 1 & -1 & 1 & 3 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 5 & 3 & -1 \\ 0 & 1 & 2 & 2 & 2 & -2 \\ 0 & 0 & 1 & 0 & 0 & -1 \end{array}\right).$$

From here you can solve by back-substitution for each RHS column, to find the three coordinate vectors (the three columns of $[T]_{\mathcal{C}}^{\mathcal{B}}$).

Or, reduce further to reduced row-echelon form $[I \mid X]$. Then the columns of X are the coordinate vectors we need; that is, $X = [T]_{\mathcal{C}}^{\mathcal{B}}$.

Exercise: Check that this gives

$$[T]^{\mathcal{B}}_{\mathcal{C}} = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

This is the solution to Example 3.9. $\diamond$

Recall that if A is invertible then the system $A\boldsymbol{x} = \boldsymbol{b}$ has the unique solution $\boldsymbol{x} = A^{-1}\boldsymbol{b}$, and that the reduced row-echelon form of $(A \mid \boldsymbol{b})$ is $(I \mid A^{-1}\boldsymbol{b})$.

In our example, A was the matrix with columns given by the elements of the basis $\mathcal{C}$ of the codomain. The inverse of this matrix is useful in finding $[T]^{\mathcal{B}}_{\mathcal{C}}$.

Example 3.10 Find the matrix of the identity map on $\mathbb{R}^3$ with the basis $\mathcal{C}$ from Example 3.9 in the domain and the standard basis $\mathcal{S}$ in the codomain.

Solution.

Let the elements of $\mathcal{C}$ be $\boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3$ in the given order.

- Applying the identity map to these vectors gives $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, $\boldsymbol{c}_3$.
- Taking coordinates of the resulting vectors with respect to $\mathcal{S}$ gives (surprise!) $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, $\boldsymbol{c}_3$.
- The required matrix has these three coordinate vectors as columns:

$$[\mathrm{id}]_{\mathcal{S}}^{\mathcal{C}} = (\boldsymbol{c}_1|\boldsymbol{c}_2|\boldsymbol{c}_3) = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & -1 \end{pmatrix}.$$

Definition 3.6
Let V be a vector space V with two bases $\mathcal{B}$ and $\mathcal{C}$.
The matrix $[\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}}$ of the identity map is called the change of basis matrix from $\mathcal{B}$ to $\mathcal{C}$.

We can use a change of basis matrix to change coordinates:

$$[\boldsymbol{v}]_{\mathcal{C}} = [\mathrm{id}]_{\mathcal{C}}^{\mathcal{B}} \, [\boldsymbol{v}]_{\mathcal{B}} \, .$$

Corollary 3.17 tells us that the inverse matrix will change the coordinates in the opposite direction.

Example 3.11

Find the change of basis matrix from the standard basis $\mathcal{S}$ of $\mathcal{P}_2(\mathbb{R})$ to the basis $\mathcal{B} = \{1, 1+t, 1+t+t^2\}$ of $\mathcal{P}_2(\mathbb{R})$.

Solution.

$\diamond$

To find the matrix of $T : V \to W$ with respect to bases $\mathcal{B}$ in the domain and $\mathcal{C}$ in the codomain, write T as

$$T = \mathrm{id}_W \circ T \circ \mathrm{id}_V$$

where id_V, id_W are the identity maps on V, W respectively.

Using $\mathcal{S}$ to denote the standard basis in V and $\mathcal{S}'$ to denote the standard basis in W:

$$[T]_{\mathcal{C}}^{\mathcal{B}} = [\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{S}'} [T]_{\mathcal{S}'}^{\mathcal{S}} [\mathrm{id}_V]_{\mathcal{S}}^{\mathcal{B}}.$$

We assume $A = [T]_{\mathcal{S}'}^{\mathcal{S}}$ is easy to find. We know $P = [\mathrm{id}_V]_{\mathcal{S}}^{\mathcal{B}}$ and $Q = [\mathrm{id}_W]_{\mathcal{S}'}^{\mathcal{C}}$ are easy to find. Finally, we need to invert Q.

$$
\begin{tikzcd}
\text{basis } \mathcal{S} & \text{basis } \mathcal{S}' \\
T : V \arrow[r, "A"] & W \\
T : V \arrow[u, "P"] \arrow[r, "M"] & W \arrow[u, "Q"'] \\
\text{basis } \mathcal{B} & \text{basis } \mathcal{C}
\end{tikzcd}
$$

In this commutative diagram:

(a) $\mathcal{S}$ and $\mathcal{S}'$ are standard bases.

(b) The vertical arrows represent identity maps, so
$P = [\mathrm{id}_V]_{\mathcal{S}}^{\mathcal{B}}$ and $Q = [\mathrm{id}_W]_{\mathcal{S}'}^{\mathcal{C}}$.

(c) The matrix $A = [T]_{\mathcal{S}'}^{\mathcal{S}}$, which we assume is easy to find.

(d) The matrix $M = [T]_{\mathcal{C}}^{\mathcal{B}}$, and

(e) $M = Q^{-1}AP$. That is, $[T]_{\mathcal{C}}^{\mathcal{B}} = [\mathrm{id}_W]_{\mathcal{C}}^{\mathcal{S}'} \, [T]_{\mathcal{S}'}^{\mathcal{S}} \, [\mathrm{id}_V]_{\mathcal{S}}^{\mathcal{B}}$.

Example 3.12 Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(\boldsymbol{x}) = \begin{pmatrix} x_1 + 2x_2 + 3x_3 \\ 4x_1 + 3x_2 + 2x_3 \end{pmatrix} .$$

Find the matrix of T with respect to bases $\mathcal{B}$ in the domain and $\mathcal{C}$ in the codomain, where

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\}, \quad \mathcal{C} = \left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \begin{pmatrix} -3 \\ 1 \end{pmatrix} \right\}.$$

Exercise: Check that you can do this - your answer should be

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \frac{1}{5} \begin{pmatrix} -4 & 3 & 2 \\ 2 & 11 & -1 \end{pmatrix}.$$

Lemma 3.18 Let $T : V \to W$ be a linear map between finite-dimensional vector spaces over $\mathbb{F}$ and let A be the matrix of T with respect to given bases in V and W.
Then

$$\mathrm{nullity}(A) = \mathrm{nullity}(T) \quad \text{and} \quad \mathrm{rank}(A) = \mathrm{rank}(T)\,.$$

Proof.

$\square$

Exercise: Write down a matrix version of Theorem 3.11 using this result.

Definition 3.7
Let V be a vector space over $\mathbb{F}$ and let $T : V \to V$ be a linear map.

If $X \leq V$ is such that $T(X) \leq X$ then we call X an invariant subspace of T.

Note:
We know from Theorem 3.8 that $T(X)$ is a subspace of V.
Therefore, if $T(X) \subseteq X$ then $T(X) \leq X$.

The trivial subspace and V are both invariant for any map T.

Exercise: Sometimes these are the only invariant subspaces.
Can you think of an example?

Given an invariant subspace X of map $T : V \to V$, we know there is a complementary subspace Y so that $V = X \oplus Y$.

If Y is also invariant then the matrix of T can be simplified.

Theorem 3.19 Let $T : V \to V$ be a linear map on a finite-dimensional vector space. Suppose that $V = X \oplus Y$ where X and Y are both invariant subspaces of T with dimensions p and q respectively.

Then there is a ordered basis $\mathcal{B}$ for V such that the matrix $[T]_{\mathcal{B}}^{\mathcal{B}}$ of T is of the form

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

where A is a $p \times p$ matrix and B is a $q \times q$ matrix, and each 0 represents a suitably-sized zero matrix.

Proof.

$\square$

The matrix $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$ is called the direct sum of A and B, and denoted $A \oplus B$.

Example 3.13

In $\mathbb{R}^3$, let $\boldsymbol{n}$ be any vector and let X be the plane through the origin normal to $\boldsymbol{n}$: the point-normal form of X is $\boldsymbol{n} \cdot \boldsymbol{x} = 0$.

Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be orthogonal projection onto X, which we know is linear.

Geometrically, we can see that $T(X) = X$, so X is an invariant subspace. (In fact, every element of X is a fixed point of T.)

Any complementary subspace to X must be a line, but to also be invariant the line must be orthogonal to X and hence project to zero. (Why?)

Hence $Y = \operatorname{span}(\{\boldsymbol{n}\})$ is the only complementary invariant subspace for X.

To find the matrix of T, suppose that $\{\boldsymbol{a}, \boldsymbol{b}\}$ is a basis for X.

Then $\mathcal{B} = \{\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{n}\}$ is a basis of $\mathbb{R}^3$ (why?) and the action of T on this basis is

$$T(\boldsymbol{a}) = \boldsymbol{a}, \quad T(\boldsymbol{b}) = \boldsymbol{b}, \quad T(\boldsymbol{n}) = \mathbf{0}.$$

Hence the matrix of T with respect to $\mathcal{B}$ is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \mathrm{I}_2 \oplus (0)$$

where I_2 is the 2×2 identity matrix. $\diamond$

More on isomorphisms

Theorem 3.20 Let $T : V \to W$ be a linear map between finite-dimensional spaces, and let A be the matrix of T (with respect to any bases).

Then T is invertible if and only if A is invertible.

If T is invertible then the matrix of T^{-1} (with respect to the same bases) is A^{-1}.

Proof.

$\square$

Example 3.14 Let $V = \{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 0\}$, which is a vector space (see Example 2.4), and let $D : V \to \mathcal{P}_2(\mathbb{R})$ be differentiation, $D(p) = p'$, which we know is linear.

The inverse of D is the map $D^{-1} : \mathcal{P}_2(\mathbb{R}) \to V$ given by

$$D^{-1}(p) = \int_0^t p(s)\, ds\,.$$

You can check that $\mathcal{B} = \{t, t^2, t^3\}$ is a basis for V and $\mathcal{C} = \{1 + 2t,\ 3 + 4t,\ 1 + 3t^2\}$ is a basis for $\mathcal{P}_2(\mathbb{R})$.

Exercise: Check that the matrices of D and D^{-1} with respect to $\mathcal{B}$ and $\mathcal{C}$ are, respectively,

$$A = \begin{pmatrix} -2 & 3 & 2 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 3 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and that $M = A^{-1}$.

The Normal Form

Theorem 3.21 Let $T : V \to W$ be a linear map between vector spaces over $\mathbb{F}$ where $\dim V = p$, $\dim W = q$ and $\operatorname{rank}(T) = r$.

Then there are bases $\mathcal{B}$ of V and $\mathcal{C}$ of W with respect to which the matrix of T takes the form

$$N_{q,p;r} = \begin{pmatrix} \mathrm{I}_r & 0 \\ 0 & 0 \end{pmatrix} \in M_{q,p}(\mathbb{F})$$

where I_r is the $r \times r$ identity matrix and each 0 is a suitably-sized zero matrix.

Proof.

$\square$

Corollary 3.22 Let $A \in M_{q,p}(\mathbb{F})$ have rank r. Then there is a $q \times q$ invertible matrix Q and a $p \times p$ invertible matrix P such that

$$Q^{-1}AP = N_{q,p;r}.$$

Proof: Apply the previous result to the (rank r) map $T : \mathbb{F}^p \to \mathbb{F}^q$ defined by $T(\mathbf{x}) = A\mathbf{x}$.

Then P is the matrix whose columns are the basis $\mathcal{B}$ of $\mathbb{F}^p$, and Q is the matrix whose columns are the basis $\mathcal{C}$ of $\mathbb{F}^q$. $\square$

Definition 3.8 The $q \times p$, rank r matrix $N_{q,p;r}$ in these results is called the normal form of the map or matrix.

Note that the bases in the theorem and the invertible matrices in the corollary are not unique.

3.5 Similarity

For a linear map $T : V \to W$, Theorem 3.21 tells us that the matrix of T can always be taken to be the normal form $\mathrm{I}_r \oplus \mathbf{0}$, where $r = \mathrm{rank}(T)$, by carefully choosing bases in V and W.

This implies that the only really important thing for such maps is the rank.

But when $V = W$ it is much more natural to choose one basis $\mathcal{B}$ for V, and use it in both the domain and the codomain.

This also leads to a lot more interesting mathematics about linear maps.

Let $T : V \to V$ be a linear map, where V is finite-dimensional, and let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two bases for V.

If the matrix of T with respect to $\mathcal{B}_1$ is M_1 and the matrix with respect to $\mathcal{B}_2$ is M_2, then our commutative diagram implies that

$$M_2 = P^{-1}\, M_1\, P \ ,$$

where P is the matrix whose columns are the basis vectors in $\mathcal{B}_2$, written as coordinate vectors with respect to $\mathcal{B}_1$.

This equation defines an important relation between matrices.

Definition 3.9 Matrices A and B in $M_{p,p}(\mathbb{F})$ are similar if there exists a matrix $P \in \mathrm{GL}(p, \mathbb{F})$ such that $B = P^{-1}AP$.

Exercise: Show that similarity is an equivalence relation on square matrices of a given size.

Theorem 3.23 Matrices A_1 and A_2 are similar if and only if they are the matrices of the same linear transformation with respect to two choices of bases.

Proof.

$\square$

Given two matrices A and B, how would you check whether they are similar?

Definition 3.10 A property of matrices is called a similarity invariant if it the same for all similar matrices.

A familiar example is the determinant:
since $\det(AB) = \det(A)\det(B)$ it follows that

$$\det(P^{-1}AP) = \underline{\hspace{8cm}}.$$

Similarity invariants can be used to prove matrices are not similar.

For example $\begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are not similar as they have different determinants.

However, the determinant cannot tell us that the matrices $\begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are not similar.

Note: the only matrix similar to the identity is the identity.

In order to use similarity invariants to prove matrices are similar we need a complete set of similarity invariants: a set of properties that are always the same for matrices that are similar and always different for matrices that are not similar.

Finding a complete set of similarity invariants is one of the main goals of this course.

Recall that the trace of a (square) matrix is the sum of its diagonal elements.

Theorem 3.24 The rank, nullity and trace of matrices are all similarity invariants.

Proof.

. . .

$\square$

Note from this proof that kernels are (in general) not similarity invariants: the spans of the set $\mathcal{B}_1$ and $\mathcal{B}_2$ in the proof will usually be different, although they have the same dimension.

The same applies to images.

Here are the similarity invariants for A that we know so far:

$$\{\det(A), \operatorname{tr}(A), \operatorname{rank}(A), \operatorname{nullity}(A)\}$$

This is not a complete set yet; for example you can check that

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \mathrm{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

have the same determinants, traces, ranks and nullities.
But we know they are not similar (I_2 is only similar to itself).

You may also remember that eigenvalues are similarity invariants. However, A and I_2 have the same eigenvalues, but are not similar, so we still don't have complete set of similarity invariants.

(We will come back to this later.)

3.6 Multilinear maps

Definition 3.11

Let V_1, V_2 and W be vector spaces over field $\mathbb{F}$. A map $T : V_1 \times V_2 \to W$ is bilinear if it is linear in each argument; that is,

$$T(\lambda \boldsymbol{v}_1 + \boldsymbol{v}_1', \boldsymbol{v}_2) = \lambda T(\boldsymbol{v}_1, \boldsymbol{v}_2) + T(\boldsymbol{v}_1', \boldsymbol{v}_2),$$
$$T(\boldsymbol{v}_1, \lambda \boldsymbol{v}_2 + \boldsymbol{v}_2',) = \lambda T(v_1, \boldsymbol{v}_2) + T(\boldsymbol{v}_1, \boldsymbol{v}_2')$$

for all suitable vectors and scalars.

If $V_2 = V_1$ then we say that T is bilinear on V_1.

This definition can be extended to

- trilinear maps (on $V_1 \times V_2 \times V_3$), and to
- general multilinear maps (on a Cartesian product of k vector spaces).

Example 3.15 The standard dot product on $\mathbb{R}^n$ is a bilinear map from $\mathbb{R}^n$ to $\mathbb{R}$.

The scalar triple product on $\mathbb{R}^3$ defined by

$$[\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}] = (\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}$$

is a trilinear map from $\mathbb{R}^3$ to $\mathbb{R}$.

The determinant on $M_{p,p}(\mathbb{F})$ is multilinear as a map from the product of p copies of $\mathbb{F}^p$ (the columns of the matrix) to $\mathbb{F}$.

Definition 3.12 A multilinear map T on V is said to be symmetric if its value on any ordered set of vectors is unchanged when any two of the vectors are swapped.

If such a swap always simply changes the sign of the image then T is called alternating.

The dot product is symmetric; the scalar triple product and the determinant are alternating.

Key points for Chapter 3

- Linear transformation, p. 2
- Linearity Test Lemma, p. 4
- Kernel and image, pp. 11, 23
- Nullity and rank, pp. 15, 43
- Rank-Nullity Theorem, pp. 16, 24.
- Isomorphism of vector spaces, p. 20.
- Matrix of a linear map, p. 29.
- Change of basis matrix, p. 38.
- Invariant subspace, p. 44.
- Similarity of matrices, p. 54.
- Similarity invariants, p. 56.
- Bilinear maps, p. 61.

MATH2601 – Higher Linear Algebra

Chapter 4. Inner Product Spaces

4.1 The dot product in $\mathbb{R}^p$

Recall the dot product of two vectors in $\mathbb{R}^p$:

$$\boldsymbol{a} \cdot \boldsymbol{b} = a_1 b_1 + \cdots + a_p b_p \,.$$

The dot product is a real-valued, symmetric, bilinear map on $\mathbb{R}^p$. The dot product is also positive definite:

Definition 4.1

Suppose that T is a bilinear $\mathbb{R}$-valued map on $\mathbb{R}^p$. We say that T is positive definite if $T(\boldsymbol{a}, \boldsymbol{a}) \geqslant 0$ for all $\boldsymbol{a} \in \mathbb{R}^p$, and $T(\boldsymbol{a}, \boldsymbol{a}) = 0$ if and only if $\boldsymbol{a} = \mathbf{0}$.

Note: Recall the length of a vector in $\mathbb{R}^p$ is $\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a} \cdot \boldsymbol{a}}$. Positive definiteness implies that the zero vector is the only vector of zero length.

We can try to use the dot product for fields other than $\mathbb{R}$, but we run into problems:

- In $\mathbb{C}^2$, let $\boldsymbol{a} = \begin{pmatrix} 1 \\ i \end{pmatrix}$. Then $1^2 + i^2 = 0$ but $\boldsymbol{a} \neq \mathbf{0}$.

- Suppose $\mathbb{F} = \mathbb{Z}_3$ and $V = \mathbb{F}^3$. Let $\boldsymbol{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

  Then $\boldsymbol{a} \cdot \boldsymbol{a} = 1 + 1 + 1 = 0$ but $\boldsymbol{a} \neq \mathbf{0}$.

We can bring back positive-definiteness in $\mathbb{C}^n$, as we will see.

But for finite fields there is no way to define something like the dot product.

So for this chapter we work with vector spaces over $\mathbb{R}$ or over $\mathbb{C}$, and always specify which field.

Theorem 4.1 (Cauchy–Schwarz Inequality in $\mathbb{R}^p$)

For any $\boldsymbol{a},\ \boldsymbol{b} \in \mathbb{R}^p$ we have

$$-\|\boldsymbol{a}\|\,\|\boldsymbol{b}\| \leq \boldsymbol{a} \cdot \boldsymbol{b} \leq \|\mathbf{a}\|\,\|\mathbf{b}\|.$$

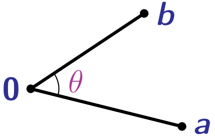
Proof:

$\square$

Corollary 4.2 If $\boldsymbol{a} \neq \mathbf{0}$ and $\boldsymbol{b} \neq \mathbf{0}$ then

$$-1 \leq \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\|\boldsymbol{a}\| \, \|\boldsymbol{b}\|} \leq 1.$$

Note: In $\mathbb{R}^2$ we have $\boldsymbol{a} \cdot \boldsymbol{b} = \|\boldsymbol{a}\| \, \|\boldsymbol{b}\| \cos\theta$, where $\theta \in [0, \pi]$ is the angle between the vectors:



Definition 4.2 If $\boldsymbol{a}, \ \boldsymbol{b} \in \mathbb{R}^n$ are non-zero then the angle θ between $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined by

$$\cos\theta = \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{\|\boldsymbol{a}\| \, \|\boldsymbol{b}\|}, \quad \theta \in [0, \pi].$$

We call non-zero vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ orthogonal if $\boldsymbol{a} \cdot \boldsymbol{b} = 0$.

Example 4.1 Let $U = \{\boldsymbol{x} \in \mathbb{R}^{17} \ : \ x_i = 0 \ \text{ for } \ 1 \leq i \leq 7\}$ and $W = \{\boldsymbol{x} \in \mathbb{R}^{17} \ : \ x_i = 0 \ \text{ for } \ 8 \leq i \leq 17\}$.

Note that every vector in U is orthogonal to every vector in W.

Furthermore $\dim U = 10$, $\dim W = 7$ and $U \cap W = \{\mathbf{0}\}$, therefore $U \oplus W = \mathbb{R}^{17}$.

We have a special name for subspaces like this:

Definition 4.3 Let $X \leq \mathbb{R}^p$ for some p. The space

$$Y = \{\boldsymbol{y} \in \mathbb{R}^p \ : \ \boldsymbol{y} \cdot \boldsymbol{x} = 0 \ \text{ for all } \ \boldsymbol{x} \in X\}$$

is called the orthogonal complement of X, denoted $X^{\perp}$.

Proposition 4.3 For any $X \leq \mathbb{R}^p$ we have $\mathbb{R}^p = X \oplus X^{\perp}$ and $(X^{\perp})^{\perp} = X$.

Proof: Deferred!! We prove this later, when we have more tools. (See Theorem 4.11.)

Definition 4.4 A set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subseteq \mathbb{R}^p$ of non-zero vectors is orthogonal if $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ whenever $i \neq j$.

We say that S is orthonormal if $\mathbf{v}_i \cdot \mathbf{v}_j = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{otherwise.} \end{cases}$

A standard example of an orthonormal set is the standard basis.

Lemma 4.4.
An orthogonal set S in $\mathbb{R}^p$ is linearly independent.

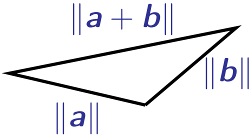
Proof.

$\square$

Theorem 4.5 (The Triangle Inequality)

For $\boldsymbol{a}$, $\boldsymbol{b} \in \mathbb{R}^p$,

$$\|\boldsymbol{a} + \boldsymbol{b}\| \leq \|\boldsymbol{a}\| + \|\boldsymbol{b}\|.$$



Proof.

$\square$

4.2 Dot Product in $\mathbb{C}^p$

As we saw earlier, the dot product in $\mathbb{C}^p$ must be different from the one we use in $\mathbb{R}^p$.

We need to make sure that $\boldsymbol{a} \cdot \boldsymbol{a}$ is real and nonnegative, but this is not always true for z^2 when $z \in \mathbb{C}$.

However, $z\overline{z} = |z|^2 \geq 0$ for all $z \in \mathbb{C}$, where $\overline{z}$ is the complex conjugate of z. Given $\boldsymbol{a} \in \mathbb{C}^p$ with $\boldsymbol{a}^T = (a_1, \ldots, a_p)$,

$$\|\boldsymbol{a}\|^2 = \sum_{i=1}^{p} a_i \overline{a_i} = \sum_{i=1}^{p} \overline{a_i} a_i = \overline{\boldsymbol{a}}^T \boldsymbol{a}.$$

Then $\|\boldsymbol{a}\|$ is real and nonnegative, and $\|\boldsymbol{a}\| = 0$ if and only if $\boldsymbol{a} = 0$.

Definition 4.5 The standard dot product on $\mathbb{C}^p$ is defined by

$$\boldsymbol{a} \cdot \boldsymbol{b} = \sum_{i=1}^{p} \overline{a_i} b_i = \overline{\boldsymbol{a}}^T \boldsymbol{b}\,.$$

As in the real case, the length or norm of a vector in $\mathbb{C}^p$ is $\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a} \cdot \boldsymbol{a}}$.

We will use $\boldsymbol{a}^*$ as a useful shorthand for $\overline{\boldsymbol{a}}^T$ from now on, so that $\boldsymbol{a} \cdot \boldsymbol{b} = \boldsymbol{a}^* \boldsymbol{b}$.

Note: Some texts (or lecturers) define the standard dot product on $\mathbb{C}^p$ as $\boldsymbol{a}^T \overline{\boldsymbol{b}}$. Make sure you check which convention (definition) is being used.

Example 4.2 In $\mathbb{C}^2$,

$$\begin{pmatrix} 1 \\ i \end{pmatrix} \cdot \begin{pmatrix} 1-i \\ 2i \end{pmatrix} = 1(1-i) + (-i)(2i) = 3 - i.$$

Lemma 4.6

The standard dot product on $\mathbb{C}^p$ has the following properties:

(a) $\boldsymbol{a} \cdot (\lambda \boldsymbol{b} + \boldsymbol{c}) = \lambda \boldsymbol{a} \cdot \boldsymbol{b} + \boldsymbol{a} \cdot \boldsymbol{c}$ for $\lambda \in \mathbb{C}$.

(b) $\boldsymbol{b} \cdot \boldsymbol{a} = \overline{\boldsymbol{a} \cdot \boldsymbol{b}}$.

(c) $(\lambda \boldsymbol{b} + \boldsymbol{c}) \cdot \boldsymbol{a} = \overline{\lambda} \boldsymbol{b} \cdot \boldsymbol{a} + \boldsymbol{c} \cdot \boldsymbol{a}$ for $\lambda \in \mathbb{C}$.

(d) $\|\boldsymbol{a}\| \geqslant 0$ for all $\boldsymbol{a} \in \mathbb{C}^p$, and $\|\boldsymbol{a}\| = 0$ if and only if $\boldsymbol{a} = \mathbf{0}$.

Exercise: Prove these statements.

We see from this lemma that the standard dot product on $\mathbb{C}^p$ is not bilinear or symmetric, but is positive definite.

We refer to property (c) in Lemma 4.6 as conjugate linearity, and property (b) as conjugate symmetry.

The term sesquilinearity is sometimes used for properties (a) and (c) together.

Next we generalise the dot product.

4.3 Inner Product Spaces

For the rest of this section, $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$. We usually state and prove theorems for $\mathbb{F} = \mathbb{C}$. (The case $\mathbb{F} = \mathbb{R}$ is analogous and may be easier.)

Definition 4.6

Let V be a vector space over $\mathbb{F}$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$. An inner product on V is a function $\langle \cdot\,, \cdot \rangle : V \times V \to \mathbb{F}$ which maps $(\boldsymbol{u}, \boldsymbol{v}) \in V \times V$ to $\langle \boldsymbol{u}, \boldsymbol{v} \rangle \in \mathbb{F}$, such that for all $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in V$ and all $\alpha \in \mathbb{F}$,

(IP1) $\langle \boldsymbol{u}, \boldsymbol{v} + \boldsymbol{w} \rangle = \langle \boldsymbol{u}, \boldsymbol{v} \rangle + \langle \boldsymbol{u}, \boldsymbol{w} \rangle$.

(IP2) $\langle \boldsymbol{u}, \alpha \boldsymbol{v} \rangle = \alpha \langle \boldsymbol{u}, \boldsymbol{v} \rangle$.

(IP3) $\langle \boldsymbol{v}, \boldsymbol{u} \rangle = \overline{\langle \boldsymbol{u}, \boldsymbol{v} \rangle}$.

(IP4) $\langle \boldsymbol{v}, \boldsymbol{v} \rangle$ is real and positive if $\boldsymbol{v} \neq \mathbf{0}$, and $\langle \mathbf{0}, \mathbf{0} \rangle = 0$.

We call V together with $\langle\ ,\ \rangle$ an inner product space.

Note: (IP4) is positive-definiteness, generalising Definition 4.1.

The norm of a vector (with respect to this inner product) is $\|\boldsymbol{v}\| = \langle \mathbf{v}, \mathbf{v} \rangle^{1/2}$.

Note: If $\mathbb{F} = \mathbb{R}$ then (IP3) is just symmetry: $\langle \boldsymbol{v}, \boldsymbol{u} \rangle = \langle \boldsymbol{u}, \boldsymbol{v} \rangle$.

Example 4.3 The standard dot products on $\mathbb{R}^p$ and $\mathbb{C}^p$ are both inner products.

Example 4.4 Recall that $C[0, 1]$ is the vector space of all continuous real-valued functions on $[0, 1]$. The function

$$\langle f, g \rangle = \int_0^1 f(t)g(t)\, dt$$

is an inner product.

Exercise: Check this. The trickiest part is (IP4) for $f \neq 0$.

Note: This example can be generalised to any interval $[a, b]$, or to real-valued functions on closed bounded subsets of $\mathbb{R}^n$.

Example 4.5 For $X, Y \in M_{p,q}(\mathbb{C})$, define

$$\langle X, Y \rangle = \operatorname{tr}(X^* Y).$$

This is an inner product on $M_{p,q}(\mathbb{C})$.

Let's check positive definiteness (the trickiest part):

$\diamond$

Example 4.6 Let $V = \mathbb{R}^2$ and define

$$\begin{aligned} \langle \boldsymbol{u}, \boldsymbol{v} \rangle &= u_1 v_1 - u_1 v_2 - u_2 v_1 + 7 u_2 v_2 \\ &= (u_1 \ u_2) \begin{pmatrix} 1 & -1 \\ -1 & 7 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}. \end{aligned}$$

We can write $\langle \boldsymbol{u}, \boldsymbol{v} \rangle$ as $\boldsymbol{u}^T A \boldsymbol{v}$ where A is a symmetric matrix.
Let's check $\langle \boldsymbol{u}, \boldsymbol{u} \rangle$:

So $\langle \boldsymbol{u}, \boldsymbol{u} \rangle$ is nonnegative, and equals zero if and only if $\boldsymbol{u} = \mathbf{0}$.

Exercise:
Check that all other properties of an inner product hold.
Hence $\langle \boldsymbol{u}, \boldsymbol{v} \rangle$ defines an inner product on $\mathbb{R}^2$.

Example 4.7

Let ℓ^2 be the vector space of real sequences $\{a_k\}$ such that $\sum_{k=1}^{\infty} a_k^2$ is finite. (These sequences are square summable.)

Show that $\langle \{a_k\}, \{b_k\} \rangle = \sum_{k=1}^{\infty} a_k b_k$ is an inner product on ℓ^2.

Solution:

$\diamond$

Lemma 4.7 Let V be an inner product space. Then for $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in V$ and $\alpha \in \mathbb{C}$,

(a) $\|\boldsymbol{u}\| > 0$ if and only if $\boldsymbol{u} \neq \mathbf{0}$.

(b) $\langle \boldsymbol{u} + \boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{u}, \boldsymbol{w} \rangle + \langle \boldsymbol{v}, \boldsymbol{w} \rangle$.

(c) $\langle \alpha \boldsymbol{u}, \boldsymbol{v} \rangle = \overline{\alpha} \langle \boldsymbol{u}, \boldsymbol{v} \rangle$ and $\|\alpha \boldsymbol{u}\| = |\alpha| \, \|\boldsymbol{u}\|$.

(d) $\langle \boldsymbol{x}, \boldsymbol{u} \rangle = 0$ for all $\boldsymbol{u}$ if and only if $\boldsymbol{x} = \mathbf{0}$.

(e) $|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \|\boldsymbol{u}\| \, \|\boldsymbol{v}\|$ (Cauchy-Schwarz inequality)

(f) $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$ (the triangle inequality).

Proof:

Notes.

(a) Example 4.6 can be generalised to $\mathbb{R}^n$ by choosing a suitable symmetric matrix $A \in M_{n,n}(\mathbb{R})$.

We can use the same idea in $\mathbb{C}^n$ by defining $\langle \mathbf{u}, \mathbf{v} \rangle = \boldsymbol{u}^* A \boldsymbol{v}$ where A satisfies $A^* = A$. We say that A is a Hermitian.

Not every symmetric (or Hermitian) matrix will define an inner product : for example, $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ does not work. (Check!)

Even if positive definiteness fails, there are still lots of interesting results as long as A is invertible – there are connections with Special Relativity. (We do not explore this.)

(b) Example 4.4 can also be generalised. For example, if $w(t) \geqslant 0$ is continuous then

$$\langle f, g \rangle = \int_0^1 f(t)g(t)w(t)\,dt$$

also gives an inner product on $C[0,1]$.

We call $w(t)$ a weighting function.

(c) If we choose the weighting function appropriately, and/or restrict the set of allowable functions, then we could create an inner product for functions over $\mathbb{R}$.

This is beyond the scope of this course (infinite-dimensional function spaces are tricky!) but forms part of functional analysis.

(d) We can use the Cauchy-Schwarz inequality

$$\left| \int_0^1 f(t)\, g(t)\, dt \right| \leq \sqrt{\int_0^1 f(t)^2\, dt \int_0^1 g(t)^2\, dt}$$

and the triangle inequality

$$\sqrt{\int_0^1 (f(t) + g(t))^2\, dt} \leq \sqrt{\int_0^1 f(t)^2\, dt} + \sqrt{\int_0^1 g(t)^2\, dt}$$

to obtain results on $C[0,1]$ which are hard to prove directly.

4.4 Orthogonality and Orthonormality

Definition 4.7

Let V be an inner product space. Non-zero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are orthogonal if $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = 0$. If $\boldsymbol{u}$ and $\boldsymbol{v}$ are orthogonal then we write $\boldsymbol{u} \perp \boldsymbol{v}$.

Let $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\} \subseteq V$ be a set of of nonzero vectors. We say that S is orthogonal if $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = 0$ for all $i \neq j$.

We say S is orthonormal if $\langle \boldsymbol{v}_i, \boldsymbol{v}_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$.

Given an orthogonal set of vectors, we can divide each vector by its length (which is nonzero) to obtain an orthonormal set.

Fact: Orthogonal sets are linearly independent.
(The proof is the same as the proof as Lemma 4.4.)

Example 4.8

Consider $M_{2,2}(\mathbb{R})$ with inner product $\langle X, Y\rangle = \mathrm{tr}(X^*Y)$.
Show that

$$\left\{ \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix},\ \frac{1}{2}\begin{pmatrix}1 & -1\\ 1 & -1\end{pmatrix},\ \frac{1}{2}\begin{pmatrix}-1 & -1\\ 1 & 1\end{pmatrix},\ \frac{1}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix} \right\}$$

is an orthonormal basis.

Solution.

$\diamond$

Example 4.9

Consider the set $C[-\pi, \pi]$ of real continuous functions on $[-\pi, \pi]$, with the inner product $\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)\,dx$.
Note that for $p, q \in \mathbb{Z}$,

$$\int_{-\pi}^{\pi} \sin(px)\sin(qx)\,dx = \frac{1}{2}\int_{-\pi}^{\pi} \cos\left((p-q)x\right) - \cos\left((p+q)x\right)\,dx$$

which is zero if $p \neq q$. So

$$S = \{\sin(x), \sin(2x), \sin(3x), \dots\}$$

is an infinite orthogonal set in this inner product space.

Exercise: Prove similarly that

$$C = \{1, \cos(x), \cos(2x), \cos(3x), \dots\}$$

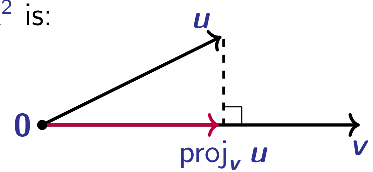
is an infinite orthogonal set in this inner product space.

Definition 4.8 In an inner product space V let $\boldsymbol{v} \neq \mathbf{0}$.
Given $\boldsymbol{u} \in V$, the projection of $\boldsymbol{u}$ onto $\boldsymbol{v}$ is defined as

$$\operatorname{proj}_{\boldsymbol{v}}(\boldsymbol{u}) = \frac{\langle \boldsymbol{v}, \boldsymbol{u} \rangle}{\langle \boldsymbol{v}, \boldsymbol{v} \rangle}\, \boldsymbol{v}.$$

Note: $\boldsymbol{u} - \alpha \boldsymbol{v} \perp \boldsymbol{v}$ if and only if $\boldsymbol{u} - \alpha \boldsymbol{v} = \boldsymbol{u} - \operatorname{proj}_{\boldsymbol{v}} \boldsymbol{u}$.

The picture in $\mathbb{R}^2$ is:



Exercise: Prove that the projection operator is linear; that is,

$$\operatorname{proj}_{\boldsymbol{v}}(\alpha \boldsymbol{x} + \boldsymbol{y}) = \alpha \operatorname{proj}_{\boldsymbol{v}}(\boldsymbol{x}) + \operatorname{proj}_{\boldsymbol{v}}(\boldsymbol{y})\,.$$

Note:
This is why we use $\langle \boldsymbol{v}, \boldsymbol{u} \rangle$ in the complex case and not $\langle \boldsymbol{u}, \boldsymbol{v} \rangle$.

Example 4.10 Consider $\mathbb{C}^2$ with the standard inner product.
Find the projection of $\boldsymbol{u} = \begin{pmatrix} -5+9i \\ -3-2i \end{pmatrix}$ onto $\boldsymbol{v} = \begin{pmatrix} 1+i \\ 2-i \end{pmatrix}$.

Solution.

$\diamond$

Note that $\boldsymbol{x} = \boldsymbol{u} - \operatorname{proj}_{\boldsymbol{v}} \boldsymbol{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ and

$$\langle \boldsymbol{x}, \boldsymbol{v} \rangle =$$

Example 4.11 In $C[-\pi, \pi]$ from Example 4.9, find the projection of $f(x) = x$ onto $g(x) = \sin(px)$ for $p \in \mathbb{Z}$.

Solution.

$\diamond$

Here $\langle g, f \rangle$ is a Fourier coefficent (up to a scale factor).

Lemma 4.8

Let $S = \{\boldsymbol{v}_1, \dots, \boldsymbol{v}_k\}$ be an orthogonal set of non-zero vectors in inner product space V.

If $\boldsymbol{v} \in \operatorname{span}(S)$ then $\quad \boldsymbol{v} = \sum_{i=1}^{k} \operatorname{proj}_{\boldsymbol{v}_i} \boldsymbol{v}$.

If S is an orthonormal set then $\boldsymbol{v} = \sum_{i=1}^{k} \langle \boldsymbol{v}_i, \boldsymbol{v} \rangle \, \boldsymbol{v}_i$.

Proof.

$\square$

Corollary 4.9 If $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$ is an orthonormal basis for V then

$$\boldsymbol{v} = \sum_{i=1}^{n} \langle \boldsymbol{e}_i, \boldsymbol{v} \rangle \, \boldsymbol{e}_i.$$

Note: These last two results illustrate one reason we like orthogonal and orthonormal bases: we can find coordinates in them without solving linear equations.

Next we look at a way of creating orthonormal bases in any (finite-dimensional) inner product space.

4.5 The Gram-Schmidt Process

Theorem 4.10. (Gram-Schmidt Process)
Every finite-dimensional inner product space has an orthonormal basis.

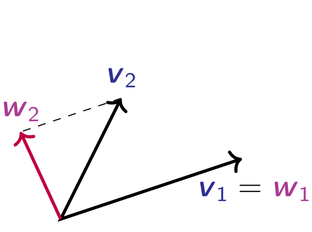
Proof: The proof is constructive: we show how to turn any given basis into an orthogonal one and then normalise.

Suppose $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ is a basis for V over $\mathbb{F}$. We define

$$\begin{aligned}
\boldsymbol{w}_1 &= \boldsymbol{v}_1, \\
\boldsymbol{w}_2 &= \boldsymbol{v}_2 - \mathrm{proj}_{\boldsymbol{w}_1}(\boldsymbol{v}_2), \\
\boldsymbol{w}_3 &= \boldsymbol{v}_3 - \mathrm{proj}_{\boldsymbol{w}_1}(\boldsymbol{v}_3) - \mathrm{proj}_{\boldsymbol{w}_2}(\boldsymbol{v}_3),
\end{aligned}$$

and in general, $\boldsymbol{w}_k = \boldsymbol{v}_k - \displaystyle\sum_{j=1}^{k-1} \mathrm{proj}_{\boldsymbol{w}_j}(\boldsymbol{v}_k)$.

You should be aware of how this looks in $\mathbb{R}^p$:



Now we use induction to prove that $\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_p\}$ is an orthogonal basis.

Example 4.12. Apply the Gram-Schmidt process to the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

in $\mathbb{R}^3$ with the standard dot product.

Solution.

$\diamond$

Example 4.13. Find an orthonormal basis of $\mathbb{C}^3$, with its standard dot product, which contains a multiple of $\begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}$.

Solution.

$\diamond$

Example 4.14.
Find an orthonormal basis for $\text{span}(\{1, t, t^2\})$ as a subspace of $C[0,1]$ with respect to $\langle f, g\rangle = \int_0^1 f(t)\,g(t)\,dt$.

Solution.

$\diamond$

4.6 Orthogonal Complements

Definition 4.10. Let $X \leq V$ for some inner product space V. The space

$$Y = \{\boldsymbol{y} \in V \ : \ \langle \boldsymbol{y}, \boldsymbol{x} \rangle = 0 \quad \text{for all} \quad \boldsymbol{x} \in X\}$$

is called the orthogonal complement of X, denoted $X^{\perp}$.

We now have a generalisation of Proposition 4.3.

Theorem 4.11. Suppose that V is a finite-dimensional inner product space and $W \leq V$. Then

$$V = W \oplus W^{\perp} \qquad \text{and} \qquad (W^{\perp})^{\perp} = W\,.$$

Proof:

$\square$

This result is not true for infinite dimensions:

Example 4.15. Returning to ℓ^2 from Example 4.7, let V be the subspace of all sequences in ℓ^2 with only finitely many non-zero terms.

This is a proper subspace: for example the sequence $\{n^{-1}\}$ belongs to ℓ^2 but not V.

Claim: $V^\perp$ is trivial:

Hence ℓ^2 is not equal to the sum of V and its orthogonal complement. $\diamond$

Infinite-dimensional inner product spaces are topics studied in courses on Functional Analysis e.g. MATH3611.

Corollary 4.12

Suppose V is a finite-dimensional inner product space, $W \leq V$ and $\boldsymbol{v} \in V$. Then there is a unique way to write

$$\boldsymbol{v} = \boldsymbol{a} + \boldsymbol{b} \quad \text{where} \quad \boldsymbol{a} \in W \;\text{ and }\; \boldsymbol{b} \in W^{\perp}.$$

We say that $\boldsymbol{a} = \operatorname{proj}_W \boldsymbol{v}$ and $\boldsymbol{b} = \operatorname{proj}_{W^\perp} \boldsymbol{v}$.

Fact: If $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ is an orthonormal basis of V such that $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ is an orthonormal basis of W then

$$\boldsymbol{a} = \sum_{i=1}^{k} \operatorname{proj}_{\boldsymbol{v}_i} \boldsymbol{v} \text{ and } \boldsymbol{b} = \boldsymbol{v} - \boldsymbol{a} = \sum_{i=k+1}^{n} \operatorname{proj}_{\boldsymbol{v}_i} \boldsymbol{v}.$$

Also, proj_W is a linear map on V for any $W \leq V$.

Corollary 4.12 implies that if V is finite-dimensional then

$$\operatorname{im}(\operatorname{proj}_W) = W \text{ and } \ker(\operatorname{proj}_W) = W^{\perp}.$$

Example 4.16. Let $W = \text{span}(1, t, t^2) \leq C[0,1]$.

Using the inner product in $C[0,1]$ from Example 4.14, find the projection of $\sqrt{t}$ onto W.

Solution.

We use the orthonormal basis for W from Example 4.14.

$\diamond$

Lemma 4.13. Let V be a finite-dimensional inner product space with $W \leq V$.

(a) If $\boldsymbol{v} \in V$ then $\boldsymbol{v} - \mathrm{proj}_W(\boldsymbol{v})$ is in $W^\perp$.

(b) If $\boldsymbol{w} \in W$ then $\mathrm{proj}_W(\boldsymbol{w}) = \boldsymbol{w}$. Hence the projection map is idempotent: that is, $(\mathrm{proj}_W) \circ (\mathrm{proj}_W) = \mathrm{proj}_W$.

(c) For all $\boldsymbol{v} \in V$ we have $\|\mathrm{proj}_W(\boldsymbol{v})\| \leq \|\boldsymbol{v}\|$.

(d) The function $\mathrm{proj}_W + \mathrm{proj}_{W^\perp}$ is the identity function on V.

Exercise: Prove these statements (using Corollary 4.12).

Lemma 4.14. Let W be a subspace of a finite-dimensional inner product space V, and let $\boldsymbol{v} \in V$. For every $\boldsymbol{w} \in W$ we have

$$\|\boldsymbol{v} - \boldsymbol{w}\| \geq \|\boldsymbol{v} - \mathrm{proj}_W(\boldsymbol{v})\|,$$

with equality if and only if $\boldsymbol{w} = \mathrm{proj}_W(\boldsymbol{v})$ (which holds if and only if $\boldsymbol{v} - \boldsymbol{w} \in W^{\perp}$).

Proof:

$\square$

Lemma 4.14 shows that projection onto a subspace solves the following optimisation problem:

Q: If $\boldsymbol{v}$ is not in W, which point in W is closest to v?

("Closest" means "closest with respect to the distance defined by the inner product".)

We come back to this in Section 4.10 where we consider least squares problems.

4.7 Adjoints

Let $\boldsymbol{x}$ be a fixed vector in an inner product space V over the field $\mathbb{F}$. Properties of the inner product tell us that the map $\boldsymbol{v} \mapsto \langle \boldsymbol{x}, \boldsymbol{v} \rangle$ is a linear map from V to $\mathbb{F}$.

Linear maps from V to the field $\mathbb{F}$ are called covectors, and the set of all covectors is called the dual space of V, denoted V^*.

From Corollary 3.15, if V is finite-dimensional then V and V^* have the same dimension. Then Theorem 3.11 says that V and V^* are isomorphic.

The following lemma will help us find a canonical (basis independent) isomorphism between V and V^* when $\mathbb{F} = \mathbb{R}$.

Lemma 4.15. If $(V, \langle\cdot,\cdot\rangle)$ is a finite-dimensional inner product space and $T : V \to \mathbb{F}$ is linear then there is a unique vector $\boldsymbol{t} \in V$ such that $T(\boldsymbol{v}) = \langle \boldsymbol{t}, \boldsymbol{v} \rangle$ for all $\boldsymbol{v} \in V$.

Proof:

$\square$

Claim: If V is a real finite-dimensional inner product space then the map which sends covector T to the corresponding vector $\boldsymbol{t}$ is an isomorphism between V and the dual space V^*.

Proof of Claim:

$\square$

In the case of $\mathbb{R}^n$ with the standard dot product, we have $\boldsymbol{w} \cdot \boldsymbol{v} = \boldsymbol{w}^T \boldsymbol{v}$.

Hence given $\boldsymbol{w} \in \mathbb{R}^n$, the covector $T : \mathbb{R}^n \to \mathbb{R}$ such that $T(\boldsymbol{v}) = \boldsymbol{w} \cdot \boldsymbol{v}$ (that is, the covector corresponding to $\boldsymbol{w}$) is just matrix multiplication by $\boldsymbol{w}^T$.

So we can think of the dual space $(\mathbb{R}^n)^*$ as row vectors.

Note: in the complex case we would have conjugate linearity, and we would end up with an isomorphism to the conjugate dual space $\overline{V^*}$. We do not prove this.

Theorem 4.16 For any linear map $T : V \to W$ between finite-dimensional inner product spaces, there is a unique linear map $T^* : W \to V$, called the adjoint of T, such that

$$\langle \boldsymbol{w}, T(\boldsymbol{v})\rangle = \langle T^*(\boldsymbol{w}), \boldsymbol{v}\rangle \tag{1}$$

for all $\boldsymbol{v} \in V$ and $\boldsymbol{w} \in W$.

Note: the inner product on the left is in W, the inner product on the right is in V.

Proof: We prove uniqueness now, and prove existence at the end of the section.

$\square$

Uniqueness implies that we do not need to use the proof to find the adjoint. If we find a linear map T^* which satisfies equation (1) then T^* must be the adjoint of T.

For example, in the case of the standard inner product on $\mathbb{R}^p$, suppose $T(\boldsymbol{v}) = A\boldsymbol{v}$ for some $p \times q$ matrix A.

Then

$$\boldsymbol{w} \cdot A\boldsymbol{v} = \boldsymbol{w}^T A\boldsymbol{v} = (A^T\boldsymbol{w})^T\boldsymbol{v} = (A^T\boldsymbol{w}) \cdot \boldsymbol{v}\,.$$

So $T^*(\boldsymbol{w}) = A^T\boldsymbol{w}$. The adjoint is just multiplication by A^T.

Similarly for $\mathbb{C}^p$, we find that $T^*(\boldsymbol{w}) = \overline{A}^T \boldsymbol{w}$, so the adjoint is multiplication by the conjugate transpose of A. We call $\overline{A}^T$ the adjoint of A, and write it as A^*.

Uniqueness also gives this easy result:

Proposition 4.17 For any inner product space V, the identity map on V is its own adjoint.

Proof: Exercise. $\square$

Example 4.18.

Consider the inner product $\langle p, q\rangle = \int_{-1}^{1} p(t)q(t)\,dt$ on both $\mathcal{P}_2(\mathbb{R})$ and $\mathcal{P}_1(\mathbb{R})$. Find the adjoint of the differentiation map $T : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_1(\mathbb{R})$, $p \mapsto p'$.

Solution.

◇

The adjoint depends on the particular inner product used.

For example, if the integration in the previous example had been over $[0,1]$ then the adjoint would have been

$$T^*(q_0 + q_1 t) = 2q_1 - 6q_0 + (12q_0 - 24q_1)t + 30q_1 t^2 \,.$$

Theorem 4.18 Suppose that V and W are finite-dimensional inner product spaces. Let S and T be linear transformations from V to W. Then

(a) $(S+T)^* = S^* + T^*$.

(b) for any scalar α we have $(\alpha T)^* = \overline{\alpha} T^*$;

(c) $(T^*)^* = T$.

(d) if $U : W \to X$ is linear then $(U \circ T)^* = T^* \circ U^*$.

Proof:

$\square$

Now we show that in an orthonormal basis, the matrix of the adjoint of a map is the adjoint (conjugate transpose) of its matrix.

Theorem 4.19 Let V and W be finite-dimensional inner product spaces with orthonormal bases $\mathcal{B}$ and $\mathcal{C}$ respectively.

If A is the matrix of the linear transformation $T : V \to W$ with respect to bases $\mathcal{B}$ and $\mathcal{C}$, then the matrix of $T^* : W \to V$ with respect to bases $\mathcal{C}$ and $\mathcal{B}$ is the adjoint of A.

Proof:

$\square$

Exercise: Prove that $\left\{ \frac{1}{\sqrt{2}}, \frac{\sqrt{6}}{2} t, \frac{3}{4}\sqrt{10}\left(t^2 - \frac{1}{3}\right) \right\}$ is an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$ for the inner product in Example 4.18, and check the formula for the adjoint we found in Example 4.18.

We finish this section by completing the proof of the existence of the adjoint. We restate the theorem below.

Theorem 4.16 For any linear map $T : V \to W$ between finite-dimensional inner product spaces, there is a unique linear map $T^* : W \to V$, called the adjoint of T, such that

$$\langle \boldsymbol{w}, T(\boldsymbol{v}) \rangle = \langle T^*(\boldsymbol{w}), \boldsymbol{v} \rangle$$

for all $\boldsymbol{v} \in V$ and $\boldsymbol{w} \in W$.

Proof (real case only): Recall we have already proved uniqueness.

$\square$

The proof for the complex case is similar, but requires some complex conjugation in a couple of places and is left as an optional exercise.

4.8 Maps with Special Adjoints

Definition 4.11. Let $T : V \to V$ be a linear map on a finite-dimensional inner product space V. Then T is said to be

- unitary if $T^* = T^{-1}$;
- an isometry if $\|T(\boldsymbol{v})\| = \|\boldsymbol{v}\|$ for all $\boldsymbol{v} \in V$;
- self-adjoint or Hermitian if $T^* = T$.

Two properties of these types of maps are easy:

Proposition 4.20

(a) A map T is unitary if and only if T^* is unitary.

(b) The set of all unitary transformations on V forms a group under composition.

Exercise: Prove these statements.

Theorem 4.21 Let T be a linear map on a finite-dimensional inner product space V. Then the following are equivalent:

(a) T is an isometry;

(b) $\langle T(\boldsymbol{v}), T(\boldsymbol{w})\rangle = \langle \boldsymbol{v}, \boldsymbol{w}\rangle$ for all $\boldsymbol{v}, \boldsymbol{w} \in V$ (that is, T preserves inner products);

(c) T is unitary (that is, $T^* \circ T$ is the identity);

(d) T^* is an isometry;

(e) if $\{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n\}$ is an orthonormal basis for V then so is $\{T(\boldsymbol{e}_1), \ldots, T(\boldsymbol{e}_n)\}$.

Proof (Outline).

Exercise: Fill in the details. $\square$

We can make similar definitions for matrices.

Definition 4.12. A matrix $A \in M_{p,p}(\mathbb{C})$ is called unitary if $A^* = A^{-1}$ and Hermitian if $A^* = A$.

A matrix $A \in M_{p,p}(\mathbb{R})$ is called orthogonal if $A^T = A^{-1}$ and symmetric if $A^T = A$.

If A is a Hermitian matrix then the map $\boldsymbol{x} \mapsto A\boldsymbol{x}$ is a Hermitian map (with the standard inner product); the same holds for unitary, symmetric etc.

Applying Theorem 4.21(e) to the standard bases gives:

Corollary 4.22 The columns of a $p \times p$ matrix are an orthonormal basis of $\mathbb{C}^p$ if and only if A is unitary.

The columns of a $p \times p$ matrix are an orthonormal basis of $\mathbb{R}^p$ if and only if A is orthogonal.

The same results apply to rows.

Example 4.19. From Examples 4.12 and 4.13 we have the orthogonal matrix Q_1 and unitary matrix Q_2, where

$$Q_1 = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 1/\sqrt{3} & 2/\sqrt{6} & 0 \\ i/\sqrt{3} & -i/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & -1/\sqrt{6} & i/\sqrt{2} \end{pmatrix}.$$

Exercise: Check that the rows of Q_1 are an orthonormal basis of $\mathbb{R}^3$, and (slightly harder) that the rows of Q_2 are an orthonormal basis of $\mathbb{C}^3$.

4.9 QR Factorisations

Theorem 4.23 Let A be a $p \times q$ matrix of rank q (so $p \geqslant q$). Then we can write $A = QR$ where Q is an $p \times q$ matrix with orthonormal columns, and R is an $q \times q$ invertible upper triangular matrix.

Proof:

Example 4.20

Returning to Examples 4.12 and 4.13 we have (Check!)

$$\begin{pmatrix} 1 & 3 & 3 \\ 2 & 1 & 1 \\ 2 & -4 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}\begin{pmatrix} 3 & -1 & 3 \\ 0 & 5 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & 1 & 0 \\ i & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ \frac{i}{\sqrt{3}} & -\frac{i}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} & \frac{i}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} \sqrt{3} & \frac{1}{\sqrt{3}} & -\frac{i}{\sqrt{3}} \\ 0 & \frac{\sqrt{6}}{3} & \frac{i}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

If A is square (and so invertible), then Q is also square and either unitary (if complex) or orthogonal (if real).

In the non-square cases, we can modify our QR factorisation to make Q square, at the expense of making R not square.

Corollary 4.24 Let $A \in M_{p,q}(\mathbb{F})$ with $p > q$ and $\operatorname{rank}(A) = q$. Then we can write $A = \widetilde{Q}\widetilde{R}$, with $\widetilde{Q}$ a $p \times p$ unitary (or orthogonal) matrix, and $\widetilde{R}$ a $p \times q$ matrix of rank q and in echelon form.

Proof:

$\square$

Example 4.21 Find both QR factorisations of

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Solution.

◇

Example 4.22 Find a QR factorisation of

$$A = \begin{pmatrix} 2i & -2 & 0 \\ 1 & 2i & -2 \\ 2 & 3i & 1 \end{pmatrix}$$

Solution:

$\diamond$

4.10 The Method of Least Squares

Suppose we have a lot of data and while trying to fit a model to the data we end up with a system of linear equations $A\mathbf{x} = \mathbf{b}$.

Also suppose that we want to take all the data into account and not introduce too many unknowns, and so we have p equations in q unknowns with $p > q$.

Typically, there will be no exact solution. So what is the "best" approximate solution?

Q: What do we mean by "best"?

A: We will minimize the sum of the squares of the errors.

So we minimize $\|A\mathbf{x} - \mathbf{b}\|$ using the standard inner product on $\mathbb{R}^p$ or $\mathbb{C}^p$. We call these solutions the least squares solutions to the system.

Now $A\boldsymbol{x}$ is in the column space of A, so Lemma 4.14 tells us that the least squares solution is the projection of $\boldsymbol{b}$ onto $\operatorname{col}(A)$.

Equivalently, we want $\boldsymbol{x}$ so that $A\boldsymbol{x} - \boldsymbol{b}$ is in $\operatorname{col}(A)^{\perp}$. But

We have proved:

Theorem 4.25. A least squares solution to the system of equations $A\boldsymbol{x} = \boldsymbol{b}$ is a solution to the equations $A^*A\boldsymbol{x} = A^*\boldsymbol{b}$, which are called the normal equations.

Q: Are there always solutions to the normal equations? If so, how many solutions are there?

A: Since $A\boldsymbol{x}$ is the projection onto the column space of A, the least squares solution to $A\boldsymbol{x} = \boldsymbol{b}$ always exists, and will be unique as long as the columns of A are independent.

If A has independent columns then this implies that A^*A is invertible (you can also prove this directly). Hence the solution to the normal equations is

$$\boldsymbol{x} = (A^*A)^{-1}A^*\boldsymbol{b} \tag{2}$$

and the projection onto $\mathrm{col}(A)$ is given by

$$\mathrm{proj}_{\mathrm{col}(A)}(\boldsymbol{x}) = A(A^*A)^{-1}A^*\boldsymbol{x}\,.$$

The most common use of least squares is finding the line of best fit to a set of points in $\mathbb{R}^2$. Given a set of points $\{(x_i, y_i) : i = 1, \ldots, n\}$ we look for a least squares solution to the equations $\{y_i = a + bx_i \ : \ i = 1, \ldots, n\}$.

In matrix form we have $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{x} = \begin{pmatrix} a \\ b \end{pmatrix}$, $\boldsymbol{b} \in \mathbb{R}^n$ given by $(y_1, \ldots, y_n)^T$ and $A \in M_{n,2}(\mathbb{R})$ given by

$$A = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}. \quad \text{Hence} \quad A^*A = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix}.$$

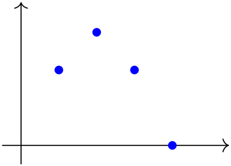
As long as the x_i are not all equal, the columns of A are independent and so there is a unique line of best fit. Then we can use equation (2) (rather than row reduction).

Example 4.23 Find the line of best fit to the points

$$(1,2),(2,3),(3,2),(4,0)$$

in the least squares sense.

Solution. Plotting the points gives



$\diamond$

Example 4.24. Consider

$$W = \{\boldsymbol{x} \in \mathbb{R}^3 \,:\, 2x_1 + 3x_2 - x_3 = 0\} \leq \mathbb{R}^3\,.$$

Find the projection matrix onto W.

Check your result by showing that the normal to W is projected to $\mathbf{0}$.

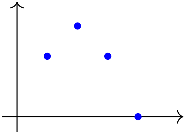
Solution:

$\diamond$

Example 4.25. Find the quadratic of best fit for the points from Example 4.23:

$$(1, 2), (2, 3), (3, 2), (4, 0).$$

Solution.



$\diamond$

Exercise: If the original equations $A\boldsymbol{x} = \boldsymbol{b}$ have an exact solution, show that the least squares method will find it.

A possible problem with the least squares procedure, if we calculate using decimal approximations, is that A^*A may be ill–conditioned. This can mean that that numerical inversion or row-reduction of the matrix can incur large rounding errors.

We can use a QR factorisation for A to help find least squares solutions (see Example 4.26). However, there can also be numerical issues with the Gram-Schmidt method (such as subtracting two numbers that are very close).

There is another method of finding a QR factorisation called the Householder algorithm that is designed to avoid numerical issues.

(We do not cover the Householder algorithm in this course, but I will post some notes on to Moodle so you can read about it if you are interested.)

Suppose we want a least squares solution to $A\boldsymbol{x} = \boldsymbol{b}$ and we have a QR factorisation of A as $A = QR$ where R is square and invertible.

Then the normal equations in the form $A^*A\boldsymbol{x} = A^*\boldsymbol{b}$ are

$$R^*Q^*QR\boldsymbol{x} = R^*Q^*\boldsymbol{b}. \quad \text{Hence} \quad R^*R\boldsymbol{x} = R^*Q^*\boldsymbol{b},$$

since orthogonality of the columns of Q means that $Q^*Q = I$ even if Q is not square.

If R is invertible then so is R^*. Then the normal equations reduce to

$$R\boldsymbol{x} = Q^*\boldsymbol{b}.$$

These equations can be solved by back-substitution.

Example 4.26 Use a QR factorisation to find the line of best fit to the points

$$(1, 2), (2, 3), (3, 2), (4, 0)$$

in the least squares sense.

Solution.

$\diamond$

Key points for Chapter 4

- Standard dot products, pp. 2, 10.
- Cauchy-Schwarz Inequality, pp. 4, 18.
- Orthogonal and orthonormal vectors, pp. 5, 6, 23.
- Orthogonal Complement, pp. 6, 38.
- Inner Product, p. 13.
- Projections, pp. 26, 73.
- Gram-Schmidt process, p. 31.
- Adjoint of linear map, p. 50.
- Unitary, Hermitian etc pp. 60, 63.
- QR factorisation, p. 65.
- Least squares, normal equations, pp. 72, 79.

MATH2601 – Higher Linear Algebra

Chapter 5. Determinants

In this chapter we shall look at several approaches to defining determinants and proving their properties.

We want you to see this theory for your background knowledge, but proofs from the definition will not be examinable.

You will be expected to know how to use properties of determinants to calculate determinants quickly, and to be able to use these properties in linear algebra proofs and computations in subsequent chapters.

Historical fact: the idea of a determinant is older than the idea of a matrix!

Determinants were invented because they determine when a square system of linear equations has a solution.

Recall that

$\det(a_{11}) = |(a_{11})| = a_{11}$ and $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$.

Also

$$\begin{aligned}
\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\
&= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} \\
&\quad + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} \\
&= \sum_{\sigma \in \mathcal{S}_3} \operatorname{sign}(\sigma)\; a_{1\sigma(1)}a_{2\sigma(2)}a_{3\sigma(3)}
\end{aligned}$$

where $\mathcal{S}_3$ is the group of permutations of 3 objects and $\operatorname{sign}(\sigma) = \pm 1$ depending on the permutation σ.

We could also have written the 2×2 determinant as

$$a_{11}a_{22} - a_{12}a_{21} = \sum_{\sigma \in \mathcal{S}_2} \operatorname{sign}(\sigma)\ a_{1\sigma(1)}a_{2\sigma(2)}\,.$$

Maybe we could generalise this idea, if we can find a good way to define $\operatorname{sign}(\sigma)$.

Recall our notation for permutations on $[n] = \{1, 2, 3, \ldots, n\}$ from Chapter 1. For example, the permutation

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

on $[3]$ swaps 2 and 3 and leaves 1 unchanged.

We will omit the first row of this matrix and write a permutation as a re-ordering of the elements of $[n]$, using the notation $p_1\, p_2\, \cdots\, p_n$ where $1 \mapsto p_1$, $2 \mapsto p_2$ and so on.

So $2\,1\,4\,3$ is the permutation on $[4]$ that swaps 1 and 2 and also swaps 3 and 4.

In a permutation $p_1\, p_2\, \cdots\, p_n$, an inversion occurs whenever a larger number precedes a smaller number.

For example, the permutation $3\,6\,2\,1\,4\,5$ on $[6]$ contains seven inversions:

- 3 precedes 1 and 2 (two inversions)
- 6 precedes 1, 2, 4 and 5 (four inversions)
- 2 precedes 1 (one inversion)

Definition 5.1 If a permutation $\sigma = p_1\, p_2\, \cdots\, p_n$ of $[n]$ contains k inversions then its sign is $\text{sign}(\sigma) = (-1)^k$.

A permutation is called even (or odd, respectively) if it has an even (respectively, odd) number of inversions.

The permutation $3\,6\,2\,1\,4\,5$ on $[6]$ has 7 inversions, so this permutation is odd.

The permutation $2\,1\,4\,3$ on $[4]$ has ___ inversions, so it is _____.

Now check the formula for the 3×3 determinant. In each term, the sign corresponds to the sign of the permutation:

$$\begin{array}{lll} \text{sign}(1\,2\,3) = 1, & \text{sign}(1\,3\,2) = -1, & \text{sign}(2\,1\,3) = -1, \\ \text{sign}(3\,2\,1) = -1, & \text{sign}(2\,3\,1) = 1, & \text{sign}(3\,1\,2) = 1. \end{array}$$

Definition 5.2 The determinant of an $n \times n$ matrix $A = (a_{ij})_{i,j=1,\dots,n}$ is

$$\det(A) = \sum_{\sigma \in \mathcal{S}_n} \operatorname{sign}(\sigma)\, a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)} \ .$$

We will need to prove some properties of the sign of a permutation. First we introduce a useful object:

Definition 5.3 Define the n'th difference product Δ_n by

$$\Delta_n = \prod_{i=1}^{n} \prod_{j=i+1}^{n} (i-j)$$

$$= (1-2)(1-3)\cdots(1-n)(2-3)(2-4)\cdots((n-1)-n).$$

For example, $\Delta_3 = (1-2)(1-3)(2-3)$.

Now define the action of a permutation $\sigma \in \mathcal{S}_n$ on Δ_n by

$$\sigma(\Delta_n) = \prod_{i=1}^{n} \prod_{j=i+1}^{n} (\sigma(i) - \sigma(j)).$$

For example, the permutation $\sigma = 2\,3\,1$ acting on Δ_3 gives

$$\sigma(\Delta_3) = (2-3)(2-1)(3-1) = \Delta_3.$$

The permutation rearranges the factors of Δ_n and changes the signs of some factors.

In fact, the number of factors whose signs are changed equals the number of inversions in the permutation!

Proposition 5.1 For any $\sigma \in \mathcal{S}_n$, $\quad \sigma(\Delta_n) = \text{sign}(\sigma)\Delta_n$.

Lemma 5.2 If $\alpha, \beta \in \mathcal{S}_n$ then $\text{sign}(\alpha \circ \beta) = \text{sign}(\alpha)\,\text{sign}(\beta)$ and $\text{sign}(\alpha^{-1}) = \text{sign}(\alpha)$.

Proof:

$\square$

This lemma tells us that $\text{sign} : \mathcal{S}_n \to \{-1, 1\}$ is a group homomorphism, where $\{-1, 1\}$ is a group under multiplication.

Definition: A transposition is a swap of two elements.

Proposition 5.3 Every transposition in $\mathcal{S}_n$ is odd.

Proof:

$\square$

Theorem 5.4 Let $A \in M_{p,p}(\mathbb{F})$. Then

(a) $\det(A) = \sum_{\sigma \in \mathcal{S}_n} \operatorname{sign}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n}$.

(b) $\det(A) = \det(A^T)$ and $\det(A^*) = \overline{\det(A)}$.

(c) If any row or column of A is zero then $\det(A) = 0$.

(d) If a permutation is applied to the rows or columns of A, then the determinant is multiplied by the sign of the permutation.

(e) If two rows of A are the same, or two columns or A are the same, then $\det(A) = 0$.

(f) If any row (or column) of A has a multiple of another row (or column) (respectively) added to it then the determinant is unchanged.

Before proving Theorem 5.4, we note one important consequence that we will prove later.

Theorem 5.5
If det is considered as a map on the rows (or columns) of a matrix (that is, from p copies of $\mathbb{F}^p$ to $\mathbb{F}$), then it is multilinear and alternating.

As a **corollary:** If any row or column of A is multiplied by a constant α, then the determinant is multiplied by α.

Proof of Theorem 5.4:

Now we prove Theorem 5.5, restated below.

Theorem 5.5

If det is considered as a map on the rows (or columns) of a matrix (that is, from p copies of $\mathbb{F}^p$ to $\mathbb{F}$), then it is multilinear and alternating.

Proof:

Definition 5.4

Given $A \in M_{p,p}(\mathbb{F})$, let A_{ij} be the matrix obtained from A by deleting row i and column j. We call A_{ij} the (i,j)-minor of A.

Lemma 5.6

Suppose that row i of matrix $A \in M_{p,p}(\mathbb{F})$ is zero except for entry a_{ij}. Then $\det(A) = (-1)^{i+j} a_{ij} \det(A_{ij})$.

Note: The same result applies if column j of A is zero except for a_{ij}.

Example:

$$\begin{vmatrix} 1 & 3 & 2 \\ 0 & 2 & 0 \\ 1 & 5 & -1 \end{vmatrix} = (-1)^{2+2} \times 2 \times \begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix}$$

Proof.

□

Lemma 5.6 and the last part of Theorem 5.4 allows us to efficiently calculate determinants as follows:

(1) Add appropriate multiples of one row (or column) to all the other rows (or columns) until one row (or column) has only one non-zero entry. This does not change the determinant.

(2) Use Lemma 5.6 to reduce the determinant to one that is one row and column smaller.

(3) Repeat steps 1 and 2 until you have a 2×2 determinant, which you can calculate.

Example 5.1. Find $\begin{vmatrix} -1 & 0 & 0 & 0 \\ 1 & 3 & 4 & 1 \\ 2 & 2 & 1 & 5 \\ 0 & 1 & 2 & 7 \end{vmatrix}$.

Another way to calculate determinants is to row reduce the matrix to upper triangular form, keeping careful track of any row swaps.

Proposition 5.7 The determinant of a square upper triangular or lower triangular matrix is the product of its diagonal elements.

Proof:

$\square$

Now we work towards proving that the determinant is multiplicative: $\det(AB) = \det(A)\det(B)$.

Definition 5.5 Recall that an elementary row operation on a matrix is one of the following:

- swapping two rows
- multiplying one row by a non-zero scalar
- adding a multiple of one row to another

An elementary matrix is a matrix obtained from an identity matrix after an elementary row operation.

Performing an elementary row operation on a matrix is the same as multiplying the matrix on the left by the equivalent elementary matrix.

Proposition 5.7 implies that $\det(\mathrm{I}) = 1$ for all identity matrices.

It follows from Theorem 5.4 and Theorem 5.5 that for an elementary matrix E and any matrix A,

(1) if E swaps two rows then $\det(EA) = -\det(A)$;

(2) if E multiplies one row by a non-zero scalar λ then

$$\det(EA) = \lambda \det(A);$$

(3) if E adds a multiple of one row to another then

$$\det(EA) = \det(A).$$

We can rewrite this as follows (Check this!):

If E is an elementary matrix then $\det(EA) = \det(E)\det(A)$ for any matrix A.

Lemma 5.8 If A is invertible then there is a sequence of elementary matrices E_1, E_2, ... E_k such that

$$A = E_1 E_2 \cdots E_k \quad \text{and} \quad \det(A) = \prod_{i=1}^{k} \det(E_i)\,.$$

Proof:

$\square$

Lemma 5.9

A square matrix A is invertible if and only if $\det(A) \neq 0$.

Proof:

$\square$

Theorem 5.10
For any $A, B \in M_{p,p}(\mathbb{F})$, $\quad \det(AB) = \det(A)\det(B)$.

Proof:

$\square$

We have already used Theorem 5.10 to show that the determinant is a similarity invariant. Another corollary of Theorem 5.10 is:

Corollary 5.11 If A is invertible then $\det(A^{-1}) = \frac{1}{\det(A)}$.

Next we highlight something from the proof of Lemma 5.9 which will be useful in the next chapter:

Lemma 5.12 A square matrix A has determinant zero if and only if it has linearly dependent columns, and hence if and only if it has linearly dependent rows.

Definition 5.6 Given a square matrix A, the number $c_{ij} = (-1)^{i+j} \det(A_{ij})$ is called the cofactor of element a_{ij}.

The formula from Lemma 5.6 can thus be rewritten as $\det(A) = a_{ij}\, c_{ij}$ for a matrix with only one non-zero entry in row i (or column j).

Theorem 5.13 For any $A \in M_{p,p}(\mathbb{F})$ and any fixed j,

$$\det(A) = \sum_{i=1}^{p} a_{ij} c_{ij}\,.$$

This is called expanding along column j.

You can also expand along rows.

Proof (sketch):

$\square$

In our example:

Note: It is usually better to perform row operations or column operations to simplify matrices.

Avoid using the cofactor expansion to calculate determinants except for the 2×2 case.

We can use cofactors as an alternative way of calculating inverses.

Definition 5.6 For $A \in M_{p,p}(\mathbb{F})$, the adjugate (also called the classical adjoint and adjunct) of A, denoted $\mathrm{adj}(A)$, is the transpose of the matrix of cofactors:

$$\mathrm{adj}(A)_{ij} = c_{ji}.$$

Theorem 5.14 If A is invertible then $A^{-1} = \frac{\mathrm{adj}(A)}{\det(A)}$.

Proof:

$\square$

In the 2×2 case this gives the familiar formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

This is the only situation where you should use the adjugate formula for the inverse.

Key points for Chapter 5

- Sign of a permutation, p. 6.
- Determinant of a matrix, p. 7.
- Properties of the determinant, pp. 11, 12, 27, 28, 29.
- Minor, p. 19.
- Elementary row operation, elementary matrix p. 24.
- Cofactor and cofactor expansion, p. 30.
- Adjugate, p. 33.

MATH2601 – Higher Linear Algebra

Chapter 6. Eigenvalues and Eigenvectors

In Chapter 3 we briefly looked at the normal form: if we are free to choose bases separately in the domain and codomain, then any linear map T has a matrix of the form $\begin{pmatrix} \mathrm{I}_r & 0 \\ 0 & 0 \end{pmatrix}$ where r is the rank of T and each 0 represents a zero matrix of the appropriate size.

If $T : V \to V$ for some vector space V then it is more natural to use the same basis for V in both the domain and codomain.

Q: Which basis would you choose to make the matrix of $T : V \to V$ as simple as possible?

We already saw that if $V = X \oplus Y$ where both X and Y are invariant under T then there is a basis of V for which the matrix of T is $A \oplus B$ for some matrices A and B.

6.1 Eigenvalues and Eigenvectors

Definition 6.1 Let $T \in L(V, V)$.

(a) if $T(\boldsymbol{v}) = \alpha \boldsymbol{v}$ for some $\boldsymbol{v} \in V$ and $\alpha \in \mathbb{F}$ with $\boldsymbol{v} \neq \mathbf{0}$ then α is an eigenvalue of T and $\boldsymbol{v}$ is an eigenvector for T corresponding to α.

(b) If λ is an eigenvalue of T then the eigenspace of T is $E_\lambda(T) = \{\boldsymbol{v} \in V \ : \ T(\boldsymbol{v}) = \lambda \boldsymbol{v}\}$.

(c) We call the set of all eigenvalues of T the spectrum of T.

Notes:

(1) Eigenvectors are never zero.

(2) $E_\lambda(T) = \{\mathbf{0}\} \cup \{\text{all eigenvectors corresponding to } \lambda\}$.

(3) Each eigenspace is invariant under T.

(4) If $A \in M_{p,p}(\mathbb{F})$ then we define the eigenvalues, eigenvectors, eigenspaces and spectrum of A to equal those of the map $T : \mathbb{F}^p \to \mathbb{F}^p$ given by $T(\boldsymbol{x}) = A\boldsymbol{x}$.

Lemma 6.1 Let $T : V \to V$ be linear. Then

(a) $E_\lambda(T) = \ker(\lambda\,\mathrm{id} - T)$.

(b) If $\lambda_1, \ldots, \lambda_k$ are distinct eigenvalues and $T(\boldsymbol{v}_i) = \lambda_i \boldsymbol{v}_i$ for $i = 1, \ldots, k$ then $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ is linearly independent.

(c) If $\lambda \neq \mu$ then $E_\lambda(T) \cap E_\mu(T) = \{\mathbf{0}\}$.

Proof:

$\square$

Note that Lemma 6.1 applies to finite-dimensional and infinite-dimensional spaces.

The matrix version of Lemma 6.1 (a) implies that $\boldsymbol{v}$ is an eigenvector of A with eigenvalue λ if and only if $\boldsymbol{v} \in \ker(\lambda \mathrm{I} - A)$ and $v \neq 0$.

Example 6.1. Let $A = \begin{pmatrix} 4 & 6 & 6 \\ -4 & -7 & -8 \\ 3 & 6 & 7 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$.

Prove that $\boldsymbol{v}$ is an eigenvector of A for eigenvalue 1.

Show that $E_1(A)$ has dimension 2 and find a second independent eigenvector for eigenvalue 1.

$\diamond$

Recall that a square matrix A is diagonal if the only non-zero entries are on the main diagonal: $a_{ij} = 0$ if $i \neq j$.

Definition 6.2 A matrix $A \in M_{p,p}(\mathbb{F})$ is diagonalisable over $\mathbb{F}$ if there is an invertible matrix $P \in \mathrm{GL}(p, \mathbb{F})$ such that $P^{-1}AP$ is diagonal (that is, A is similar over $\mathbb{F}$ to a diagonal matrix).

A linear map $T : V \to V$ is diagonalisable if there is a basis of V with respect to which the matrix of T is diagonal.

Remark. For a finite-dimensional vector space V, a linear map $T : V \to V$ is diagonalisable if and only if its matrix with respect to any basis is diagonalisable.

Theorem 6.2

Suppose that V is a finite-dimensional vector space over $\mathbb{F}$ and $T \in L(V, V)$, then T is diagonalisable if and only if V has a basis whose elements are all eigenvectors of T.

Similarly, $A \in M_{p,p}(\mathbb{F})$ is diagonalisable if and only if $\mathbb{F}^p$ has a basis consisting of eigenvectors of A.

Proof:

$\square$

Lemma 6.1 (c) and Theorem 6.2 give the useful special case:

Corollary 6.3

A $p \times p$ matrix with p distinct eigenvalues is diagonalisable.

Note: The converse of this corollary is not true: there are $p \times p$ matrices with fewer than p different eigenvalues which are diagonalisable.

Example 6.2

Let $V = \{\text{infinitely differentiable functions} : \mathbb{R} \to \mathbb{R}\}$ and define the linear map $D : V \to V$ by $f \longmapsto \frac{df}{dx}$.

Then

$$Df = \lambda f \quad \text{if and only if} \quad f = Ae^{\lambda x} \text{ for some } A \in \mathbb{R},$$

so each eigenspace is 1-dimensional. The spectrum of D is $\mathbb{R}$.

Example 6.3.
Following on from Example 6.2, consider the linear map D^2 defined by $f \mapsto \frac{d^2f}{dx^2}$. The eigenvectors (eigenfunctions) of D^2 are solutions to the differential equation $D^2f = \lambda f$.

Fact: The form of the eigenvectors f depend on the sign of the eigenvalue:

(a) If $\lambda = 0$ then $f(x) = Ax + B$.

(b) If $\lambda > 0$ then with $\lambda = \nu^2$ we have $f(x) = Ae^{\nu x} + Be^{-\nu x}$.

(c) If $\lambda < 0$ then with $\lambda = -\omega^2$ we have
$$f(x) = A\sin\omega x + B\cos\omega x.$$

In each case A, B are real constants which are not both zero. So all eigenspaces are 2-dimensional.

The spectrum of D^2 is $\mathbb{R}$.

Example 6.4

Following on from Example 6.3, let W be the subspace of V consisting of functions that are zero at both 0 and π.

Solving the same differential equation as in Example 6.3, the boundary conditions exclude the cases where $\lambda \geq 0$.

In fact, the only eigenfunctions that survive are the sines, and to make the function to vanish at π then the eigenvalues must be $-n^2$ for $n \in \mathbb{Z}$.

So in W, the spectrum of D^2 is $\{-1, -4, -9, \dots\}$, and the eigenvectors of $-n^2 \in \mathbb{Z}$ have the form $x \mapsto A\sin(nx)$ where A is a nonzero constant.

Here each eigenspace is 1-dimensional.

6.2 The Characteristic Polynomial

In order to compute the eigenvalues of a $p \times p$ matrix, note that by definition,

Definition 6.3 If A is a $p \times p$ matrix over $\mathbb{F}$ then the characteristic polynomial of A is

$$\mathrm{cp}_A(t) = \det(t\,\mathrm{I} - A)\,.$$

In practice, we usually calculate $\det(A - t\,\mathrm{I}) = (-1)^p\,\mathrm{cp}_A(t)$. (Why?)

Notes.

- You can check that $\det(t\mathrm{I} - A)$ really is a polynomial in t.
  If A is a $p \times p$ then $\mathrm{cp}_A(t)$ has degree p.
- The spectrum of A is the same as the set of zeros of $\mathrm{cp}_A(t)$.
- By definition, the characteristic polynomial is always monic (that is, the leading coefficient is 1).

In first year and elsewhere, the characteristic polynomial is defined to be $\det(A - t\mathrm{I})$, which equals our definition if p is even, and is opposite in sign if p is odd.

Lemma 6.4
The characteristic polynomial is a similarity invariant.

Proof:

$\square$

This result means we can make the following definition.

Definition 6.4 If V is finite-dimensional and $T \in L(V, V)$ then the characteristic polynomial of T, denoted $\mathrm{cp}_T(t)$, is defined to be $\mathrm{cp}_A(t)$ for any matrix A of T.

Lemma 6.5 Suppose $T \in L(V, V)$ and $\dim V = n$. Then

(a) λ is an eigenvalue if and only if $\mathrm{cp}_T(\lambda) = 0$.

(b) if $W \leq E_\lambda(T)$ then W is T-invariant.

(c) λ is an eigenvalue if and only if $\mathrm{nullity}(T - \lambda\,\mathrm{id}) > 0$.

Proof:

$\square$

Theorem 6.6 Let $T : V \to V$ be a linear map on finite-dimensional vector space V, and let $\mathcal{B}$ be a basis for V. Let A be the matrix of T with respect to $\mathcal{B}$.

The following are equivalent:

- $\boldsymbol{v}$ is an eigenvector of T corresponding to eigenvalue λ;
- the coordinate vector $[\boldsymbol{v}]_{\mathcal{B}}$ is an eigenvector of A corresponding to eigenvalue λ.

Proof:

$\square$

By the Rank–Nullity Theorem, part (c) of Lemma 6.5 is equivalent to the condition $\operatorname{rank}(T - \lambda \operatorname{id}) < n$.

We can (sometimes) use part (c) of Lemma 6.5 to spot an eigenvalue.

If subtracting a number λ from each entry of the diagonal makes a matrix obviously not full rank then λ is an eigenvalue.

For example, let $A = \begin{pmatrix} 2 & 1 & 2 \\ 2 & 3 & 3 \\ 0 & 0 & 2 \end{pmatrix}$.

- Since $A - 2\mathrm{I}$ has a row of zeros, we know that 2 is an eigenvalue of A.
- Similarly, $A - \mathrm{I}$ has its first two columns the same, so 1 is also an eigenvalue of A.

Diagonalisation Algorithm. Given an $n \times n$ matrix A:

(1) Calculate $\mathrm{cp}_A(t)$.

(2) Find all roots α of $\mathrm{cp}_A(t)$.

(3) For each such α, calculate $\dim\left(E_\alpha(A)\right)$.

(4) If $\sum_\alpha \dim\left(E_\alpha(A)\right) = n$ then A is diagonalizable.

(5) If A is diagonalisable, let P be a matrix whose columns are a basis of V consisting of eigenvectors of A.
Then $D = P^{-1}AP$ is diagonal.

The diagonal entries in D will be the eigenvalues of A, and their order will match the order of the eigenvectors in P.

(Which order you use in P and D does not usually matter, so long as they match.)

Example 6.5. Let $A = \begin{pmatrix} -1 & -12 & 0 \\ 2 & 5 & 4 \\ 0 & 4 & -1 \end{pmatrix}$.

Given that $\mathrm{cp}_A(t) = (t-3)(t^2-1)$, show that A is diagonalisable and find a diagonalising matrix P.

$\diamond$

We know from Corollary 6.3 that a 3×3 matrix with 3 distinct eigenvalues is diagonalisable.

What if some eigenvalues are repeated?

Example 6.6. The two matrices

$$A = \begin{pmatrix} 1 & -1 & -3 \\ -3 & -1 & -7 \\ 2 & 2 & 7 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 4 & -10 \\ 1 & 0 & 5 \\ 1 & -2 & 7 \end{pmatrix}$$

both have characteristic polynomial $(t-2)^2(t-3)$.

Explain why B is diagonalisable but A is not.

$\diamond$

Whether or not a matrix is diagonalisable depends on the field we work in.

Example 6.7. Consider

$$A = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}, \quad \text{so} \quad \mathrm{cp}_A(t) = \begin{vmatrix} t-1 & 1 \\ -2 & t+1 \end{vmatrix} = t^2 + 1.$$

- If we work in $\mathbb{R}$ (or $\mathbb{Q}$) then $\mathrm{cp}_A(t)$ has no roots, and so A has no eigenvalues.
- Over $\mathbb{C}$ (or $\mathbb{Q}(i)$), the matrix A has eigenvalues $i, -i$ and so A is diagonalisable.
  Exercise: find a diagonalising matrix P.
- If we work in $\mathbb{Z}_7$ then checking squares,
  $$0^2 = 0,\ 1^2 = 6^2 = 1,\ 2^2 = 5^2 = 4 \text{ and } 3^2 = 4^2 = 2.$$
  Hence $t^2 + 1 = 0$ (equivalently, $t^2 = 6$) has no solutions.
  So A is not diagonalisable over $\mathbb{Z}_7$ as it has no eigenvalues ☹

- Over $\mathbb{Z}_2$ the equation $t^2 + 1$ has only one solution $t = 1$, and A has only one eigenvector (check!). Hence A is not diagonalisable.
- But over $\mathbb{Z}_5$:

It turns out that this matrix A is diagonalisable over $\mathbb{Z}_p$ for p prime if and only if $p \equiv 1 \pmod 4$.

Questions such as "in which $\mathbb{Z}_p$ does a given polynomial have a solution" are covered in number theory courses such as MATH3431.

6.3 Multiplicities

Recall the matrices from Example 6.6:

$$A = \begin{pmatrix} 1 & -1 & -3 \\ -3 & -1 & -7 \\ 2 & 2 & 7 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 4 & -10 \\ 1 & 0 & 5 \\ 1 & -2 & 7 \end{pmatrix},$$

both with characteristic polynomial $(t-2)^2(t-3)$.

We saw that A was not diagonalisable because the eigenvalue 2 occurs twice as a root of the characteristic polynomial, but the eigenspace $E_2(A)$ only has dimension 1.

The matrix B is diagonalisable because each eigenspace has the necessary dimension to allow us to find a basis of eigenvectors of B. In particular, $\dim\left(E_2(B)\right) = 2$.

Definition 6.5 Let T be a linear map on finite-dimensional vector space V. Suppose further that λ is an eigenvalue of T.

- The geometric multiplicity (g.m.) of λ is $\dim\left(E_\lambda(T)\right)$.
- The algebraic multiplicity (a.m.) of λ is the multiplicity of $t - \lambda$ as a factor of $\mathrm{cp}_T(t)$.

Remark: The geometric multiplicity of eigenvalue λ is the nullity of $T - \lambda\,\mathrm{id}$, which is the maximum number of linearly independent eigenvectors of λ.

The algebraic multiplicity of λ is a if and only if $(t-\lambda)^a$ divides $\mathrm{cp}_T(t)$ but $(t-\lambda)^{a+1}$ does not divide $\mathrm{cp}_T(t)$.

Exercise: For matrices A and B from Example 6.6:

	a.m. in A	g.m. in A	a.m. in B	g.m. in B
$\lambda = 2$				
$\lambda = 3$				

The algebraic and geometric multiplicities depend on the field.

In particular, to get p eigenvalues for a $p \times p$ matrix, we need to work in a field in which $\mathrm{cp}_A(t)$ splits, that is, where $\mathrm{cp}_A(t)$ has p roots (counting multiplicities).

This often involves working over $\mathbb{C}$, as $\mathbb{C}$ is algebraically closed: every complex polynomial of degree p has p roots in $\mathbb{C}$ (counted according to multiplicity).

Theorem 6.7. Let $A \in M_{p,p}(\mathbb{C})$. Then A has p eigenvalues $\alpha_1, \ldots, \alpha_p$ counting algebraic multiplicities. Also

$$\det(A) = \prod_{i=1}^{p} \alpha_i \quad \text{and} \quad \mathrm{tr}(A) = \sum_{i=1}^{p} \alpha_i\,.$$

Proof:

Notes.

- For a 2×2 matrix, if you know the determinant and trace then you can find the eigenvalues.

Recall, in Example 6.7 we had $\text{tr}(A) = 0$ and $\det(A) = 1$.
The only two numbers with sum 0 and product 1 are $i, \ -i$.
So over $\mathbb{C}$, the eigenvalues of A are $i, -i$.

- Similarly, for a 3×3 matrix, it is enough to know the determinant, trace and one eigenvalue, or the trace and two eigenvalues, etc.

Recall the matrix $A = \begin{pmatrix} 2 & 1 & 2 \\ 2 & 3 & 3 \\ 0 & 0 & 2 \end{pmatrix}$ from page 17.

We noticed earlier 2 and 1 are eigenvalues of A.

Since $\text{tr}(A) = 1 + 2 + \lambda_3 = 7$ the third eigenvalue must be $\lambda_3 = 4$.

- If $\mathbb{F}$ is any field then

$$\mathrm{cp}_A(t) = t^p - \mathrm{tr}(A)\, t^{p-1} + \cdots + (-1)^p \det(A).$$

If $\mathbb{F}$ is a splitting field of $\mathrm{cp}_A(t)$ (that is, $\mathrm{cp}_A(t)$ splits over $\mathbb{F}$; in other words, $\mathrm{cp}_A(t)$ can be written as a product of linear factors over $\mathbb{F}$) then the trace is the sum of the eigenvalues and the determinant is the product of the eigenvalues.

- In fact, all the coefficients of the characteristic polynomial can be found from the traces of power of A.

  For example the coefficient of t^{p-2} is

$$\frac{1}{2}\left(\big(\mathrm{tr}(A)\big)^2 - \mathrm{tr}(A^2)\right).$$

  But $\mathrm{tr}(A)$ and $\det(A)$ are the most important and useful.

Theorem 6.8 Let $T : V \to V$ be a linear map on the finite-dimensional space V, and let λ be an eigenvalue of T. Then

$$1 \leq \text{geometric multiplicity } \lambda \leq \text{algebraic multiplicity } \lambda.$$

Proof:

$\square$

Theorem 6.9 Let $T : V \to V$ be a linear map on the finite-dimensional space V. The following four statements are equivalent:

(a) T is diagonalizable.

(b) There is a basis for V consisting of eigenvectors for T.

(c) $V = E_{\lambda_1}(T) \oplus E_{\lambda_2}(T) \oplus \cdots \oplus E_{\lambda_k}(T)$ where $\lambda_1, \ldots, \lambda_k$ are the distinct eigenvalues of T.

(d) The sum of the geometric multiplicities of the distinct eigenvalues is $\dim V$. That is,

$$\sum_{j=1}^{k} \dim E_{\lambda_j}(T) = \dim V.$$

Proof: We have already proved most parts.
Exercise: Put together the parts we have already proved, and prove the rest. $\square$

An easy corollary of this is a rewording of Corollary 6.3.

Corollary 6.10 A linear map T on a p-dimensional space V over $\mathbb{F}$ is diagonalizable if $\mathrm{cp}_T(t)$ has p distinct roots in $\mathbb{F}$.

Example 6.8. Let

$$\boldsymbol{v} = \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 6 & -2 & 3 & 1 \\ 3 & -1 & 9 & 3 \\ -2 & 4 & -1 & -2 \\ -1 & 2 & -3 & 4 \end{pmatrix}.$$

Show $\boldsymbol{v}$ is an eigenvector of A and find with minimal calculation all eigenvalues of A and their multiplicities.
Is A diagonalisable?

Solution:

Exercise. Repeat the previous example with

$$\boldsymbol{v} = \begin{pmatrix} 3 \\ 1 \\ -1 \\ 0 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 1 & 3 & -6 & -3 \\ 1 & 3 & 2 & 1 \\ 1 & -1 & 6 & 1 \\ 2 & -2 & 4 & 6 \end{pmatrix}.$$

Example 6.9 Show that for any constant a, the $n \times n$ matrix

$$A = \begin{pmatrix} 1+a & 1 & \cdots & 1 \\ 1 & 1+a & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1+a \end{pmatrix}$$

is diagonalisable.

6.4 Normal Operators

Let V be an inner product space with inner product $\langle\ ,\ \rangle : V \times V \to \mathbb{F}$.

Recall the adjoint T^* of a linear map $T : V \to V$ from Chapter 4: for all $\boldsymbol{v}, \boldsymbol{w} \in V$,

$$\langle T^*(\boldsymbol{v}), \boldsymbol{w}\rangle = \langle \boldsymbol{v}, T(\boldsymbol{w})\rangle.$$

Definition 6.6

A linear map T on an inner product space is normal if and only if $T^* \circ T = T \circ T^*$.

An $n \times n$ matrix A is normal if and only if $A^* A = A A^*$.

(That is, normal maps or matrices commute with their adjoints.)

Recall from Chapter 4 that a square matrix over $\mathbb{C}$ is unitary if $A^* = A^{-1}$, and is Hermetian if $A^* = A$. Over $\mathbb{R}$ we say orthogonal and symmetric, respectively.

- All unitary or Hermitian matrices over $\mathbb{C}$ are normal.
  All orthogonal or symmetric matrices over $\mathbb{R}$ are normal.
  Exercise: prove this.

- Let $A = \begin{pmatrix} 1 & 1 \\ i & 3+2i \end{pmatrix}$. Then $A^* = \begin{pmatrix} 1 & -i \\ 1 & 3-2i \end{pmatrix}$ and

$$A\,A^* = \begin{pmatrix} 2 & 3-3i \\ 3+3i & 14 \end{pmatrix} = A^*\,A\,.$$

  (Exercise: check this!)

  But A is neither unitary or Hermitian.

Theorem 6.11. Let V be a finite-dimensional inner product space and T a normal linear map on V. Then:

(a) $||T(\boldsymbol{v})|| = ||T^*(\boldsymbol{v})||$ for all $\boldsymbol{v} \in V$.

(b) The map $T - \alpha\,\mathrm{id}$ is normal for any scalar α.

(c) If T has eigenvalue λ then $\overline{\lambda}$ is an eigenvalue of T^*.

(d) $E_\lambda(T) = E_{\overline{\lambda}}(T^*)$.

(e) If $\lambda \neq \mu$ then $E_\lambda(T) \perp E_\mu(T)$.

Proof:

Theorem 6.12.
Let V be a finite-dimensional vector space and $T \in L(V, V)$. If T is normal with eigenvalue λ then the algebraic multiplicity of λ equals the geometric multiplicity of λ.

Proof:

Theorem 6.13
(The Spectral Theorem for Normal Operators)

If V is a finite-dimensional inner product space over $\mathbb{C}$ and $T \in L(V, V)$ is normal then there is an orthonormal basis $\mathcal{B}$ of V consisting of eigenvectors of T.

Hence if $A \in M_{p,p}(\mathbb{C})$ is normal then there is a unitary matrix P such that $P^{-1}AP = P^*AP$ is diagonal.

Conversely, if $[T]_{\mathcal{B}}^{\mathcal{B}} = A = PDP^{-1}$, where D is diagonal and P is unitary (so $\mathcal{B}$ is orthonormal), then both A and T are normal.

Proof:

Corollary 6.14 If A is $n \times n$ and normal with eigenvalues $\lambda_1, \ldots, \lambda_n$ then there are matrices $P_1, \ldots, P_n$ such that

$$A = \sum_{i=1}^{n} \lambda_i P_i, \qquad P_i^2 = P_i = P_i^* \quad \text{for} \quad i = 1, \ldots, n,$$

$$P_i P_j = 0 \quad \text{if } i \neq j, \qquad \sum_{i=1}^{n} P_i = \mathrm{I}\,.$$

Proof:

$\square$

These ideas are very important since, for example, they generalize to operators on infinite-dimensional inner product spaces.

6.5 Self-Adjoint Maps and Matrices

Theorem 6.15. Let V be a finite-dimensional inner product space over the field $\mathbb{F}$ (where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$) and $T \in L(V, V)$.

(a) If T is self-adjoint (that is, Hermitian if $\mathbb{F} = \mathbb{C}$ or symmetric if $\mathbb{F} = \mathbb{R}$) then the eigenvalues of T are real.

(b) If $\mathbb{F} = \mathbb{C}$ and T is Hermitian there is a orthonormal basis for V consisting of eigenvectors of T.

(c) If $\mathbb{F} = \mathbb{R}$ and T is symmetric then there is a real orthonormal basis of V consisting of eigenvectors of T.

Proof:

The matrix version of part (b) of Theorem 6.15 says that for a Hermitian (self-adjoint) matrix A there is a unitary matrix P such that P^*AP is real and diagonal.

Example 6.10. Unitarily diagonalise the Hermitian matrix

$$A = \begin{pmatrix} 3 & -i \\ i & 3 \end{pmatrix}.$$

Solution.

$\diamond$

The matrix version of part (c) of Theorem 6.15 says that for a real symmetric matrix A there is a real orthogonal matrix P with $P^T AP$ diagonal.

Example 6.11 Orthogonally diagonalise the symmetric matrix

$$A = \begin{pmatrix} 2 & -4 & 2 \\ -4 & 2 & -2 \\ 2 & -2 & -1 \end{pmatrix},$$

given that -2 is an eigenvalue of A.

Solution.

$\diamond$

Conic sections

We can use matrix methods to classify certain curves and surfaces.

Rather than describe the method in general, we will illustrate it with an example.

Example 6.12 Identify and sketch the curve

$$8x^2 + 12xy + 3y^2 = 48 \ .$$

Find the points, if any, on the curve which maximise or minimise the distance to the origin.

Solution: We can rewrite the equation for the curve as

$$\boldsymbol{x}^T A \boldsymbol{x} = 48 \quad \text{where} \quad \boldsymbol{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 8 & 6 \\ 6 & 3 \end{pmatrix} \ .$$

The eigenspaces of A are

$$E_{12} = \text{span}\left\{\begin{pmatrix}3\\2\end{pmatrix}\right\} \quad \text{and} \quad E_{-1} = \text{span}\left\{\begin{pmatrix}-2\\3\end{pmatrix}\right\}.$$

Since A is symmetric, these subspaces are orthogonal.

Let $Q = \frac{1}{\sqrt{13}}\begin{pmatrix}3 & -2\\2 & 3\end{pmatrix}$. Then Q orthogonally diagonalises A:

that is, $Q^T A Q = \begin{pmatrix}12 & 0\\0 & -1\end{pmatrix} = D$.

Define a new coordinate system $\boldsymbol{X} = \begin{pmatrix}X\\Y\end{pmatrix}$ by $\boldsymbol{X} = Q^T\boldsymbol{x}$.

Since $A = QDQ^T$, the equation of the curve becomes $\boldsymbol{X}^T D\boldsymbol{X} = 48$, which we rewrite as

$$12X^2 - Y^2 = 48\,.$$

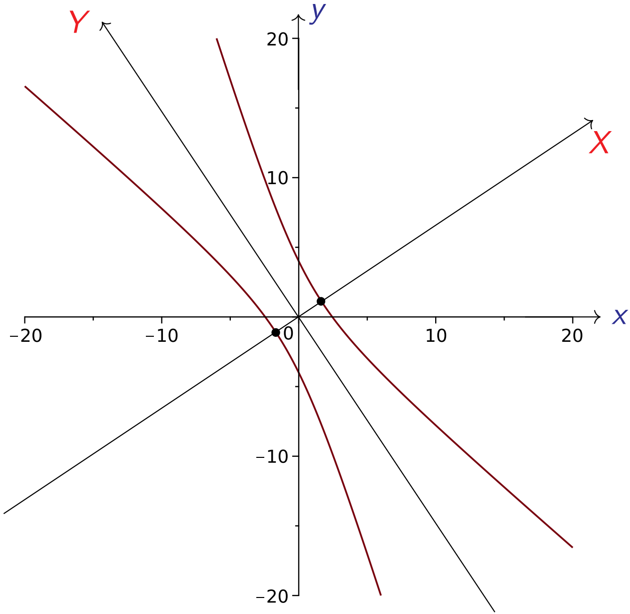
The transformation of coordinates given by Q is an isometry, so the new coordinate directions (X and Y) are still orthogonal, and we can now see that the curve is a hyperbola.

We can find the points closest to the origin in (X, Y) coordinates (which is easy) and map them to (x, y) coordinates.

The points $(X, Y) = \pm(2, 0)$ are closest to the origin, and these points are

$$\mathbf{x} = Q\mathbf{X} =$$

In a hyperbola there no points at maximal distance from the origin.


Y
y
20
X
10
x
−20
−10
0
10
20
−10
−20

Notes.

- The curve is not an ellipse, even though the inital equation has positive coefficients for both x^2 and y^2.
- To identify the curve you just need to know the signs of the eigenvalues.
  - *Both positive* $\Rightarrow$ ellipse,
  - *One positive, one negative* $\Rightarrow$ hyperbola.
- There are other choices for the matrix A such that $\boldsymbol{x}^T A\boldsymbol{x} = 48$, for example $A = \begin{pmatrix} 8 & 0 \\ 12 & 3 \end{pmatrix}$.

  However, we need a symmetric matrix so that the new axes X, Y will be orthogonal.

The eigenvectors of A are called the principal axes of the conic section (in other words, the curve).

- By multiplying one column of Q by -1 if necessary, we can assume that $\det(Q) = 1$. Then X and Y are right-handed Cartesian coordinates formed by rotating the (x, y)-axes.

A similar process can be used to analyse quadric surfaces, involving polynomials in x, y and z of degree 2.

signs of eigenvalues	surface
$+ \; + \; +$	ellipsoid
$+ \; + \; -$	hyperboloid of one sheet
$+ \; - \; -$	hyperboloid of two sheets
$+ \; + \; 0$	elliptic cylinder
$+ \; - \; 0$	hyperbolic cylinder
$+ \; 0 \; 0$	parallel planes

The process for quadric surfaces is NOT EXAMINABLE in 2024. (That is the material below the line on this page.)

6.6 Unitary and Orthogonal Maps and Matrices

Lemma 6.16. Suppose that V is a p-dimensional inner product space over the field $\mathbb{F}$ (where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$) and let $T \in L(V, V)$ be a unitary map.

Then the eigenvalues of T lie on the unit circle in $\mathbb{C}$. That is, each eigenvalue has the form $e^{i\alpha}$ for some $\alpha \in \mathbb{R}$.
Furthermore, V has an orthonormal basis of eigenvectors of T.

Proof:

$\square$

Exercise: State and prove the matrix version of Lemma 6.16.

We can say more about the eigenvalues of real isometries (or orthogonal matrices but we have to be careful because the characteristic polynomial may have non-real roots.

Suppose that $T \in L(V, V)$ is an isometry on a real finite-dimensional inner product space. By Lemma 6.16, any real eigenvalue of T must be 1 or -1. We can span the orthogonal eigenspaces $E_1(T)$ and $E_{-1}(T)$ with orthonormal bases of real vectors.

Secondly, if λ is any non-real root of $\text{cp}_T(t)$ then $|\lambda| = 1$ and there must be a complex eigenvector $\boldsymbol{v}$ such that $T(\boldsymbol{v}) = \lambda \boldsymbol{v}$. Since $\text{cp}_T(t)$ is a real polynomial, the non-real eigenvalues of T occur in conjugate pairs $\lambda, \overline{\lambda}$.

Taking the complex conjugate of $T(\boldsymbol{v}) = \lambda \boldsymbol{v}$ shows that eigenvectors of $\overline{\lambda}$ are the complex conjugate of eigenvectors of λ. By Theorem 6.13 the eigenspaces E_λ, $E_{\overline{\lambda}}$ are orthogonal.

Suppose that $\boldsymbol{v}$ is a complex eigenvector for eigenvalue $e^{i\alpha}$ of the real isometry T, and write $\boldsymbol{v} = \boldsymbol{x} + i\boldsymbol{y}$ for real vectors $\boldsymbol{x}$ and $\boldsymbol{y}$.

Then (assuming T is linear over complex scalars too),

Considering the real and imaginary parts separately shows that the real subspace spanned by $\boldsymbol{x}$ and $\boldsymbol{y}$ is invariant under T.

Next, the orthogonality of eigenvectors $\boldsymbol{v}$ and $\overline{\boldsymbol{v}}$ implies that $\boldsymbol{x}$ and $\boldsymbol{y}$ are orthogonal and have equal lengths:

By scaling if necessary, we can assume that $\{\boldsymbol{x}, \boldsymbol{y}\}$ is an orthonormal basis of the invariant space they span.

From the previous calculation, if we restrict T to $\mathrm{span}\left(\{\boldsymbol{x}, \boldsymbol{y}\}\right)$ and use the basis $\{\boldsymbol{x}, -\boldsymbol{y}\}$, the restricted map has matrix

$$R(\alpha) = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}.$$

In other words, T acts a rotation on each of the 2-dimensional invariant subspaces defined by its complex eigenvectors.

Each of these invariant 2-dimensional spaces is orthogonal to all the others, and also orthogonal to the eigenspaces for eigenvalues 1 and -1, by Theorem 6.11.

The above discussion is a sketch of the proof of the following result.

Theorem 6.17 Let T be an isometry on a real inner product space V. Then its characteristic polynomial is of the form

$$\mathrm{cp}_T(t) = (t-1)^a\,(t+1)^b\,\prod_{j=1}^{k}(t - e^{i\alpha_j})(t - e^{-i\alpha_j})$$

where $a, b, k \in \mathbb{N}$ satisfy $a + b + 2k = \dim(V)$ and $\alpha_j \in (0, \pi)$ for $j = 1, \ldots, k$.

Also, there is an orthonormal basis of V with respect to which the matrix of T has the form

$$\mathrm{I}_a \oplus (-\mathrm{I}_b) \bigoplus_{j=1}^{k} R(\alpha_j)\,. \tag{1}$$

Note: The matrix version of this result is that any orthogonal matrix is orthogonally similar to a matrix with the form given in equation (1).

What more do we need to do to make this sketch proof rigorous? Here are the problem areas:

- We just assumed that the linear map T on the real vector space V would be linear over complex scalars too.
- We also assumed that the real inner product on V would be conjugate linear over complex scalars.

We can make this rigorous using the complexification of the real inner product space V. The complexification of V is the the complex vector space $\widehat{V} = \{\boldsymbol{u} + i\boldsymbol{v} \,:\, \boldsymbol{u}, \boldsymbol{v} \in V\}$.

We will not go over the details in lectures, but I will post some notes about complexification (prepared by John Steele) on Moodle in case you would like to see these details.

Orthogonal Matrices in $\mathbb{R}^3$

Suppose A is a 3×3 orthogonal matrix. From Chapter 4 we know that the columns of A form an orthonormal basis of $\mathbb{R}^3$.

Since $AA^T = I$ it follows that $\det(A) \in \{1, -1\}$ and the group of orthogonal matrices $O(3)$ splits into two parts: the subgroup $SO(3)$ of matrices with determinant 1, and the rest.

From the matrix version of Theorem 6.17, an element of $SO(3)$ is either the identity, or similar to $(1) \oplus (-1) \oplus (-1)$, or similar to

$$R(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in (0, \pi).$$

In fact, we can cover all cases for $SO(3)$ by saying that A is similar to $R(\theta)$ where $\theta \in [0, \pi]$.

Hence the map $\boldsymbol{x} \mapsto A\boldsymbol{x}$ is a rotation around an eigenvector of eigenvalue 1 through an angle θ.

Note that $\operatorname{tr}(A) = \operatorname{tr}(R(\theta)) = 1 + 2\cos\theta$, so we can find the angle of rotation θ from the trace.

Similarly, if $A \in O(3)$ has determinant -1, then it is either $-\mathrm{I}$ or is similar to a matrix of the form

$$F(\theta) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in [0, \pi)\,.$$

In this case, the map $\boldsymbol{x} \mapsto A\boldsymbol{x}$ is a reflection in the plane perpendicular to the eigenvector for eigenvalue -1 composed with a rotation of that same plane through an angle θ.

Here the angle of rotation θ satisfies $-1 + 2\cos\theta = \operatorname{tr}(A)$, so again we can find θ from the trace.

Example 6.14 Give a geometric description of the action of $\boldsymbol{x} \mapsto A\boldsymbol{x}$ when

$$A = \frac{1}{9}\begin{pmatrix} 4 & 7 & -4 \\ 1 & 4 & 8 \\ 8 & -4 & 1 \end{pmatrix}.$$

Solution.

$\diamond$

Example 6.15 Give a geometric description of the action of $\boldsymbol{x} \mapsto B\boldsymbol{x}$ when

$$B = \frac{1}{9}\begin{pmatrix} 4 & -7 & -4 \\ 1 & -4 & 8 \\ 8 & 4 & 1 \end{pmatrix}.$$

Solution.

$\diamond$

Remark: a 3×3 orthogonal matrix A belongs to $SO(3)$ if and only if its columns form a right-handed orthonormal basis: this means that the third column of A is equal to the cross product of the first two columns.

If you already know that A is orthogonal then you only need to check the sign of one entry of this cross product to decide whether the columns are right-handed.

For example, given the orthogonal matrices

$$A = \frac{1}{17}\begin{pmatrix} 9 & -12 & 8 \\ 8 & 12 & 9 \\ 12 & 1 & -12 \end{pmatrix} \text{ and } B = \frac{1}{17}\begin{pmatrix} 8 & 12 & 9 \\ 9 & -12 & 8 \\ 12 & 1 & -12 \end{pmatrix},$$

6.7 The Singular Value Decomposition

There are several different ways to factor a matrix, depending on what you are trying to do. So far we have seen the QR-factorization (Chapter 4) and the spectral decomposition.

We now consider the singular value decomposition (SVD), which combines aspects of both of these.

The SVD has practical applications in many fields including image compression and weather prediction. (See the Wikipedia page for more applications.)

Definition 6.7 Let $A \in M_{p,q}(\mathbb{C})$.
A singular value decomposition of A is a factorization of A as $A = U\Sigma V^*$ where U and V are square and unitary and Σ is a $p \times q$ matrix with all off-diagonal entries equal to zero, and with diagonal entries called the singular values which satisfy

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_q \geq 0\,.$$

The columns of U are called left singular vectors and the columns of V are called right singular vectors.

When applied to a real matrix the matrices U and V are orthogonal but the rest of the construction is the same.

Lemma 6.18 Let $A \in M_{p,q}(\mathbb{F})$ where $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$. Then

(a) $\ker(A^*A) = \ker(A)$.

(b) $\operatorname{rank}(A^*A) = \operatorname{rank}(A)$.

Proof:

$\square$

Theorem 6.19 (Singular Value Decomposition)
For any matrix A, a singular value decomposition of A exists and the singular values are unique.

Proof: The proof is constructive and rests on the unitary diagonalisation of A^*A.

Example 6.16. Find a singular value decomposition for

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix} .$$

Solution.

$\diamond$

Definition 6.8 Suppose that $A \in M_{p,q}(\mathbb{C})$ has rank k. The reduced singular value decomposition of A is the decomposition

$$A = \widehat{U}\,\widehat{\Sigma}\,\widehat{V}^*$$

where $\widehat{U}$ and $\widehat{V}$ have orthonormal columns and $\widehat{\Sigma}$ is a $k \times k$ invertible diagonal matrix.

The reduced SVD of A is obtained from the SVD of A by deleting the last $q - k$ columns from V, deleting the last $p - k$ columns from U and using the matrix $\widehat{\Sigma}$ from the proof of Theorem 6.19.

Example 6.17. The reduced SVD is of the matrix A from Example 6.16 is

$\diamond$

The following lemma follows from the definition of the SVD:

Lemma 6.20 For any $p \times q$ matrix A of rank k:

(a) The last $q - k$ right singular vectors form an orthonormal basis of $\ker(A)$.

(b) The first k left singular vectors form an orthonormal basis of $\mathrm{im}(A)$.

(c) $\displaystyle\sum_{i=1}^{k} \sigma_i^2 = \mathrm{tr}(A^* A) = \sum_{i=1}^{p} \sum_{j=1}^{q} |a_{ij}|^2$.

Proof. Exercise!

Note: Part (c) says that the sum of the squares of the singular values equals the norm that we get from the standard inner product $\langle A, B \rangle = \mathrm{tr}(A^* B)$ on matrices.

One interesting use of the reduced SVD is the following:

Definition 6.9 For an arbitrary matrix A with reduced SVD $\widehat{U}\widehat{\Sigma}\widehat{V}^*$, the pseudoinverse of A, denoted A^+, is defined by by

$$A^+ = \widehat{V}\widehat{\Sigma}^{-1}\widehat{U}^* .$$

Theorem 6.21 The least squares solution to $A\boldsymbol{x} = \boldsymbol{b}$ for a matrix A of full rank is given by $\boldsymbol{x} = A^+\boldsymbol{b}$.

Proof:

$\square$

The SVD and approximation

The SVD allows us to approximate matrices with matrices of lower rank. This can be useful for numerical calculations.

For matrices that arise in practice, many of the singular values may be quite small: they might only be non-zero because of round-off error, for example.

If we set all appropriately small non-singular values to be zero, we can replace the matrix by a lower rank approximation, and only need to keep the reduced SVD of this approximation.

For example, MATLAB calculates the rank of a matrix by finding the singular values and only counting those that are "large enough".

Suppose you have a 512×512 pixel black and white image, which you store as a 512×512 matrix A of greyscales.

In general, this matrix will have rank 512, and requires us to store $512^2 = 262\,144$ numbers.

Now suppose that A is well-approximated by a rank 64 approximation.

To store this reduced matrix we will need to keep the appropriate 64 left singular vectors and 64 right singular vectors (all in $\mathbb{R}^{512}$) and the 64 singular values. This requires us to store $64 \times 2 \times 512 + 64 = 65\,600$ numbers, which is about a quarter of the storage.

There are online demonstrations using colour images, for example:

`https://timbaumann.info/svd-image-compression-demo/`

To calculate the singular value decomposition we need to be able to find eigenvalues and eigenvectors of Hermitian matrices.

This is an important numerical problem in its own right. The usual numerical method for finding eigenvalues relies on another decomposition of a matrix called the Schur Factorization.

In turn, this relies on a result called Schur's Lemma that says that every matrix is unitarily similar to an upper triangular matrix.

Schur's Lemma is also one way of approaching our next topic, the Jordan Canonical Form. However, we will take a different approach.

Key points for Chapter 6

- Eigenvectors, eigenvalues, eigenspace, spectrum, p. 3.
- Diagonalisable, pp. 7, 8, 31.
- Characteristic polynomial, pp. 12,14.
- Geometric and algebraic multiplicity, p. 24.
- Normal maps and matrices, pp. 34, 36.
- Spectral Theorem (normal maps), p. 40.
- Spectral Theorem (self-adjoint maps), p. 43.
- Conic sections, pp. 47.
- Spectral Theorem (unitary maps), p. 57.
- Singular Value Decomposition, pp. 65, 67, 71.
- Pseudoinverse, p. 73.

MATH2601 – Higher Linear Algebra

Chapter 7. Canonical Forms

Question: How simple can we make the matrix of a linear map $T : V \to V$ if T is not diagonalisable?

While answering this, we will also answer the question asked in Chapter 3: is there a finite set of properties of a matrix that can determine whether or not two matrices are similar?

Note: In this chapter we always work over the field $\mathbb{C}$ so that the characteristic polynomial splits (can be written as a product of linear factors).

Hence, when working over $\mathbb{C}$, every linear map or matrix has at least one eigenvalue.

7.1 Similarity Invariants

Recall the definition of a similarity invariant from Chapter 3: it is a property of a matrix that is the same for every similar matrix.

We are looking for a complete set of similarity invariants which capture similarity exactly: two matrices are similar if and only if they agree on all similarity invariants in the complete set.

We proved in Chapter 6 that the characteristic polynomial is a similarity invariant, and thus so is the determinant, trace, all eigenvalues and their algebraic multiplicities.

We also showed in Chapter 3 that the rank and nullity are similarity invariants. It follows that the geometric multiplicities of the eigenvalues are similarity invariants.

However, this is <u>still not enough</u>! For example, the matrices

$$A = \begin{pmatrix} -1 & -1 & 1 & 0 \\ 2 & 1 & -1 & -1 \\ 2 & 0 & 0 & -2 \\ -1 & -1 & 1 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} -5 & -1 & 6 & 4 \\ 3 & 1 & -4 & -2 \\ -1 & 0 & 1 & 1 \\ -4 & -1 & 5 & 3 \end{pmatrix}$$

both have characteristic polynomial t^4, and eigenvalue 0 has geometric multiplicity 2 for both A, B, but we claim that A and B are not similar!

Exercise:

If matrices X, Y are similar then then $p(X)$ and $p(Y)$ are similar for any polynomial $p(t) = c_0 + c_1 t + \cdots + c_r t^r$, where $p(M) = c_0 I + c_1 M + \cdots + c_r M^r$ for any square matrix M.

You can check that B^2 is the zero matrix and A^2 is not. Hence A and B are not similar.

This example suggests that we can get more information about the similarity class of A by looking at $\ker(A - \lambda \mathrm{I})^k$ for $k \in \mathbb{Z}^+$.

Definition 7.1 Let $A \in M_{p,p}(\mathbb{C})$ and let λ be an eigenvalue of A. The nonzero vector $\boldsymbol{v} \in \mathbb{C}^p$ is a generalised eigenvector for eigenvalue λ if and only if

$$\boldsymbol{v} \in \ker(A - \lambda \mathrm{I})^k \text{ for some positive integer } k.$$

The generalised eigenspace of λ, denoted $GE_\lambda(A)$, is the set of all generalised eigenvectors of λ together with the zero vector.

The corresponding definitions for a linear map $T : \mathbb{C}^p \to \mathbb{C}^p$ involve $\ker(T - \lambda \, \mathrm{id})^k$ instead of $\ker(A - \lambda \mathrm{I})^k$.

Exercise: Prove that $GE_\lambda(T)$ is a subspace of $\mathbb{C}^p$.

Note: Every eigenvector of A is a generalised eigenvector of A.

Fact: The dimensions of the generalised eigenspaces for each eigenvalue are similarity invariants. To prove this, adapt the proof that nullity is a similarity invariant (see Theorem 3.24).

For the 4×4 matrices P and Q and their only eigenvalue $\lambda = 0$, we have:

k	$\text{nullity}(P^k)$	$\text{nullity}(Q^k)$
1	2	2
2	3	4
≥ 3	4	4

We will prove that the collection of all the integers $\text{nullity}(A - \lambda \mathrm{I})^k$ for all the eigenvalues λ and all integers $k \geq 1$ is a complete set of similarity invariants for A.

7.2 Generalised Eigenspaces

First we introduce some convenient notation.

Definition 7.2. Let $T \in L(V, V)$ and suppose that λ is an eigenvalue of T. For each positive integer j define the space $V_j(\lambda) = \ker(T - \lambda\,\mathrm{id})^j$.

Note: If $(T - \lambda\,\mathrm{id})^k(\mathbf{v}) = \mathbf{0}$ then $(T - \lambda\,\mathrm{id})^{k+\ell}(\mathbf{v}) = \mathbf{0}$ for any positive integers k, ℓ. Hence

$$V_1(\lambda) \leq V_2(\lambda) \leq V_3(\lambda) \leq \cdots .$$

If V is finite-dimensional then these spaces are also finite-dimensional, so their dimensions cannot increase indefinitely and at some point they reach a maximum.

Lemma 7.1 Let $T \in L(V, V)$ have eigenvalue λ.
Suppose that $V_{m+1}(\lambda) = V_m(\lambda)$ for some positive integer m.
Then $V_{m+\ell}(\lambda) = V_m(\lambda)$ for all integers $\ell \geq 1$.

Proof:

$\square$

Theorem 7.2.
Let λ be an eigenvalue of $T \in L(V, V)$, where $\dim(V) = p$ is finite. Then there is a least positive integer h, the height of λ, with $h \leq p$ such that

$$GE_\lambda(T) = \ker(T - \lambda\,\mathrm{id})^h\,.$$

Proof:

$\square$

Following directly from the proof of this theorem, we have

Corollary 7.3 Let λ be an eigenvalue of $T \in L(V, V)$, where $\dim(V) = p$ is finite. Define

$$d_k = \dim(V_k(\lambda)) = \text{nullity}(T - \lambda\,\text{id})^k\,.$$

Then

$$0 = d_0 < d_1 < d_2 < \cdots < d_{h-1} < d_h = d_{h+\ell} \leq \dim(V)\,,$$

for all positive integers ℓ.

For the rest of this chapter, we will also assume that the complex vector space V is finite-dimensional. (After all, we are thinking about similarity invariants for matrices.)

By Corollary 7.3, we can find non-trivial subspaces $Z_k(\lambda)$ of V such that

$$V_k(\lambda) = V_{k-1}(\lambda) \oplus Z_k(\lambda) \quad \text{for } k = 2, 3, \ldots, h\,.$$

We define $Z_1(\lambda) = V_1(\lambda) = \ker(T - \lambda\,\mathrm{id})$.

The $Z_k(\lambda)$ are not unique, but in the notation of Corollary 7.3,

$$\dim(Z_k) = d_k - d_{k-1}\,.$$

Once we have chosen the subspaces $Z_k(\lambda)$ we can write

$$GE_\lambda(T) = Z_1(\lambda) \oplus Z_2(\lambda) \oplus \cdots \oplus Z_h(\lambda) = \bigoplus_{k=1}^{h} Z_k(\lambda)\,,$$

where h is the height of the eigenvalue λ.

Lemma 7.4 With the previous notation, for $k > 1$,

if $\boldsymbol{v} \in Z_k(\lambda) - \{\mathbf{0}\}$ then $(T - \lambda\,\mathrm{id})(\boldsymbol{v}) \in V_{k-1}(\lambda) - V_{k-2}(\lambda)$.

(That is, applying $T - \lambda\,\mathrm{id}$ to a generalised eigenvector takes you exactly "one step down" in the $V_k(\lambda)$.)

Proof:

$\square$

Lemma 7.5. With the same notation as Corollary 7.3

$$d_1 - d_0 \geq d_2 - d_1 \geq d_3 - d_2 \geq \cdots .$$

That is,

$$\dim(Z_1(\lambda)) \geq \dim(Z_2(\lambda)) \geq \dim(Z_3(\lambda)) \geq \cdots .$$

Proof. Later! We defer the proof of this useful but technical lemma for a while.

Note: Corollary 7.3 and Lemma 7.5 tell us that if the sequence of nullities of the $V_j(\lambda)$ is non-decreasing, but their differences (the dimensions of the $Z_j(\lambda)$) must be non-increasing.

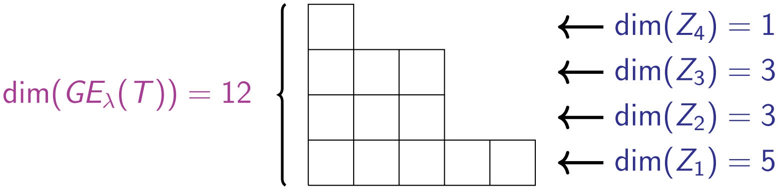
For example,

$$d_1, d_2, d_3, d_4, d_5, d_6, \cdots = 5\,,\ 8\,,\ 11\,,\ 12\,,\ 12\,,\ 12\,, \ldots$$

is a possible sequence, but these two are not possible:

$$5\,,\ 8\,,\ 7\,,\ 12\,,\ 12\,,\ 12\,,\ldots \quad \text{and} \quad 5\,,\ 8\,,\ 9\,,\ 12\,,\ 12\,,\ 12\,,\ldots$$

We can visualise a generalised eigenspace with a block diagram. For example, with the nullities $5\,,\ 8\,,\ 11\,,\ 12\,, 12\,, \ldots$ we have the diagram



The non-increasing size of the Z_j leads to the “staircase” shape of the diagram.

Now we give the deferred proof of Lemma 7.5, which we restate below.

Lemma 7.5. With the same notation as Corollary 7.3

$$d_1 - d_0 \geq d_2 - d_1 \geq d_3 - d_2 \geq \cdots .$$

That is,

$$\dim(Z_1(\lambda)) \geq \dim(Z_2(\lambda)) \geq \dim(Z_3(\lambda)) \geq \cdots .$$

Proof:

Recall the nullity sequence $5\,,\ 8\,,\ 11\,,\ 12\,,12\,,\ldots$ from page 14, with the block diagram:

$\boldsymbol{v}_4$				
$\boldsymbol{v}_3$				
$\boldsymbol{v}_2$				
$\boldsymbol{v}_1$				

By Lemma 7.4, if we choose a vector $\boldsymbol{v}_4 \in Z_4(\lambda)$, we can create a chain of (non-zero) vectors by successively applying $T - \lambda\,\mathrm{id}$:

$$\boldsymbol{v}_4 \xrightarrow{T-\lambda\,\mathrm{id}} \boldsymbol{v}_3 \xrightarrow{T-\lambda\,\mathrm{id}} \boldsymbol{v}_2 \xrightarrow{T-\lambda\,\mathrm{id}} \boldsymbol{v}_1\,,$$

where each $\boldsymbol{v}_j \in V_j(\lambda) - V_{j-1}(\lambda)$, and $\boldsymbol{v}_1$ is an eigenvector. In fact, we can assume that $\boldsymbol{v}_j \in Z_j(\lambda)$ for each j.

These vectors $v_1, \ldots, v_4$ are all generalised eigenvectors, and they have this nice property:

$$T(\boldsymbol{v}_1) = \lambda \boldsymbol{v}_1, \qquad T(\boldsymbol{v}_j) = \lambda \boldsymbol{v}_j + \boldsymbol{v}_{j-1} \quad \text{for} \quad j = 2, 3, 4.$$

Definition 7.3 A ordered set of non-zero vectors $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots \boldsymbol{v}_k\}$ such that

$$T(\boldsymbol{v}_1) = \lambda \boldsymbol{v}_1 \;\text{ and }\; T(\boldsymbol{v}_i) = \lambda \boldsymbol{v}_i + \boldsymbol{v}_{i-1} \quad \text{for} \;\; i = 2, \ldots, k$$

is called a Jordan chain (of length k) for the eigenvalue λ.

We call a Jordan chain maximal if it is not part of a longer Jordan chain.

Note: that there can be more than one maximal Jordan chain for a given eigenvalue.

Since $GE_\lambda(T)$ is the direct sum of the $Z_j(\lambda)$, every Jordan chain is a linearly independent set.

Let $\mathcal{B} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\}$ be a Jordan chain. The matrix of T restricted to span($\mathcal{B}$) with respect to $\mathcal{B}$ has the simple form

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & \cdots & 0 & 0 \\ 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & \cdots & 0 & \lambda \end{pmatrix}.$$

with λ on the diagonal, 1 on the super-diagonal and zero everywhere else.

Definition 7.4. The matrix $J_k(\lambda)$ above is called a Jordan block (for eigenvalue λ).

Back to the block diagram from p.19:

$\boldsymbol{v}_4$				
$\boldsymbol{v}_3$	$\boldsymbol{v}_7$			
$\boldsymbol{v}_2$	?			
$\boldsymbol{v}_1$	?			

Can we choose some more Jordan chains? Maybe five chains (one for each column), of lengths 4, 3, 3, 1 and 1, each ending in an eigenvector.

Suppose we have chosen our first chain of length 4 $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_4\}$, and now we choose a vector $\boldsymbol{v}_7 \in Z_3(\lambda)$, independent of $\boldsymbol{v}_3$.

If we use Lemma 7.4 to form a new Jordan chain of length 3 starting from $\boldsymbol{v}_7$, will it be (linearly) independent of the previous chain?

Is there a basis for $GE_\lambda(T)$ consisting of independent Jordan chains? The corresponding matrix of T restricted to $GE_\lambda(T)$ would be a direct sum of Jordan blocks for eigenvalue λ.

Next, if V can be written as the direct sum of the generalised eigenspaces then there would be a basis of V in which the matrix of T would be a direct sum of Jordan blocks for the various eigenvalues.

Definition 7.5 A matrix consisting of a direct sum of Jordan blocks is called a Jordan matrix.

A basis with respect to which the matrix of $T \in L(V, V)$ is a Jordan matrix is called a Jordan basis of T.

Aim: We want to prove that every linear map on a finite-dimensional complex vector space has a Jordan basis. In the matrix setting, we want to prove that every complex matrix is similar to a Jordan matrix.

Example 7.1 Consider the matrix $A = \begin{pmatrix} 2 & -9 \\ 1 & 8 \end{pmatrix}$.

Lemma 7.6. Let $U = \text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\}$ be the space spanned by a Jordan chain of length k for eigenvalue λ.

Then U is invariant under T. That is, $T(U) \subseteq U$.

Furthermore, if $U = X \oplus Y$ for some subspaces X and Y which are both invariant under T then either X or Y is trivial.

(We say that U is indecomposable under T.)

Proof:

$\square$

Example 7.2 When is $\mathbb{R}^2$ decomposable with respect to a linear transformation T on $\mathbb{R}^2$?

$\diamond$

7.3 Decomposition of a Space

Proposition 7.7 Let S and T be linear maps on a vector space V, and suppose that $S \circ T = T \circ S$. Then the kernel and image of S are invariant under T.

Proof:

$\square$

The proofs of the next two results are left as an exercise.

Proposition 7.8 Let $T : V \to V$ be a linear map and let $U \leq V$ be an invariant subspace.

If U is indecomposable with respect to T then it is indecomposable with respect to the restriction of T to U, which we denote by $T|_U$.

Proposition 7.9 Let $T : V \to V$ be a linear map and let λ be a scalar. Define $S = T - \lambda\,\mathrm{id}$. Then $U \leq V$ is:

(a) invariant under S if and only if it is invariant under T;

(b) decomposable with respect to S if and only if it is decomposable with respect to T.

Lemma 7.10 (The Decomposition Lemma).
Let T be a linear map on a finite-dimensional nontrivial vector space V. Then for some positive integer m there exist T-invariant subspaces $U_1, \ldots, U_m$, each indecomposable with respect to T, such that

$$V = U_1 \oplus \cdots \oplus U_m \,.$$

Proof:

$\square$

We call the subspaces in this lemma splitting spaces for T.

We proved in Chapter 3 (Theorem 3.19) that if $V = X \oplus Y$ with both X and Y invariant subspaces of T, there is a basis of T in which the matrix of T is $A \oplus B$.

We can extend this to say that if

$$V = U_1 \oplus \cdots \oplus U_m$$

for invariant subspaces U_i, then there a basis of V in which the matrix of T is

$$A_1 \oplus \cdots \oplus A_m \,,$$

where A_i is the matrix of T restricted to U_i in a suitable basis of U_i.

We want to show that each splitting space (Lemma 7.10) is the kernel of some power of $T - \lambda\,\mathrm{id}$ for some scalar λ, and is spanned by a Jordan chain.

Since T commutes with T^m for any positve integer m, Proposition 7.7 implies that $\ker(T^m)$ and $\mathrm{im}(T^m)$ are invariant under T.

Lemma 7.11 (Kernel–Image Decomposition) If T is a linear map on a finite-dimensional space V then there exists a positive integer m such that

$$V = (\ker T^m) \oplus (\mathrm{im}\, T^m) \, .$$

Proof:

$\square$

Note: We could restate this lemma for matrices rather than linear maps.

Example 7.3. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & 0 & -1 \end{pmatrix}. \quad \text{Then} \quad A^2 = \begin{pmatrix} 0 & 6 & 6 \\ 0 & 6 & 6 \\ 0 & -2 & -2 \end{pmatrix}$$

and $A^3 = 2A^2$.

Lemma 7.10 says that we can study a map $T : V \to V$ by looking at how it behaves on each of the splitting spaces, which are indecomposable and invariant.

We can recover the action of T on the whole space V, since every vector in V is a unique sum of one vector from each subspace.

Suppose that $T = U \oplus W$ where both U and W are invariant under T and U is indecomposable with respect to T. The restriction $T\big|_U$ of T to U is a well-defined linear map on U.

Since the field is $\mathbb{C}$, the characteristic polynomial of $T\big|_U$ has a root, so $T\big|_U$ has at least one eigenvector. Suppose that

$$T(\boldsymbol{e}) = \lambda \boldsymbol{e} \qquad \text{for some } \lambda \in \mathbb{C}\,, \quad \boldsymbol{e} \in U\,, \quad \boldsymbol{e} \neq \mathbf{0}\,.$$

Consider the linear transformation $S : U \to U$ defined by

$$S(\boldsymbol{u}) = T\big|_U(\boldsymbol{u}) - \lambda \boldsymbol{u}\,.$$

By Lemma 7.11 there is a positive integer m with

$$U = (\ker S^m) \oplus (\operatorname{im} S^m) \tag{1}$$

However $\ker S^m$ and $\operatorname{im} S^m$ are invariant under S, and U is indecomposable with respect to S by Proposition 7.9.

So equation (1) implies that either $\ker S^m$ or $\operatorname{im} S^m$ is the trivial subspace. But $\ker S^m$ contains the nonzero vector $\boldsymbol{e}$, so

$$\operatorname{im} S^m = \{\mathbf{0}\} \quad \text{and} \quad \ker S^m = U\,.$$

That is, $S^m : U \to U$ is the zero map.

Let k be the smallest positive integer such that S^k is the zero map on U. Choose $\boldsymbol{u}$ such that $S^{k-1}(\boldsymbol{u}) \neq \mathbf{0}$, and write

$$\mathcal{B} = \{\boldsymbol{u}, S(\boldsymbol{u}), S^2(\boldsymbol{u}), \ldots, S^{k-1}(\boldsymbol{u})\}.$$

We claim that $\mathcal{B}$ is a basis for U, and thus $\dim(U) = k$.

Hence $\mathcal{B}$ is linearly independent.

Next, we need to prove that $\mathcal{B}$ spans U. This part of the proof is very technical and we defer it until later.

For now, suppose we have proved this. Then any indecomposable, T-invariant subspace of V has a basis of the form

$$\mathcal{B} = \{\boldsymbol{u}, S(\boldsymbol{u}), S^2(\boldsymbol{u}), \dots, S^{k-1}(\boldsymbol{u})\},$$

where $S = T - \lambda\,\mathrm{id}$ for some scalar λ and $S^k(\boldsymbol{u}) = \mathbf{0}$.

Claim: the basis $\mathcal{B}$ is a maximal Jordan chain.

We have proved that each splitting space U_i for T has a basis $\mathcal{B}$ consisting of a maximal Jordan chain.

The choice of $\mathcal{B}$ is not unique, but $|\mathcal{B}|$ (the length of the Jordan chain) equals the dimension of U_i.

We are now ready to prove the one of the main results of this chapter.

Theorem 7.12 (The Jordan Form)
Let T be a linear map on a nontrivial, finite-dimensional complex vector space V. Then there is a basis of V with respect to which the matrix of T is a Jordan matrix.

Proof:

$\square$

Definition 7.6 The matrix of T with respect to a Jordan basis (see Definition 7.5) is called a Jordan (canonical) form of T.

The matrix version of Theorem 7.12 says that every complex matrix is similar to a Jordan matrix.

We have not proved any uniqueness results about the Jordan Form yet. First we introduce one more set of similarity invariants.

Theorem 7.13 The number and lengths of the maximal Jordan chains for each eigenvalue are similarity invariants.

Proof:

$\square$

How does Theorem 7.12 relate to the block diagrams as discussed on page 22?

Can we build a Jordan basis by choosing linearly independent Jordan chains, one for each column of the block diagram?

Lemma 7.14. Let $T : V \to V$ be linear where V is a nontrivial finite-dimensional complex vector space.

The number of splitting spaces in the Decomposition Lemma (Lemma 7.10) is the same as the maximum number of linearly independent eigenvectors of T, which is the same as the number of maximal Jordan chains in any Jordan basis.

Proof:

$\square$

We emphasise that when we talk about "the" Jordan form of a matrix or map, we ignore the order of the Jordan blocks. All that matters is the size of each block and the eigenvalue that it belongs to.

Let us apply this to the example on page 22. We have already chosen the first Jordan chain ending in eigenvector $\boldsymbol{v}_1$.

$\boldsymbol{v}_4$				
$\boldsymbol{v}_3$	$\boldsymbol{v}_7$			
$\boldsymbol{v}_2$	$\boldsymbol{v}_6$			
$\boldsymbol{v}_1$	$\boldsymbol{v}_5$			

Let $\boldsymbol{v}_5$ be an eigenvector which is linearly independent from $\boldsymbol{v}_1$. Then $\boldsymbol{v}_5$ belongs to a splitting space U, and there is a Jordan chain $\boldsymbol{v}_5, \boldsymbol{v}_6, \boldsymbol{v}_7$ that spans U. The chain starts with $\boldsymbol{v}_7 \in U \cap Z_3(\lambda)$, which is 1-dimensional.

We can continue until we have identified $\dim E_\lambda(T)$ maximal Jordan chains for $GE_\lambda(T)$.Their union over all eigenvalues λ gives a Jordan basis, which determines the Jordan form up to reorderings of the Jordan blocks.

Theorem 7.15 The Jordan form is a complete similarity invariant for complex matrices.

Proof:

□

Next we give the deferred technical proofs.

The Technical Proofs: non-examinable

Recall our definitions and results:

- For $T : V \to V$, the subspace $U \leq V$ is indecomposable and T-invariant;
- Hence $T\big|_U : U \to U$ is well-defined and has an eigenvalue λ;
- $S : U \to U$ is defined as $S = T\big|_U - \lambda\,\mathrm{id}$;
- k is the smallest positive integer such that S^k is the zero map on U;
- $\boldsymbol{u} \in U$ is some vector with $S^{k-1}(\boldsymbol{u}) \neq \mathbf{0}$;
- $\mathcal{B} = \{\boldsymbol{u}, S(\boldsymbol{u}), S^2(\boldsymbol{u}), \ldots, S^{k-1}(\boldsymbol{u})\}$ is linearly independent.
- $W = \mathrm{span}(\mathcal{B})$.

We wish to prove that $U = W$.

Lemma 7.16.
With the given notation, if X is a subspace of U such that

$$S(X) \subseteq X \ , \quad W \cap X = \{\mathbf{0}\} \ , \quad W \oplus X \neq U \ ,$$

then there exists $\boldsymbol{v} \in U$ such that

$$\boldsymbol{v} \notin W \oplus X \quad \text{and} \quad S(\boldsymbol{v}) \in X \ .$$

Proof:

Lemma 7.17. With the above notation, $W = U$.

Proof:

7.4 More on the Jordan Form

Some **facts** that follow from our results on the Jordan form of $T \in L(V, V)$.

(a) The eigenvalues of T are the diagonal entries λ_i in its Jordan form, and conversely.

(b) Since each splitting space for T admits exactly one independent eigenvector, the number of Jordan blocks for each eigenvalue λ_i is the geometric multiplicity of λ_i.

(c) Suppose the sum of the sizes of the Jordan blocks for eigenvalue λ_i is a_i.
Calculating the characteristic polynomial of the Jordan form gives $\mathrm{cp}_T(t) = \prod_{i=1}^{m}(t - \lambda_i)^{a_i}$. That is, the algebraic multiplicity of an eigenvalue is the sum of the sizes of all the Jordan blocks for that eigenvalue.

(d) The height h_i of eigenvalue λ_i is the size of the largest Jordan block for λ_i, and $(T - \lambda_i \operatorname{id})^{h_i}$ is the zero map on each splitting space for λ_i. Note: $h_i \leq a_i$.

(e) It follows from (d) that each of the splitting spaces for T consists of generalised eigenvectors (and the zero vector).

(f) Using the notation of (c) and (d), $\ker(T - \lambda_i \operatorname{id})^k$ is the generalised eigenspace of λ_i for any $k \geq h_i$.

(g) Lemma 7.14 implies that the Jordan chains are linearly independent, so $GE_\lambda(T)$ is spanned by the union of the Jordan chains for eigenvalue λ, and $\dim GE_\lambda(T)$ is the algebraic multiplicity of λ.

(h) V is the direct sum of all the generalised eigenspaces for $T \in L(V, V)$.

We can make equivalent statements about matrices.

Matrix Polynomials

Given any polynomial $f \in \mathcal{P}(\mathbb{F})$, we can substitute any square matrix A to get $f(A)$, using $A^0 = \mathrm{I}$ for the constant coefficient.

Theorem 7.18 (The Cayley-Hamilton Theorem)
For any matrix $A \in M_{p,p}(\mathbb{C})$, the matrix $\mathrm{cp}_A(A)$ is zero.

Proof:

$\square$

The Cayley–Hamilton Theorem implies that the space spanned by

$$\{\mathrm{I}, A, A^2, A^3, \dots\}$$

is at most p-dimensional, as every power of A greater than or equal to p is a linear combination of lower powers.

If A is invertible then we can extend this to negative powers.

Example 7.4. The characteristic polynomial of $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ is $\mathrm{cp}_A(t) = \det(t\mathrm{I} - A) = t^2 - 5t - 2$. We can calculate:

$\diamond$

By result (d) on page 58, $(A - \lambda_i \mathrm{I})^{h_i} \boldsymbol{v} = \mathbf{0}$ where h_i is the size of the largest Jordan block for eigenvalue λ_i and $\boldsymbol{v}$ is a generalised eigenvector for λ_i.

Our proof of the Cayley-Hamilton Theorem implies that
$$\prod_{i=1}^{m} (A - \lambda_i \mathrm{I})^{h_i}$$
is the zero matrix.

Definition 7.7 The polynomial $\mathrm{mp}_A(t) = \prod_{i=1}^{m} (t - \lambda_i)^{h_i}$ is called the minimal polynomial of A.

Fact: If $f(t)$ is a monic polynomial with smaller degree than $\mathrm{mp}_A(t)$ then $f(A)$ cannot be zero.

7.5 Calculating the Jordan Form

Given an $n \times n$ complex matrix A, how can we find an invertible matrix P and Jordan matrix J such that $A = PJP^{-1}$?

We need to calculate:

- the eigenvalues of A, and
- for each eigenvalue λ_i, the spaces $V_j(\lambda_i) = \ker(A - \lambda_i \mathrm{I})^j$ for $j = 1, \ldots, h_i$, where h_i is the height of the eigenvalue λ_i.

Then we can find the Jordan chains for each eigenvalue by choosing suitable vectors in the spaces Z_j, where $V_j = V_{j-1} \oplus Z_j$.

The matrix version of Theorem 7.2 implies that we only have to calculate $\ker(A - \lambda \mathrm{I})^j$ until there is no change in dimension.

Recall that $d_k = \text{nullity}(A - \lambda\mathrm{I})^k$ for each eigenvalue λ.

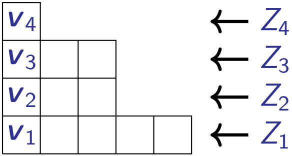
Corollary 7.3 and Lemma 7.5 tell us precisely how many Jordan chains we have for each eigenvalue.

- $d_1 - d_0 = d_1 = \text{nullity}(A - \lambda\mathrm{I})$ is the number of Jordan blocks for eigenvalue λ.
- The value of $d_2 - d_1$ equals the number of chains of length at least 2.
- Similarly, for all $k \geq 2$, there are $d_k - d_{k-1}$ chains of length at least k.

Recall (the matrix version of) the example from page 14. For an eigenvalue λ of A, the dimensions $d_k = \text{nullity}(A - \lambda \mathrm{I})^k$ are

$$5\,,\ 8\,,\ 11\,,\ 12\,,\ 12\,,\ 12\,, \ldots$$

and we have the block diagram:



Recall we used Lemma 7.4 to choose a non-zero vector $\boldsymbol{v}_4 \in Z_4$ and create a chain of (non-zero) vectors by successively applying $A - \lambda \mathrm{I}$:

$$\boldsymbol{v}_4 \xrightarrow{A-\lambda\mathrm{I}} \boldsymbol{v}_3 \xrightarrow{A-\lambda\mathrm{I}} \boldsymbol{v}_2 \xrightarrow{A-\lambda\mathrm{I}} \boldsymbol{v}_1\,,$$

where each $\boldsymbol{v}_j \in Z_j$, so $\boldsymbol{v}_1$ is an eigenvector.

Our theory confirms that if we choose $\boldsymbol{v}_7 \in Z_3$ which is linearly independent of $\boldsymbol{v}_3$, then the chain

$$\boldsymbol{v}_7 \xrightarrow{A-\lambda \mathrm{I}} \boldsymbol{v}_6 \xrightarrow{A-\lambda \mathrm{I}} \boldsymbol{v}_5$$

belongs to a distinct splitting space, and so is linearly independent of the previous chain.

Hence, in the block diagram, the horizontal layers correspond to the spaces Z_j and the columns correspond to the Jordan chains.

In this example, we have one chain of length 4, two of length 3 and two of length 1. The Jordan form of A is

$$J_4(\lambda) \oplus J_3(\lambda) \oplus J_3(\lambda) \oplus J_1(\lambda) \oplus J_1(\lambda).$$

We sometimes just write (λ) instead of $J_1(\lambda)$.

We state one more useful result and then look at some examples.

Lemma 7.19. If the matrix $A \in M_{p,p}(\mathbb{C})$ has only one eigenvalue λ then $GE_\lambda(A) = \mathbb{C}^p$.

Furthermore, if A has this property and is diagonalisable then $A = \lambda \mathrm{I}$.

Proof:

$\square$

Example 7.5.
Suppose that A is a 17×17 complex matrix with characteristic polynomial $\mathrm{cp}_A(t) = (t-13)^{17}$. Further, with $d_k = \dim(V_k) = \mathrm{nullity}(A - 13\mathrm{I})^k$, suppose that

$$\begin{aligned} d_1 &= 4, \quad d_2 = 7, \quad d_3 = 10, \quad d_4 = 12, \\ d_5 &= 14, \quad d_6 = 16 \quad d_7 = 17. \end{aligned}$$

What is the Jordan form and minimal polynomial of A?

Solution:

$\diamond$

Note that $17 = 4 + 3 + 3 + 2 + 2 + 2 + 1$ and $17 = 7 + 6 + 3 + 1$ are called conjugate partitions of 17.

Example 7.6 Suppose that

$$A - 7\mathrm{I} = \begin{pmatrix} -1 & -1 & 1 & 0 \\ 2 & 1 & -1 & -1 \\ 2 & 0 & 0 & -2 \\ -1 & -1 & 1 & 0 \end{pmatrix}.$$

Find the Jordan form for A and calculate the minimal polynomial of A. Find a Jordan basis for A.

Solution: You can check that $\mathrm{cp}_A(t) = (t-7)^4$, so $GE_7(A) = \mathbb{C}^4$. (Compare with the example on page 4.)

We calculate

$$(A - 7\mathrm{I})^2 = \begin{pmatrix} 1 & 0 & 0 & -1 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}.$$

Example 7.7 Suppose that A is a 15×15 matrix with only one eigenvalue 2, and that the nullities of $(A - 2\mathrm{I})^k$ are

$$4\,,\ 8\,,\ 11$$

for $k = 1, 2, 3$ respectively. Find all possible completions of the sequence of nullities, and hence all possible Jordan forms of A and the respective minimal polynomials.

Solution:

In cases where there are multiple eigenvalues, we treat each generalised eigenspace in turn, and concatenate the Jordan chains and Jordan blocks as appropriate.

The Jordan blocks in the Jordan form J can be arranged in any order, but they must match the order of the Jordan chains in the corresponding basis (that is, the columns of the change-of-basis matrix).

The order of the vectors within each chain is fixed by the definition of the chain.

Example 7.8 Find a Jordan form and a Jordan basis for

$$A = \begin{pmatrix} 7 & -1 & -1 \\ -2 & 7 & 2 \\ 5 & -3 & 1 \end{pmatrix},$$

given that A has eigenvectors $\boldsymbol{v} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $\boldsymbol{w} = \begin{pmatrix} -1 \\ 3 \\ -7 \end{pmatrix}$.

Solution:

◇

Key points for Chapter 7

- Generalised eigenvector, generalised eigenspace, p. 5.
- Jordan chain, p. 20.
- Jordan block, p. 21.
- Jordan matrix and Jordan basis, p. 23.
- Indecomposible subspace, p. 26.
- The Decomposition Lemma and splitting spaces p. 30.
- Jordan form, p. 42.
- Cayley-Hamilton Theorem, p. 59.
- Minimal polynomial, p. 62.

MATH2601 – Higher Linear Algebra

Chapter 8. Matrix Functions and ODEs

8.1 Background

You have probably studied linear differential equations, for example

$$\frac{dy}{dt} = a\,y(t), \quad a \in \mathbb{R}\,. \tag{1}$$

The solution to this homogeneous differential equation is $y(t) = Ce^{at}$ for some constant C.

We can solve the inhomogeneous case

$$\frac{dy}{dt} = ay(t) + f(t), \quad a \in \mathbb{R} \tag{2}$$

by the method of variation of parameters.

Let C in the solution to (1) be a function and substitute:

$$\frac{d}{dt}\left(C(t)\,e^{at}\right) = C'e^{at} + Cae^{at} = ay + C'e^{at}\;.$$

We have a solution to (2) if and only if $C'(t) = e^{-at}f(t)$.

You may have also studied systems of differential equations, for example

$$\begin{aligned} y_1' &= -3y_1 + 4y_2, \qquad & y_1(0) &= 4, \\ y_2' &= 8y_1 + y_2, \qquad & y_2(0) &= -1. \end{aligned}$$

We can write this in matrix form as $\boldsymbol{y}' = A\boldsymbol{y}$ where

$$A = \begin{pmatrix} -3 & 4 \\ 8 & 1 \end{pmatrix} \quad \text{and} \quad \boldsymbol{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

Note that the matrix A is constant, which means it is not a function of t.

A system like this can be solved using eigenvectors if the matrix A is diagonalisable. What can we do in general?

It would be great if we could write the solution of $\boldsymbol{y}' = A\boldsymbol{y}$ as

$$\boldsymbol{y}(t) = e^{At}\,\boldsymbol{y}_0$$

where $\boldsymbol{y}_0 = \boldsymbol{y}(0)$.

Maybe then we could solve inhomogeneous systems like

$$\boldsymbol{y}' = A\boldsymbol{y} + \boldsymbol{b}(t)$$

for some vector-valued function $\boldsymbol{b}(t)$, by setting

$$\boldsymbol{y} = e^{At}\,\boldsymbol{z}(t), \quad \text{where} \quad \boldsymbol{z}' = e^{-At}\,\boldsymbol{b}\,.$$

We can try to define e^{tA} for a matrix A using power series;

$$e^{tA} = \mathrm{I} + tA + \frac{1}{2}t^2A^2 + \frac{1}{3!}t^3A^3 + \cdots$$

Can we make this work?

8.2 Powers and the Jordan Form

Before we look at power series we will look at polynomials, and in particular, the powers of a matrix.

From the Practice Problems (Q91), we know that if A and B are similar then so are their powers. In fact, if $A = PBP^{-1}$ then $A^n = PB^nP^{-1}$.

Exercise: Check that if $M = A \oplus B$ then $M^n = A^n \oplus B^n$.

This means that we can find powers of a matrix if we can find powers of Jordan blocks.

Note: We can give an easy proof of the Cayley–Hamilton Theorem for diagonalisable matrices:

A Jordan block can be written as

$$J_k(\lambda) = \lambda \mathrm{I} + N \quad \text{with} \quad N = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Note $N = J_k(0)$. Powers of N are easy to calculate:

We say that the matrix N is nilpotent.

Theorem 8.1
(Binomial Theorem for Commuting Matrices)

Let A and B be $p \times p$ matrices such that $AB = BA$. Then for any integer $n \geq 0$ we have

$$(A+B)^n = A^n + \binom{n}{1} A^{n-1} B + \binom{n}{2} A^{n-2} B^2 + \cdots + B^n \;. \quad (3)$$

Proof:

$\square$

Also note that if $A\boldsymbol{v} = \lambda \boldsymbol{v}$ then $A^n \boldsymbol{v} = \lambda^n v$.

Example 8.1 Since multiples of the identity matrix commute with every matrix, we have

$$\begin{aligned} J_4(\lambda)^n &= (\lambda \mathrm{I} + N)^n \\ &= \lambda^n \mathrm{I} + \binom{n}{1}\lambda^{n-1}N + \binom{n}{2}\lambda^{n-2}N^2 + \binom{n}{3}\lambda^{n-3}N^3 \,. \end{aligned}$$

The series stops at this point since N^4 is zero.

Hence

$$J_4(\lambda)^n = \begin{pmatrix} \lambda^n & \binom{n}{1}\lambda^{n-1} & \binom{n}{2}\lambda^{n-2} & \binom{n}{3}\lambda^{n-3} \\ 0 & \lambda^n & \binom{n}{1}\lambda^{n-1} & \binom{n}{2}\lambda^{n-2} \\ 0 & 0 & \lambda^n & \binom{n}{1}\lambda^{n-1} \\ 0 & 0 & 0 & \lambda^n \end{pmatrix} \,.$$

Make sure you understand the pattern for blocks of other sizes.

Example 8.2

Consider the Jordan matrix $J = J_2(7) \oplus J_1(7) \oplus J_1(3)$.
Find a formula for J^n for $n \geq 2$

Solution.

$\diamond$

Example 8.3 Find a formula for A^n if $A = \begin{pmatrix} 2 & -9 \\ 1 & 8 \end{pmatrix}$.

Solution.

$\diamond$

Note that putting $n = 0$ gives the identity matrix and putting $n = 1$ gives A.

These form a useful check of the formula, but do not prove it.

Example 8.4 Solve the coupled recurrence system

$$a_{n+1} = 2a_n - 9b_n, \quad b_{n+1} = a_n + 8b_n, \quad n \geq 0, \quad a_0 = b_0 = 1\,.$$

Solution:

$\diamond$

8.3 Power Series of Matrices

If we want to consider power series of matrices $\sum_{i=0}^{\infty} a_i A^i$ then we need to address questions of convergence.

Definition 8.1
Let $A^{(k)} = \left(a_{ij}^{(k)}\right)$ for $k = 1, 2, 3, \ldots$ be a sequence of $p \times q$ matrices. We say that $A(k)$ converges entrywise to A as $k \to \infty$ if and only if $a_{ij}^{(k)} \to a_{ij}$ as $k \to \infty$ for all i, j.

There are other ways to define convergence which sometimes make calculations easier.

First we define a norm (measure of "length") for matrices, and then use the norm to define convergence.

There are several different norms on matrices.

Definition 8.2 Let $A \in M_{p,p}(\mathbb{C})$.

The ∞-norm of A is defined by

$$\|A\|_\infty = \max\{|a_{ij}| \,:\, 1 \le i, j \le p\}.$$

The operator norm of A is defined by

$$\|A\|_{\mathrm{op}} = \max\{\|A\boldsymbol{v}\| \,:\, \boldsymbol{v} \in \mathbb{C}^p \text{ and } \|\boldsymbol{v}\| = 1\}\ ,$$

where the norm appearing on the right is the standard norm in $\mathbb{C}^p$.

The Frobenius norm of A is defined as $\|A\|_F = \sqrt{\mathrm{tr}(A^*A)}$, which is the same as $\sqrt{\sum_{i,j} |a_{ij}|^2}$.

These norms are equivalent in the sense that convergence in one norm implies convergence in the other. We see this from the lemmas below: the proofs are omitted.

Lemma 8.2 Suppose that $A \in M_{p,p}(\mathbb{C})$ has non-zero singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$.

Then $\|A\|_{\mathrm{op}} = \sigma_1$ and $\|A\|_F = \sqrt{\sigma_1^2 + \cdots \sigma_k^2}$.

Hence for all such A,

$$\|A\|_{\mathrm{op}} \leq \|A\|_F \leq \sqrt{p}\,\|A\|_{\mathrm{op}}\,.$$

Lemma 8.3 For any matrix $A \in M_{p,p}(\mathbb{C})$ we have

$$\|A\|_F \geq \|A\|_\infty \geq \frac{1}{p}\|A\|_F\,.$$

Definition 8.3
Let N be a norm on $p \times p$ matrices and let $A^{(k)}$ be a sequence of $p \times p$ matrices. We say that $A^{(k)}$ converges to A in the norm N if and only if $N(A^{(k)} - A) \to 0$ as $k \to \infty$.

Note: Convergence in the norm means the distance between the matrices (as measured by that norm) tends to zero.

Lemma 8.4 A sequence of $p \times p$ matrices $A^{(k)}$ converges to A in the ∞-norm if and only if it converges to A in the Frobenius norm if and only if it converges to A in the operator norm.

Proof:

$\square$

Lemma 8.5. A sequence of matrices $A^{(k)}$ tends to A entrywise if and only if it converges to A in the ∞-norm.

Proof:

$\square$

Hence by Lemma 8.4, entrywise convergence is equivalent to convergence in either the operator norm or Frobenius norm.

We can choose whichever definition of convergence is most convenient for any particular problem.

We can define the convergence of a series of matrices now that we have the idea of convergence for sequences of matrices.

We will also need to check that convergence of sequences is preserved under similarity. We will need the following result.

Lemma 8.6 Let A and B be $p \times p$ complex matrices. Then

$$\|AB\|_\infty \leq p \|A\|_\infty \|B\|_\infty .$$

Proof:

$\square$

Theorem 8.7
Suppose that the sequence $A^{(k)} \in M_{p,p}(\mathbb{C})$ converges to A.
Let P be a $p \times p$ invertible matrix.
Then the sequence $B^{(k)} = P^{-1}A^{(k)}P$ converges to
$B = P^{-1}AP$.

Proof:

$\square$

Now we can use our earlier work on powers and Jordan forms to calculate power series of matrices.

Theorem 8.8

Suppose that $f : \mathbb{C} \to \mathbb{C}$ has a power series expansion
$f(t) = \sum_{k=0}^{\infty} a_k \, t^k$ with a radius of convergence R.

(Here R may equal ∞ if f converges everywhere.)

Given $A \in M_{p,p}(\mathbb{C})$, if $p||A||_\infty < R$ then $\sum_{k=0}^{\infty} a_k A^k$ converges.

The limit is called $f(A)$.

Proof:

$\square$

Lemma 8.9

Suppose $f : \mathbb{C} \to \mathbb{C}$ has a power series expansion.

(a) If $f(A)$ is defined and $B = PAP^{-1}$ then $f(B)$ is defined and $f(B) = P\,f(A)P^{-1}$.

(b) If $M = A \oplus B$ and $f(A)$ and $f(B)$ are defined, then $f(M)$ is defined and $f(M) = f(A) \oplus f(B)$.

(c) If $f(A)$ is defined and $A\boldsymbol{v} = \lambda\boldsymbol{v}$ then $f(\lambda)$ is defined and $f(A)\boldsymbol{v} = f(\lambda)\boldsymbol{v}$.

Proof:

$\square$

8.4 The Matrix Exponential

We will use the power series expansion $\exp(t) = \sum_k \frac{t^k}{k!}$ which converges for all $t \in \mathbb{C}$.

Definition 8.4. Let $A \in M_n(\mathbb{C})$. Then

$$e^A = \exp(A) = \mathrm{I} + A + \tfrac{1}{2!}A^2 + \cdots = \sum_{k=0}^{\infty} \tfrac{1}{k!}A^k.$$

Since the exponential converges everywhere, Theorem 8.8 implies that $\exp(A)$ is defined for all matrices A. In particular, $\exp(0)$ equals the identity matrix, where 0 is the zero matrix.

Lemma 8.9 implies that we can find $\exp(A)$ if we can calculate $\exp(J)$, where J is a Jordan block.

Suppose that $f : \mathbb{C} \to \mathbb{C}$ is defined by a power series. Then for a fixed matrix $A \in M_{p,p}(\mathbb{C})$ we can define $F : \mathbb{C} \to M_{p,p}(\mathbb{C})$ by $F(t) = f(tA)$ for all values of t such that $f(tA)$ converges.

We say that F is continuous (or differentiable) if the component functions $F(t)_{ij} = f(tA)_{ij}$ are all continuous (or differentiable), respectively.

We say that $G : \mathbb{C} \to M_{p,p}(\mathbb{C})$ is the derivative of $F(t)$ if and only if

$$\lim_{h \to 0} \frac{F(t+h) - F(t) - hG(t)}{h}$$

is the zero matrix. We write the derivative of F as $\frac{dF}{dt}$ or $F'(t)$.

Fact: The derivative is the matrix-valued function whose components are the derivatives of the component functions.

Example 8.5. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $f(t) = t^2$. Then

$$F(t) = f(tA) = t^2 A^2 = \begin{pmatrix} 2t^2 & 0 \\ 0 & 2t^2 \end{pmatrix}.$$

We would expect that $F'(t) = 4t\mathrm{I}$.

$\diamond$

Theorem 8.10 For any $A \in M_{p,p}(\mathbb{C})$ the function $\exp(tA)$ is differentiable and

$$\frac{d}{dt}\exp(tA) = A\exp(tA) = \exp(tA)A\,.$$

Proof:

$\square$

Corollary 8.11 Let $A \in M_{p,p}(\mathbb{C})$ and $\boldsymbol{c} \in \mathbb{C}^p$. Then

$$\boldsymbol{y}(t) = e^{tA}\,\boldsymbol{c}$$

is a solution of the initial value problem

$$\boldsymbol{y}' = A\boldsymbol{y}, \quad \boldsymbol{y}(0) = \boldsymbol{c}\,.$$

Proof. Exercise! $\square$

The next result will imply that the solution to the initial value problem in Corollary 8.11 is unique.

Theorem 8.12. For each $t \in \mathbb{R}$ and any square matrix A, the matrix e^{tA} is invertible and has inverse e^{-tA}.

Proof:

$\square$

Theorem 8.13 Let $A \in M_{p,p}(\mathbb{C})$. Then

(a) If $\boldsymbol{y}' = A\boldsymbol{y}$ and $\boldsymbol{y}(0) = \mathbf{0}$ then $\boldsymbol{y}(t) = \mathbf{0}$ for all t.

(b) The set of solutions of $\boldsymbol{y}' = A\boldsymbol{y}$ is a vector space of dimension p, and the columns of e^{tA} form a basis for this space.

(c) The initial value problem $\boldsymbol{y}' = A\boldsymbol{y}$, $\boldsymbol{y}(0) = \boldsymbol{c}$ has a unique solution $\boldsymbol{y}(t) = e^{tA}\boldsymbol{c}$.

Proof:

Theorem 8.14 If A, B in $M_{p,p}(\mathbb{C})$ commute (that is, if $AB = BA$) then

$$\exp(A + B) = \exp(A) \cdot \exp(B)\,.$$

Proof:

$\square$

We now have the results we need to find $\exp(tJ)$ for a Jordan block J.

Firstly, for an $r \times r$ block we can write

$$J_r(\lambda) = \lambda \mathrm{I}_r + N_r \,,$$

which is a sum of two commuting matrices. Then by Theorem 8.14,

$$\exp(tJ_r(\lambda)) = \exp(\lambda t \mathrm{I}_r)\, \exp(tN_r)\,.$$

But $(\lambda t\, \mathrm{I}_r)^k = (\lambda t)^k \mathrm{I}_r$, and so

$$\exp(\lambda t \mathrm{I}_r) = e^{t\lambda}\, \mathrm{I}_r\,.$$

On the other hand, since N_r^r is zero,

$$\exp(tN_r) = \mathrm{I} + tN_r + \tfrac{1}{2!} t^2 N_r^2 + \cdots + \tfrac{1}{(r-1)!} t^{r-1} N_r^{r-1}\,.$$

We investigated powers of N_r in Section 8.2.

Thus

$$\exp(tJ_r(\lambda)) = e^{\lambda t} \begin{pmatrix} 1 & t & \frac{1}{2!}t^2 & \cdots & \frac{1}{(r-1)!}t^{r-1} \\ 0 & 1 & t & \cdots & \frac{1}{(r-2)!}t^{r-2} \\ 0 & 0 & 1 & \cdots & \frac{1}{(r-3)!}t^{r-3} \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & t \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Each row of e^{tN_r} contains the terms in the series for e^t starting at the diagonal and continuing until you reach the last column.

Now we can find the exponential of any matrix, assuming we can find its Jordan Form.

Example 8.6 The 3×3 matrix $A = \begin{pmatrix} 8 & 4 & -1 \\ -2 & 3 & 2 \\ 1 & 1 & 4 \end{pmatrix}$ satisfies $A = PJP^{-1}$ with $P = \begin{pmatrix} 3 & 4 & 0 \\ -2 & -2 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ and $J = J_3(5)$. Then $e^{tJ} = e^{5t} \begin{pmatrix} 1 & t & \frac{1}{2}t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$ and

$$e^{tA} = Pe^{tJ}P^{-1} = e^{5t} \begin{pmatrix} 1+3t & 4t+\frac{3}{2}t^2 & -t+3t^2 \\ -2t & 1-2t-t^2 & 2t-2t^2 \\ t & t+\frac{1}{2}t^2 & 1-t+t^2 \end{pmatrix}.$$

$\diamond$

8.5 The Column Method

Calculating e^{tA} is a lot of work, but our main application is to solve systems of ordinary differential equations (ODEs). For this, we only need to calculate $e^{tA}\boldsymbol{c}$ for initial vectors $\boldsymbol{c}$.

It turns out that we can calculate $e^{tA}\boldsymbol{c}$ without calculating the matrix e^{tA}.

For example, by Lemma 8.9 (c), if $A\boldsymbol{c} = \lambda\boldsymbol{c}$ then $e^{tA}\,\boldsymbol{c} = e^{\lambda t}\boldsymbol{c}$.

If A is diagonalisable then we have a basis of eigenvectors $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ with respective eigenvalues $\lambda_1, \ldots, \lambda_p$.

But any vector $\boldsymbol{c}$ can be written (uniquely) as $\sum_{i=1}^{p} c_i\boldsymbol{v}_i$, and then linearity implies that $e^{tA}\boldsymbol{c} = \sum_{i=1}^{p} e^{\lambda_i t} c_i\boldsymbol{v}_i$.

We can generalise this to any matrix A using a basis of generalised eigenvectors.

If $\boldsymbol{v}$ is a generalised eigenvector for eigenvalue λ then $(A-\lambda\mathrm{I})^k\mathbf{v}=\mathbf{0}$ for some k.

Next, we can write $A=\lambda\mathrm{I}+(A-\lambda\mathrm{I})$. Since $\lambda\mathrm{I}$ commutes with $(A-\lambda\mathrm{I})$, by Theorem 8.14 we have

$$e^{tA}=e^{\lambda t\mathrm{I}}\,e^{t(A-\lambda\mathrm{I})}=e^{\lambda t}e^{t(A-\lambda\mathrm{I})}\,.$$

But

$$\begin{aligned}e^{t(A-\lambda\mathrm{I})}\boldsymbol{v}&=\left(\mathrm{I}+t(A-\lambda\mathrm{I})+\tfrac{1}{2}t^2(A-\lambda\mathrm{I})^2+\cdots\right)\boldsymbol{v}\\&=\boldsymbol{v}+t(A-\lambda\mathrm{I})\boldsymbol{v}+\cdots+\tfrac{1}{(k-1)!}t^{k-1}(A-\lambda\mathrm{I})^{k-1}\boldsymbol{v}\end{aligned}$$

which is a finite sum.

Hence, for a generalised eigenvector $\boldsymbol{v}$, finding $e^{tA}\boldsymbol{v}$ only involves a finite summation.

We can calculate $e^{tA}\boldsymbol{c}$ for any vector $\boldsymbol{c}$ by writing $\boldsymbol{c}$ as a sum of generalised eigenvectors.

Example 8.7 Remember the matrix

$$A = \begin{pmatrix} 8 & 4 & -1 \\ -2 & 3 & 2 \\ 1 & 1 & 4 \end{pmatrix},$$

from Example 8.6. We saw that A has one eigenvalue $\lambda = 5$.

Therefore $GE_5(A) = \mathbb{C}^3$, so the standard basis of $\mathbb{C}^3$ is a basis of generalised eigenvectors.

This is the same matrix we found before, but with a lot less effort. The only part of the calculation we skipped was finding the characteristic polynomial. $\diamond$

If $A \in M_{p,p}(\mathbb{R})$ then $\exp(tA)$ is also a real matrix.
If the eigenvalues are complex then the method works over $\mathbb{C}$, as long as we choose conjugate eigenvectors.

Example 8.8 Find $\exp(tA)$ for $A = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$.

Solution.

8.6 Systems of Homogeneous Linear ODEs

We can apply our method to systems of linear ODEs.

Example 8.9 Let

$$A = \begin{pmatrix} -3 & 8 & -7 \\ -1 & 4 & -1 \\ 2 & -3 & 5 \end{pmatrix} \quad \text{and} \quad \boldsymbol{c} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} .$$

It is given that A has only one eigenvalue.

Solve the initial value problem $\boldsymbol{y}' = A\boldsymbol{y}$ with $\boldsymbol{y}(0) = \boldsymbol{c}$.

Solution.

◇

Definition 8.5
Let A be a square matrix. Any square matrix whose columns are

- linearly independent as functions of t, and
- solutions of $\boldsymbol{y}' = A\boldsymbol{y}$

is called a a fundamental matrix, denoted by $\Phi(t)$.

Fact: The matrix e^{tA} is a fundamental matrix for A.
Exercise: check this! Why are the columns independent?

The general solution to $\boldsymbol{y}' = A\boldsymbol{y}$ is $\boldsymbol{y} = \Phi(t)\boldsymbol{c}$.

For an initial value problem we also have $\boldsymbol{y}(0) = \Phi(0)\boldsymbol{c}$, so $\boldsymbol{c} = \Phi^{-1}(0)\,\boldsymbol{y}(0)$ and hence

$$\boldsymbol{y}(t) = \Phi(t)\,\Phi^{-1}(0)\,\boldsymbol{y}(0)\,.$$

The column method for the exponential produces a fundamental matrix.

Let $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ be a basis for $\mathbb{C}^n$ consisting of generalised eigenvectors of A. Then $\{\exp(tA)\boldsymbol{v}_1, \ldots, \exp(tA)\boldsymbol{v}_n\}$ is a set of linearly independent solutions of $\boldsymbol{y}' = A\boldsymbol{y}$, so let

$$\Phi(t) = [\exp(tA)\boldsymbol{v}_1 \mid \cdots \mid \exp(tA)\boldsymbol{v}_n].$$

This method can be applied to higher order linear differential equations, or higher order systems.

Example 8.10. Solve $y''' - 4y'' + 5y' - 2y = 0$.

Solution.

$\diamond$

This explains the process that you've been taught for solving a linear differential equation when the characteristic polynomial has a repeated root: keep multiplying by t until you get enough solutions.

8.7 Inhomogeneous Systems of ODEs

For $A \in M_{p,p}(\mathbb{C})$, let

$$\boldsymbol{y}' = A\boldsymbol{y} + \boldsymbol{b}(t) \tag{4}$$

be an inhomogeneous system of ODEs.

If $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ are two solutions to (4) then

$$(\boldsymbol{y}_1 - \boldsymbol{y}_2)' = A(\boldsymbol{y}_1 - \boldsymbol{y}_2),$$

so $\boldsymbol{y}_1 - \boldsymbol{y}_2$ is a solution to the corresponding homogeneous system $\boldsymbol{y}' = A\boldsymbol{y}$. Hence the general solution to (4) is

$$\boldsymbol{y} = \boldsymbol{y}_h + \boldsymbol{y}_p$$

where $\boldsymbol{y}_h$ is the general solution of the homogeneous system $\boldsymbol{y}' = A\boldsymbol{y}$, and $\boldsymbol{y}_p$ is any particular solution of (4).

Method of variation of parameters: Let $\boldsymbol{y}(t) = \Phi(t)\,\boldsymbol{z}(t)$, where $\Phi(t)$ is a fundamental matrix for $\boldsymbol{y}' = A\boldsymbol{y}$.

Note: The entries of $\boldsymbol{z}$ can vary with t, hence the name of the method. (Compare with $\boldsymbol{y}_h(t) = \Phi(t)\,\boldsymbol{c}$ where $\boldsymbol{c}$ is constant.)

If $\boldsymbol{y}(t) = \Phi(t)\boldsymbol{z}(t)$ then

If $\boldsymbol{y}(t)$ is a solution of (4) then $\boldsymbol{y}' = A\boldsymbol{y} + \boldsymbol{b}(t)$. Hence

$$\Phi(t)\boldsymbol{z}'(t) = \boldsymbol{b}(t), \quad \text{so} \quad \boldsymbol{z}'(t) = \Phi^{-1}(t)\boldsymbol{b}(t)\,. \tag{5}$$

But then

$$\boldsymbol{y}(t) = \Phi(t)\boldsymbol{z}(t) = \Phi(t)\boldsymbol{c} + \Phi(t)\int_0^t \Phi^{-1}(s)\boldsymbol{b}(s)ds. \tag{6}$$

The first term on the right hand side of Equation (6) is $\boldsymbol{y}_h(t) = \Phi(t)\,\boldsymbol{c}$, and the second term is $\boldsymbol{y}_p(t)$.

Also (6) implies that $\boldsymbol{y}(0) = \Phi(0)\boldsymbol{c}$. If an initial value $\boldsymbol{y}(0)$ is given then we can solve for $\boldsymbol{c}$.

Another formulation: Suppose that we use the fundamental matrix $\Phi(t) = e^{tA}$ in (5). Then the general solution to (4) becomes

$$\boldsymbol{y} = e^{tA}\,\boldsymbol{z}(t) \quad \text{where} \quad \boldsymbol{z}'(t) = e^{-tA}\,\boldsymbol{b}(t)\,. \tag{7}$$

This is perfect for the column method: we only ever need to know e^{tA} times a vector, without needing to know e^{tA} itself.

If we have an initial value problem with $\boldsymbol{y}(0) = \boldsymbol{y}_0$ then $\boldsymbol{z}(0) = \boldsymbol{y}(0) = \boldsymbol{y}_0$.

Example 8.11

Solve $\mathbf{y}' = \begin{pmatrix} 2 & -3 \\ -2 & 1 \end{pmatrix} \mathbf{y} + \begin{pmatrix} e^t \\ e^t \end{pmatrix}, \quad \mathbf{y}(0) = \begin{pmatrix} 4 \\ -1 \end{pmatrix}.$

Solution.

Example 8.12 Solve

$$x' = x - y + 1$$
$$y' = 2x - y$$

where $x = x(t)$, $y = y(t)$ and $x(0) = y(0) = 0$.

Solution.

Example 8.13 Solve $\boldsymbol{y}' = A\boldsymbol{y} + \boldsymbol{b}$ where

$$A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 0 \\ -1 & -1 & 3 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} e^{2t} \\ e^{2t} \\ 0 \end{pmatrix} \quad \text{and} \quad \boldsymbol{y}(0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

It is given that A has only one eigenvalue.

Solution.

◇

Example 8.14 (2005 exam, slightly modified)
Let

$$A = \begin{pmatrix} -2 & 1 & 0 \\ -4 & 0 & -1 \\ 0 & -3 & -2 \end{pmatrix}, \quad \boldsymbol{v} = \begin{pmatrix} -1 \\ 0 \\ 4 \end{pmatrix}, \quad \boldsymbol{w} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}.$$

(a) Show that $\boldsymbol{v}$ is an eigenvector of A and give its eigenvalue.

(b) Show that $\boldsymbol{w} \in GE_{-1}(A)$.

(c) Give the Jordan form J of A and find a matrix P such that $J = P^{-1}AP$.

(d) Solve the system of differential equations

$$\boldsymbol{y}'(t) = A\boldsymbol{y} + e^{-t}\boldsymbol{w}$$

subject to the initial conditions $\boldsymbol{y}(0) = \boldsymbol{v}$.

◇

Key points for Chapter 8

- Homogeneous and inhomogeneous differential equations, p. 2.
- Coupled recurrence example, p. 11.
- Entrywise convergence, p. 12.
- ∞-norm, operator norm and Frobenius norms of matrices, p. 13.
- Convergence in a given norm, p. 15.
- Matrix exponential, p. 22.
- Fundamental matrix, p. 45.